Limits of Distributed Quantum Computing



Francesco d'Amore

Based on [STOC '24, STOC '25a, STOC '25b, SODA '26]

Joint works with Amirreza Akbari, Alkida Balliu, Sebastian Brandt, Filippo Casagrande, Xavier Coiteux-Roy, Massimo Equi, Rishikesh Gajjala, Barbara Keller, Fabian Kuhn, François Le Gall, Henrik Lievonen, Darya Melnyk, Augusto Modanese, Dennis Olivetti, Shreyas Pai, Marc-Olvier Renou, Václav Rozhon, Gustav Schmid, Jukka Suomela, Lucas Tendick, Isadora Veeren

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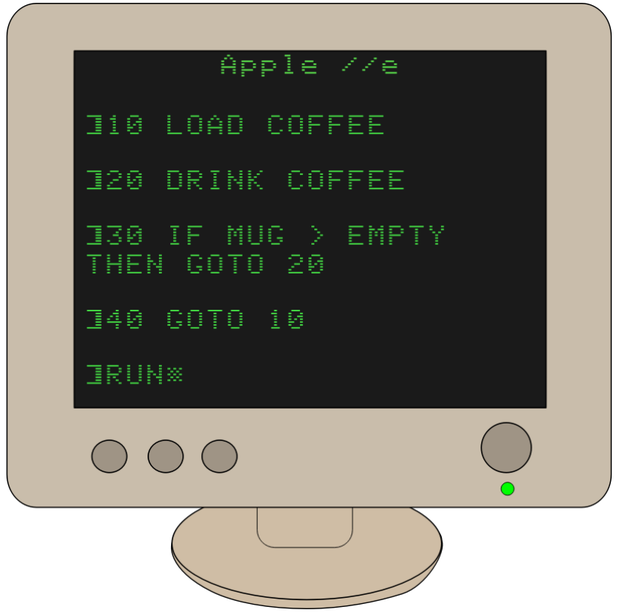
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2. **Classical lower bounds**: the indistinguishability argument
3. **Properties of distributed algorithms**: independence and non-signaling
4. **Super-quantum models**: bounded-dependence and non-signaling model
5. **State of the art results**
6. **Quantum advantage**

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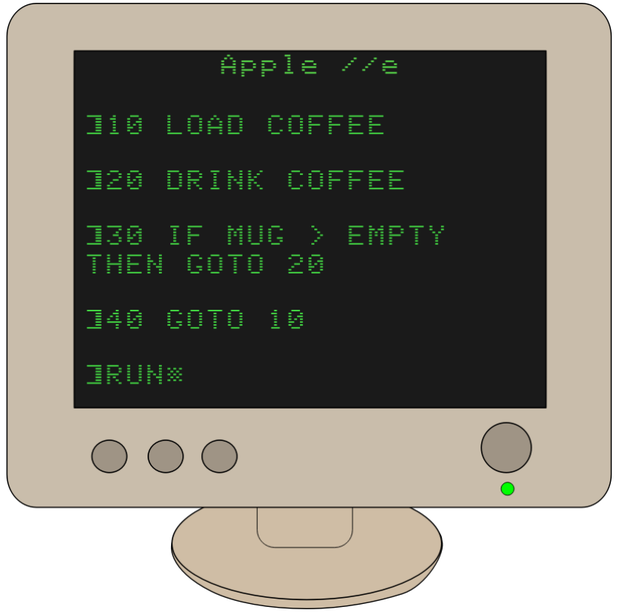
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  - use commands like *send a message through this communication port*, etc.



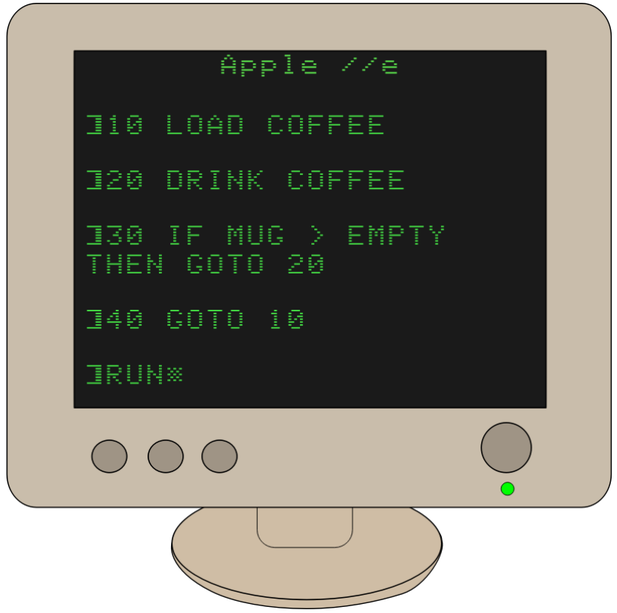
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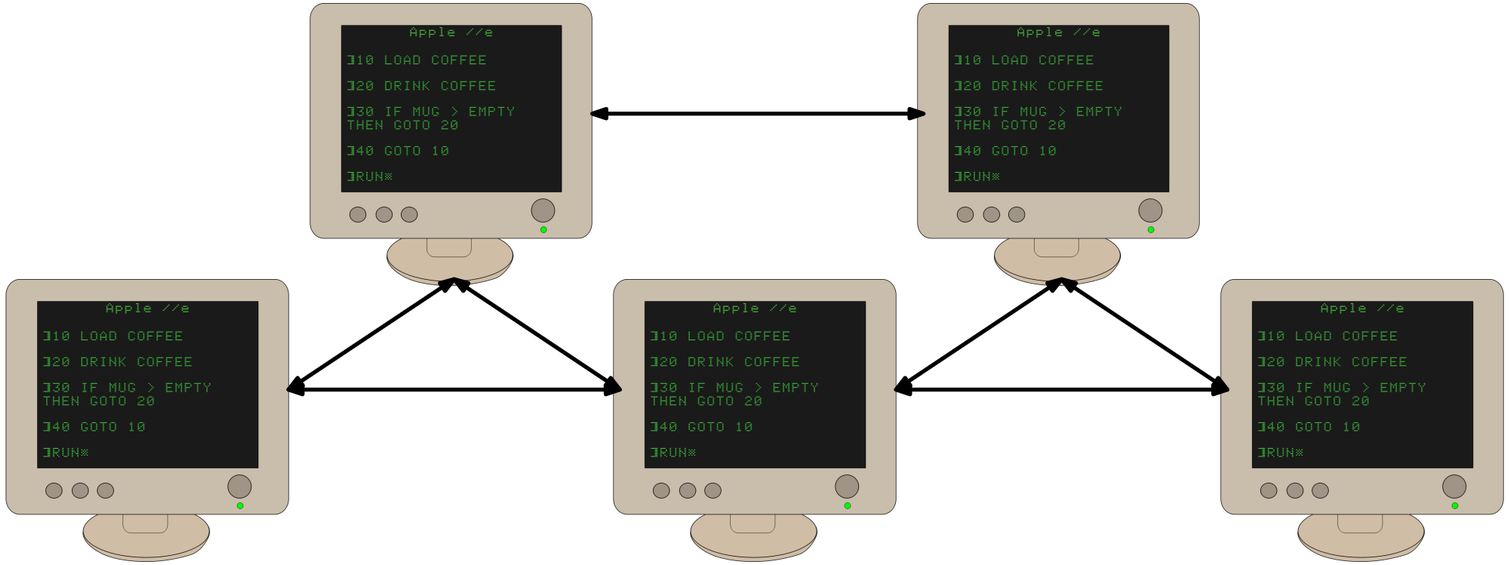
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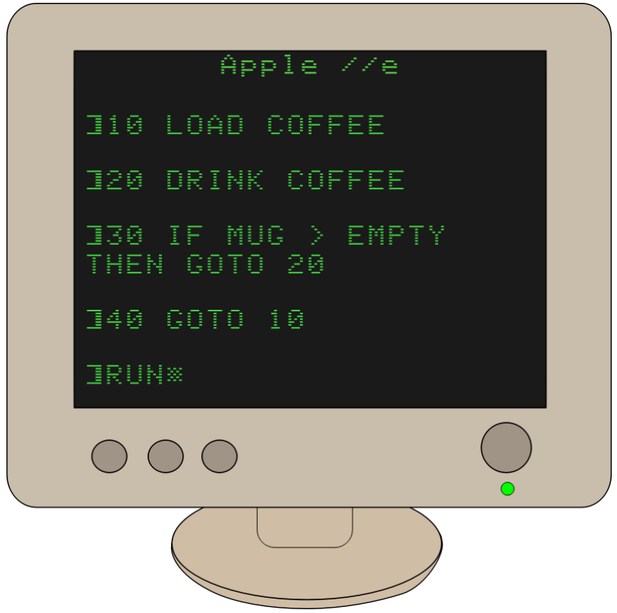
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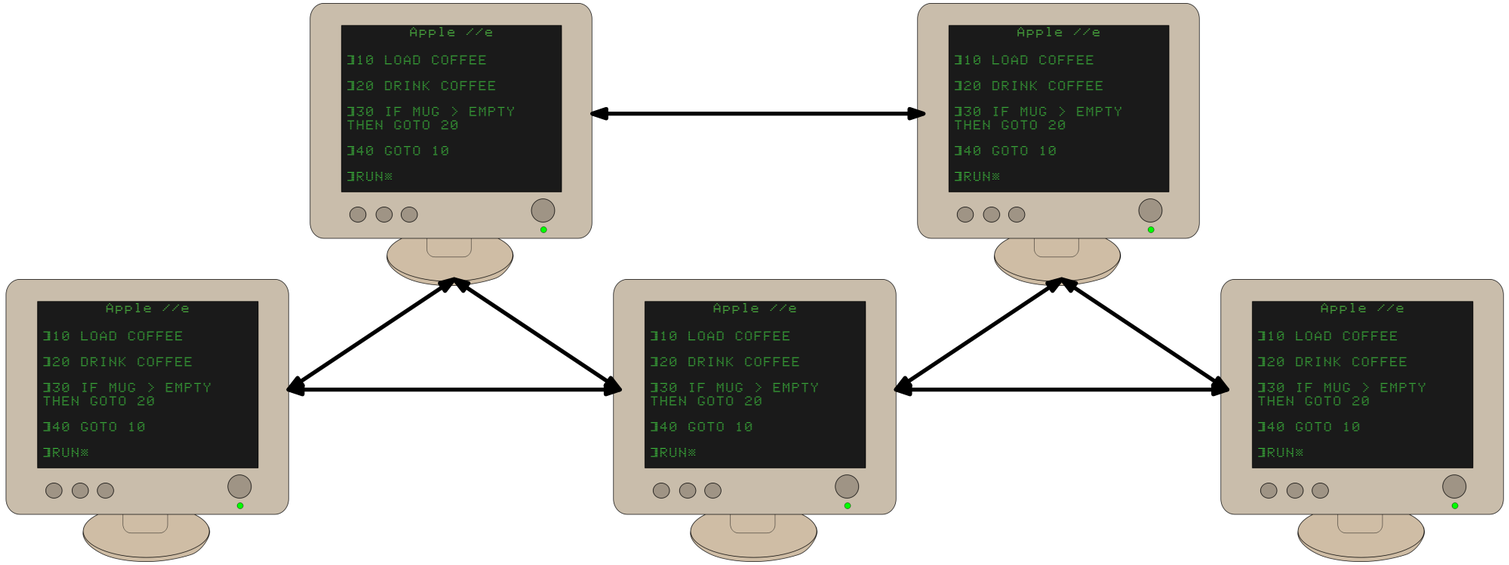




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- **Switch everything on**, see what happens

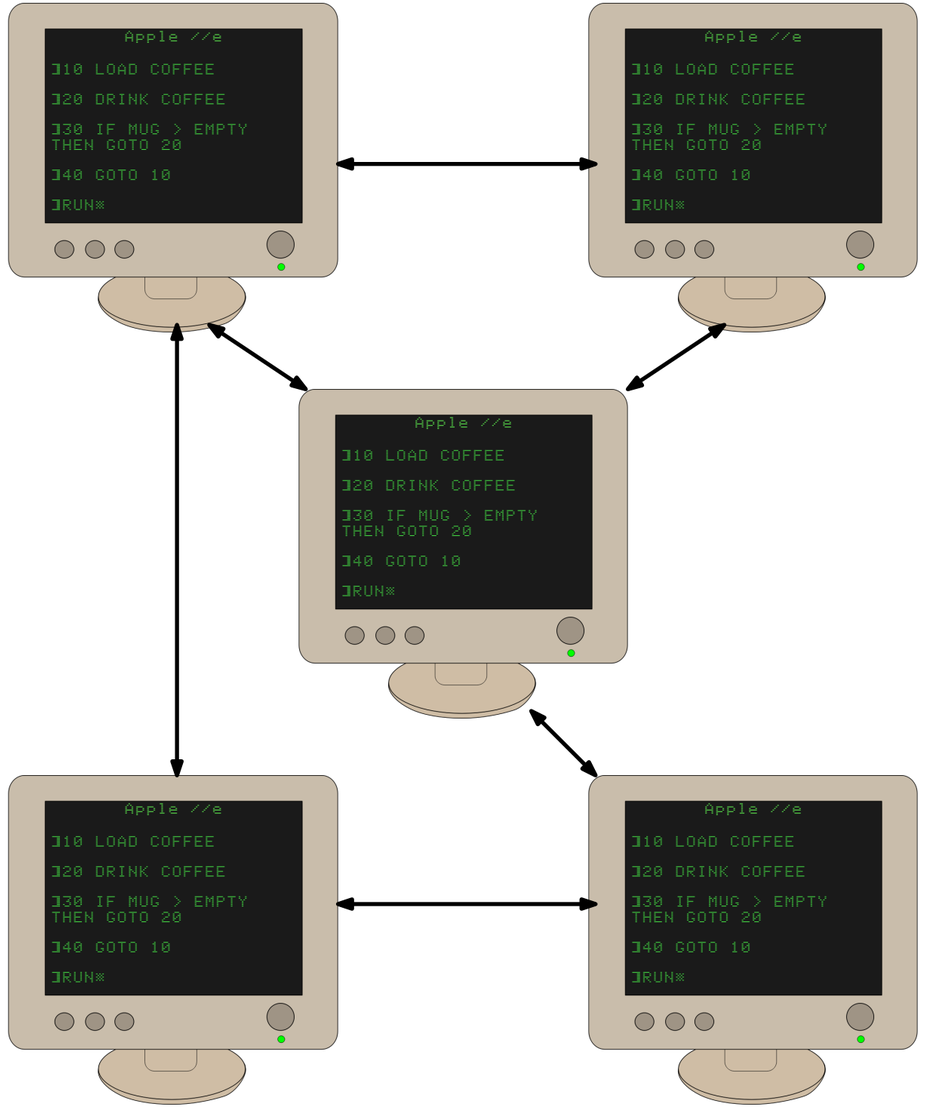




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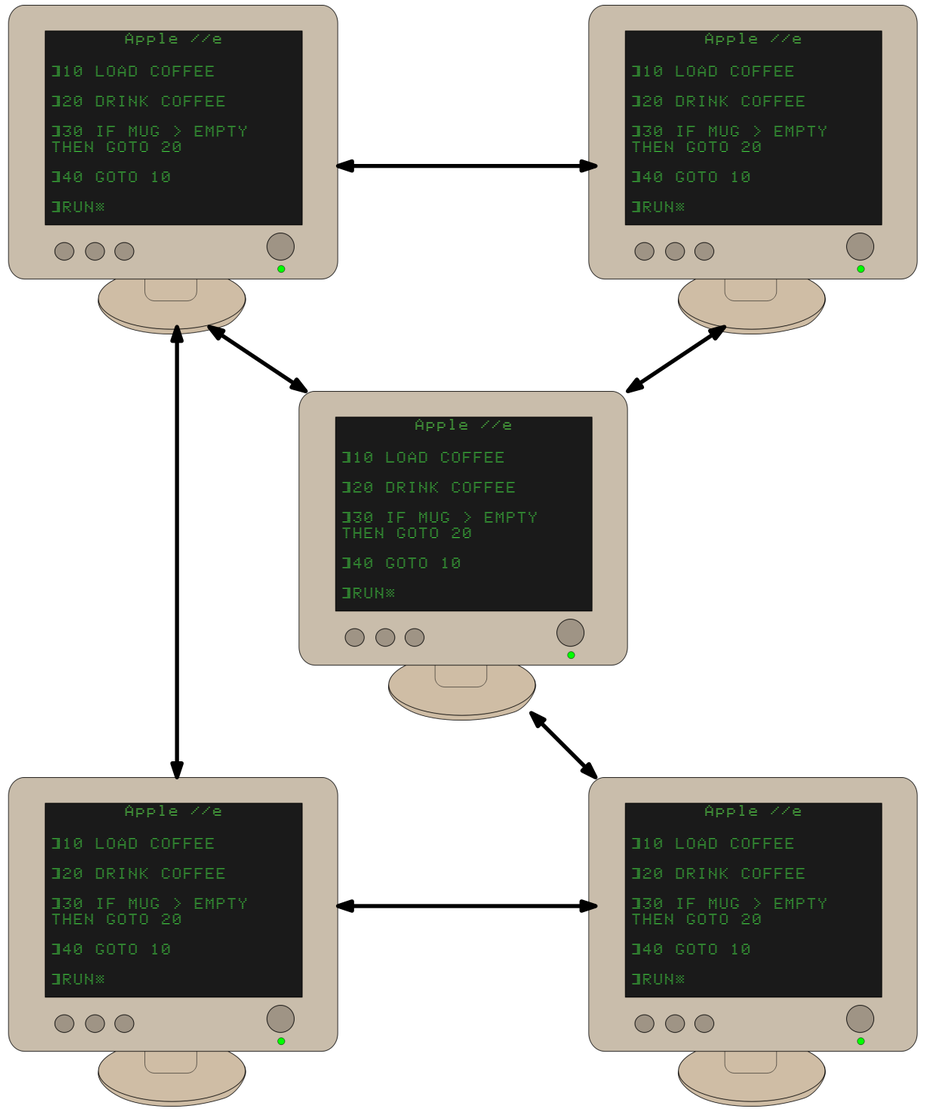
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  - identical computers



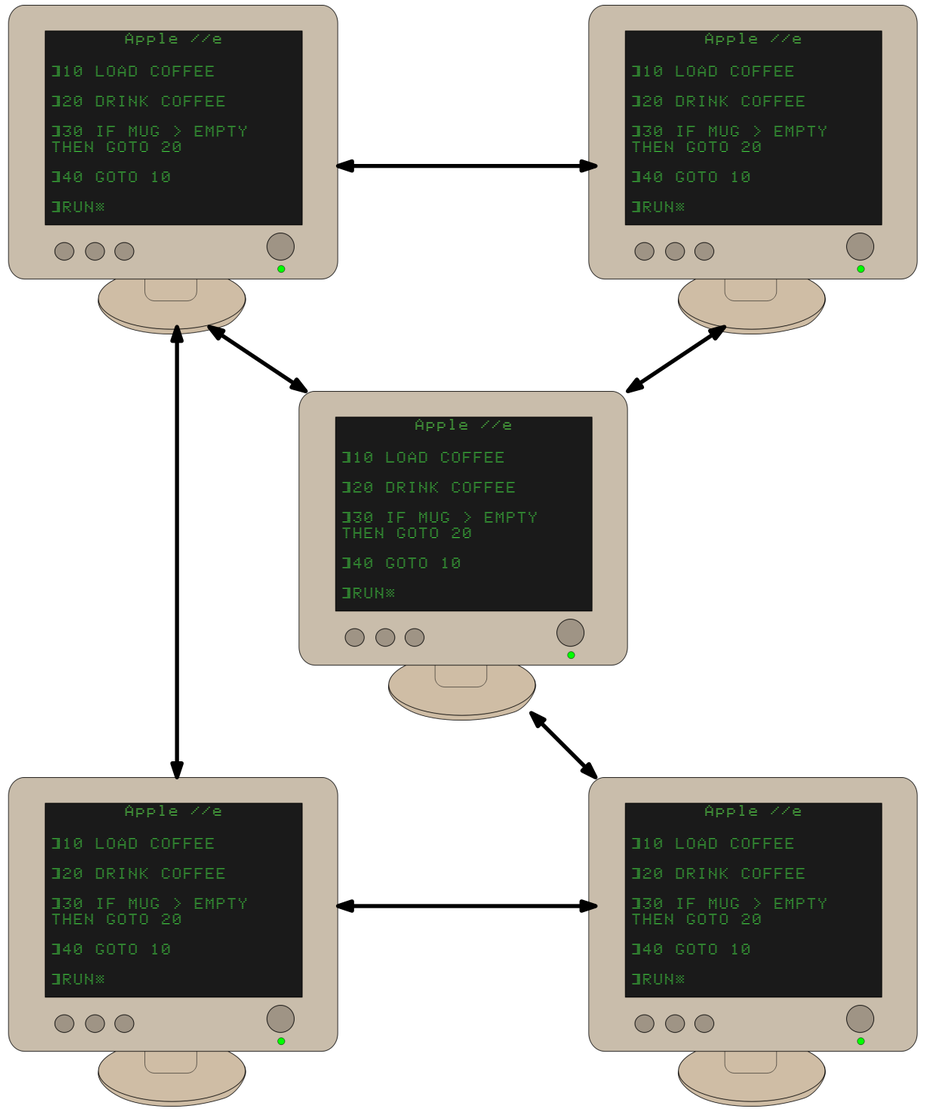
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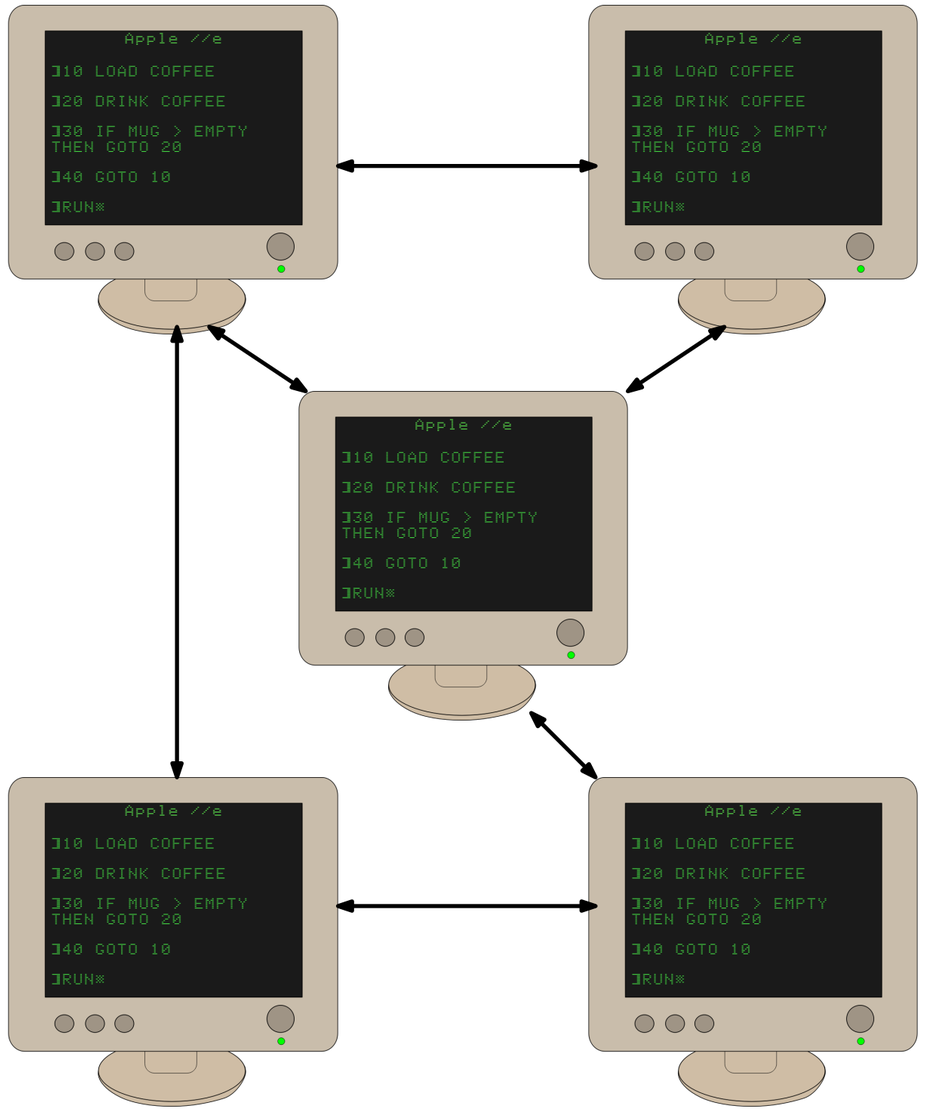
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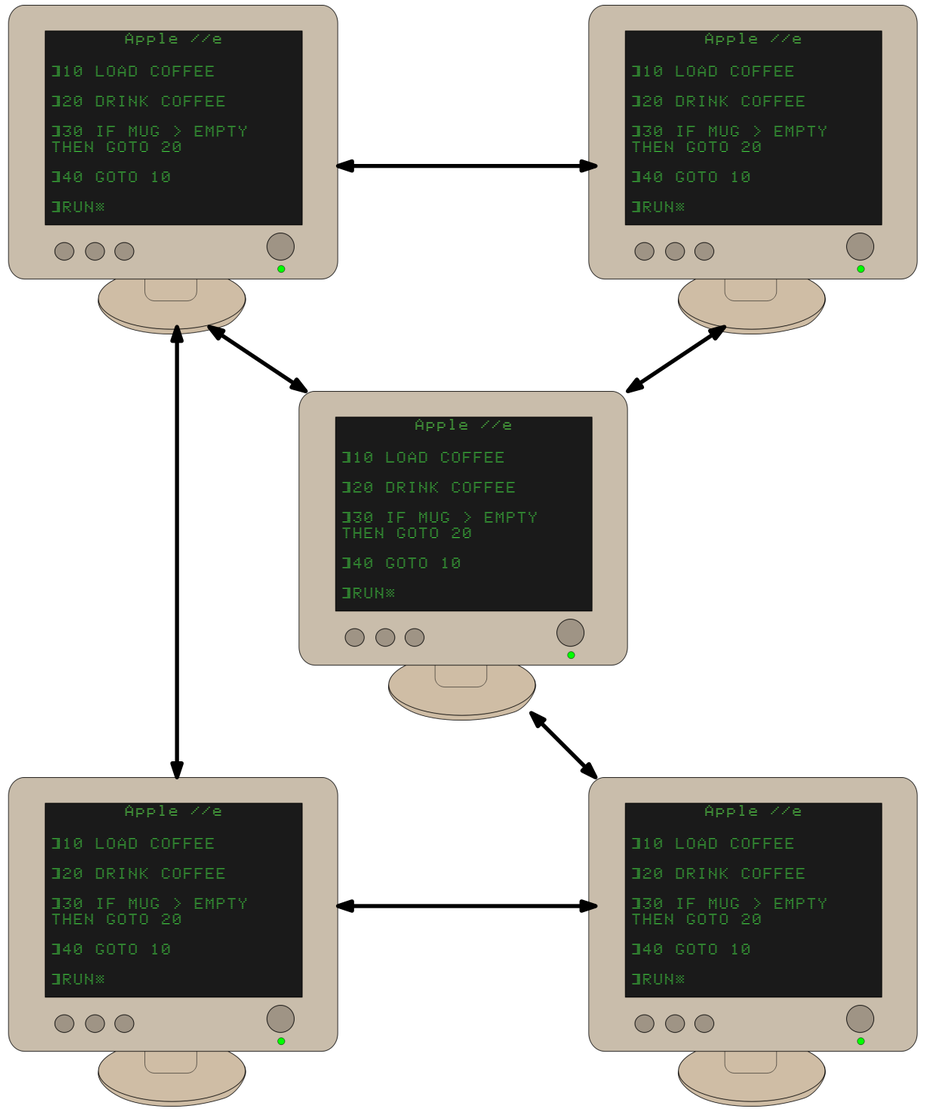
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- **Challenge**: what to do in the middle of a network?



Computability with identical computers

- **Problem**: $\mathbf{2}$-coloring $\mathbf{2}$-paths

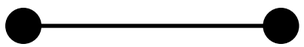
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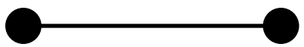
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- *What program?*

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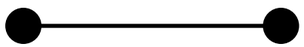
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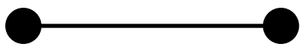
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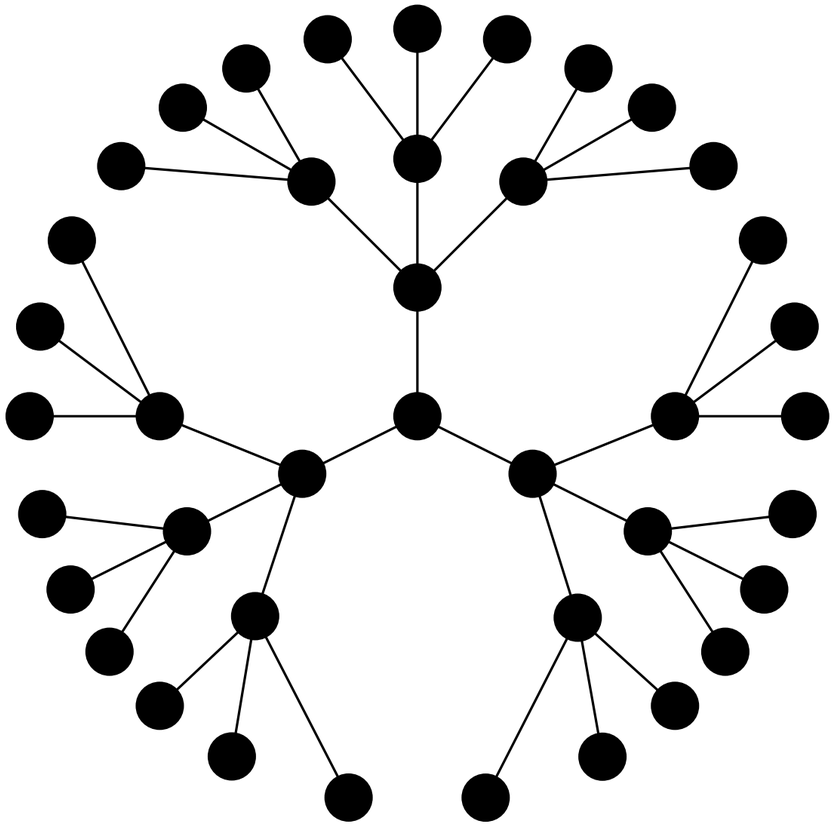


- *What program?* * **Deterministically impossible** if the two nodes are truly identical
- **Assumption**: *randomness* or *unique identifiers*
- **Algorithm**: - exchange identifier
  - if my identifier > received identifier, output blue
  - otherwise output red

The LOCAL model

[Linial FOCS ’87 & SICOMP ’92]

- **Distributed network** of n processors/nodes
  - graph $G = (V, E)$ with $|V| = n$
  - E: communication links
  - each node in V runs the same algorithm

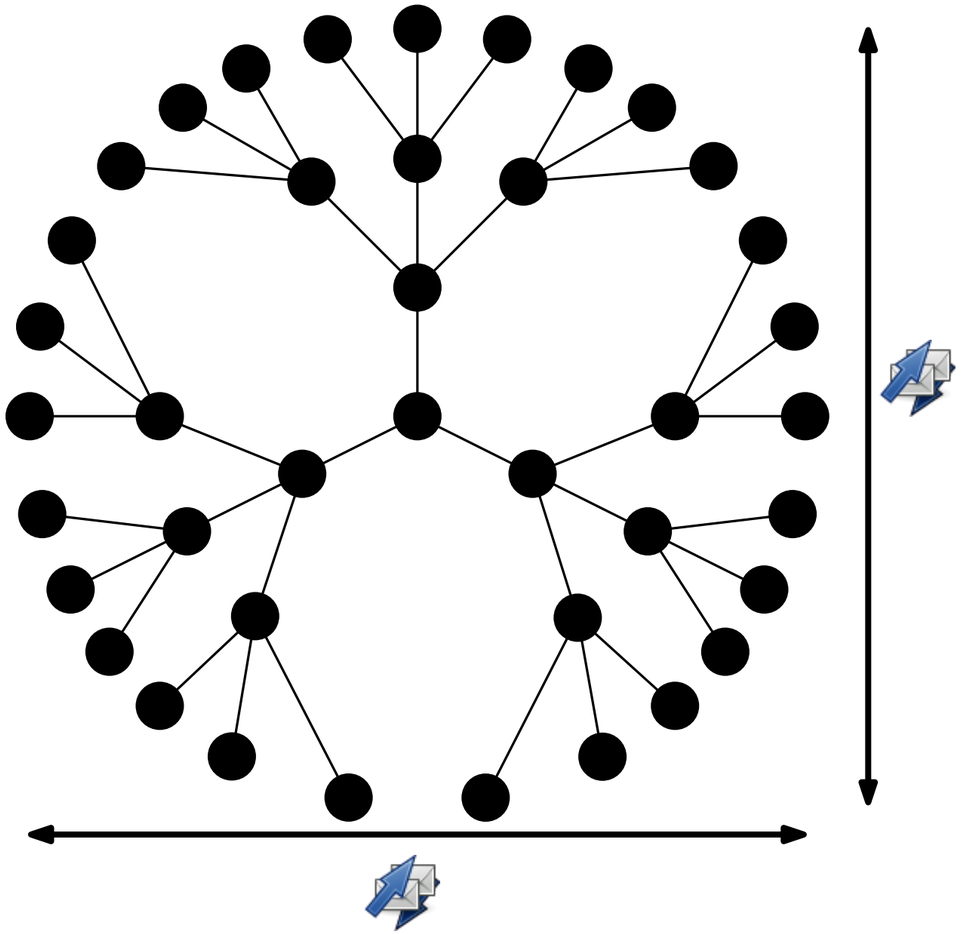


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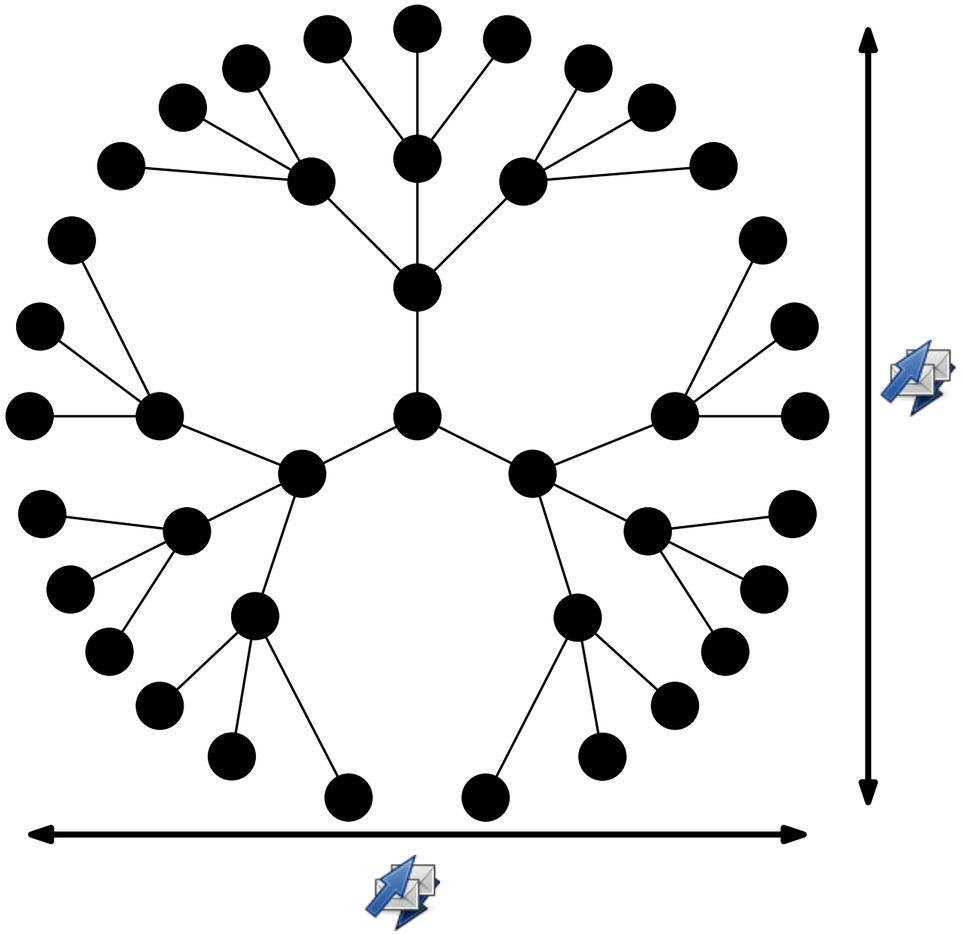
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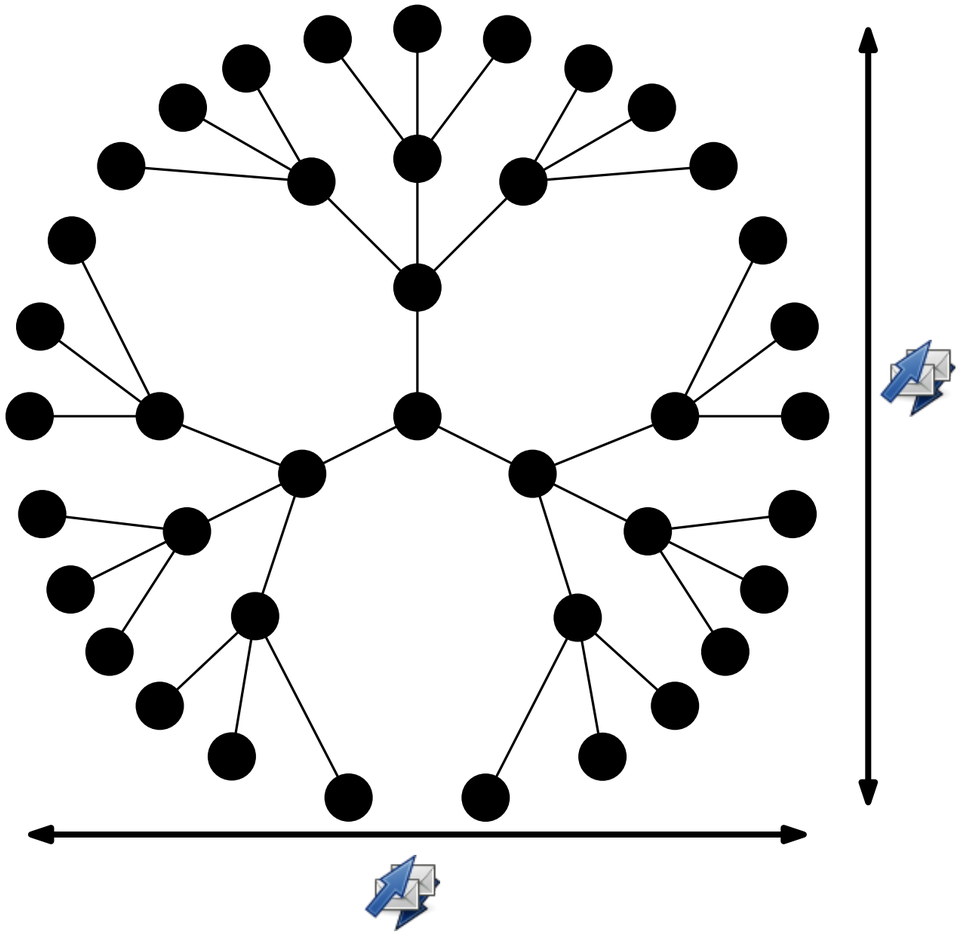
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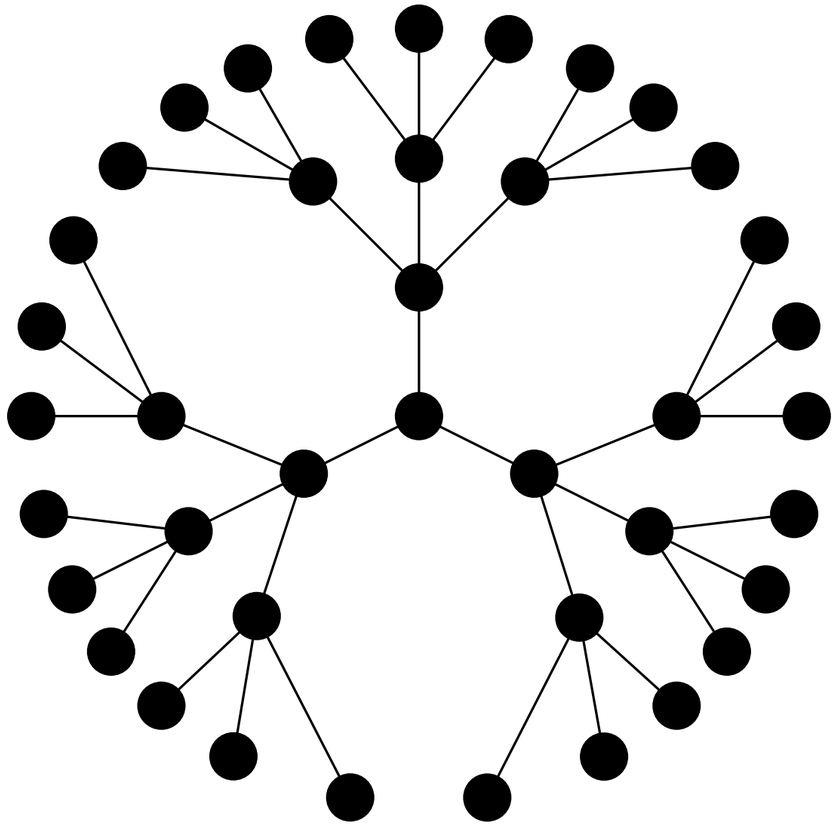
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Local view

Complexity measure: number of communication rounds

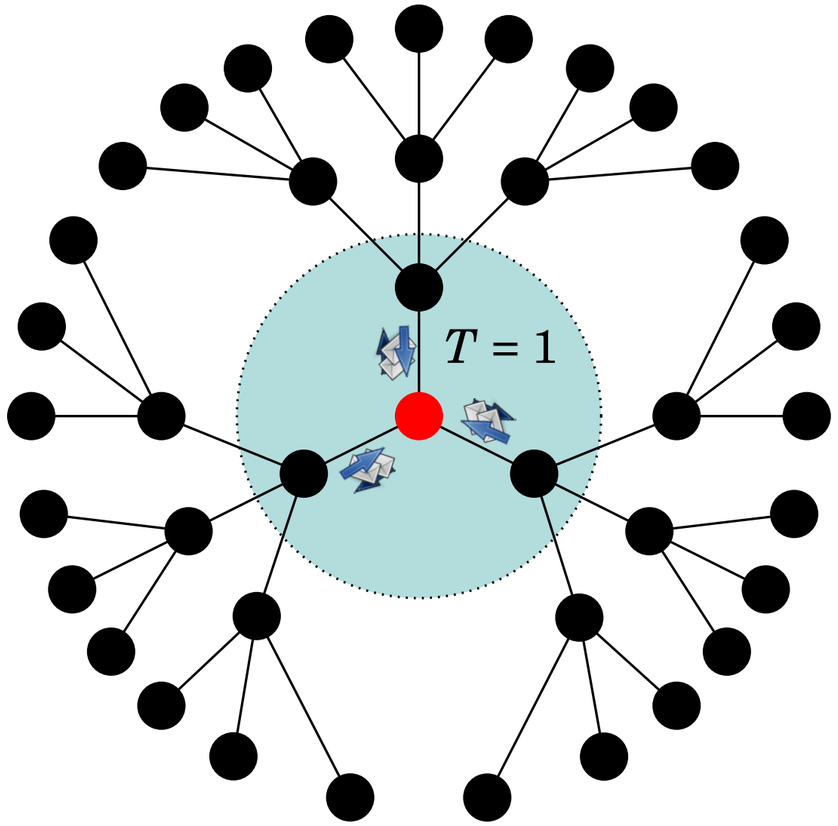


- What do we know after T rounds?

Local view

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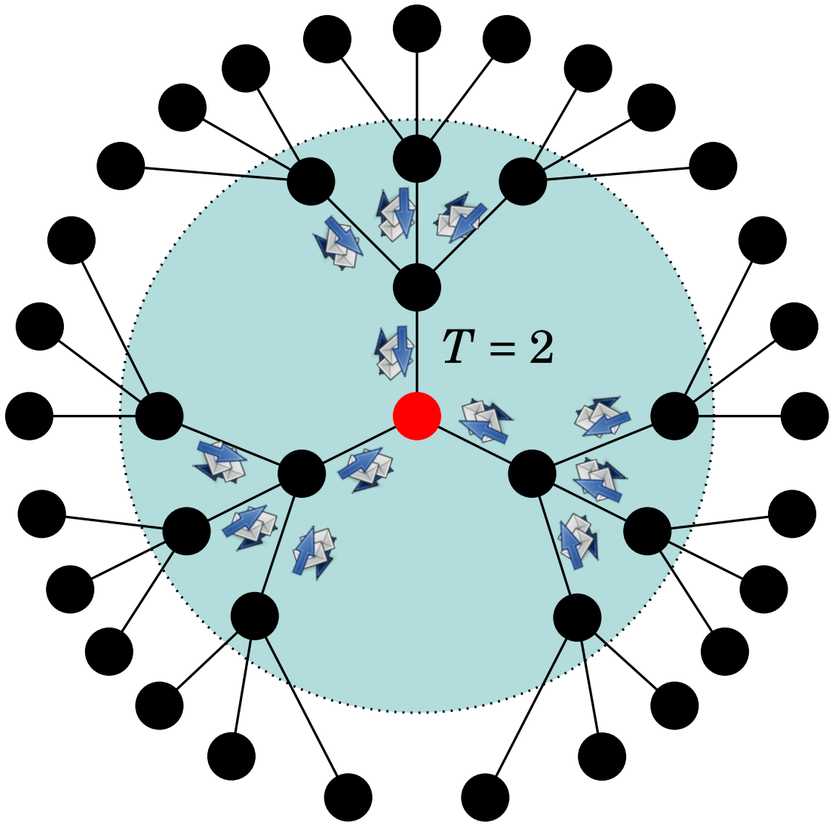


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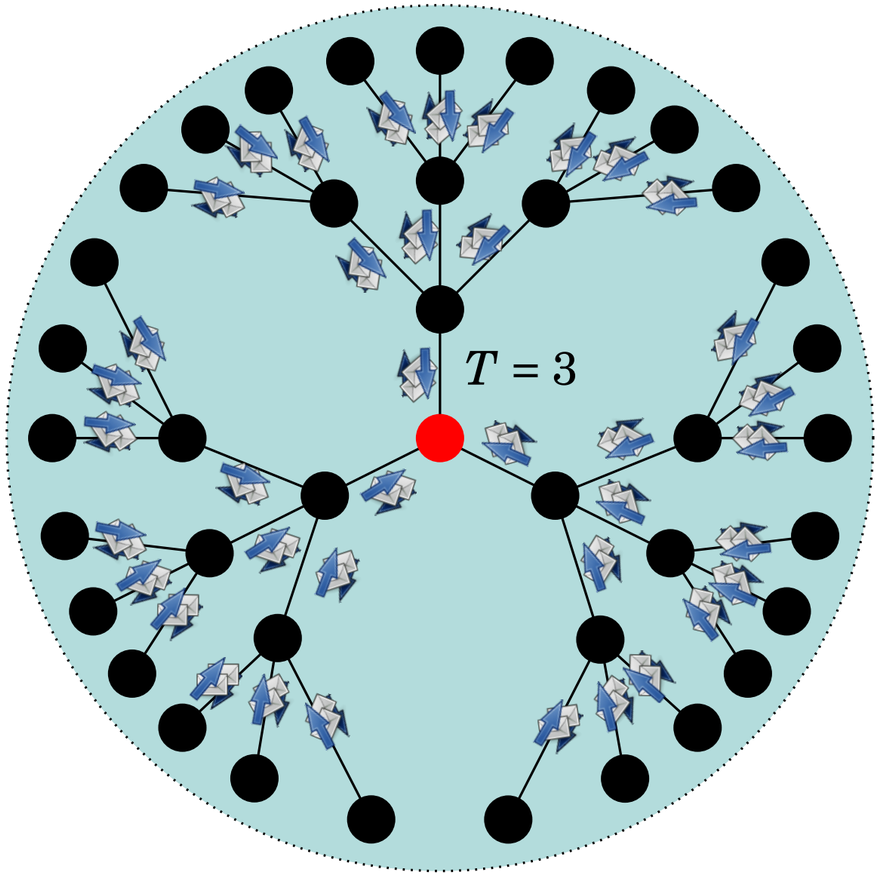


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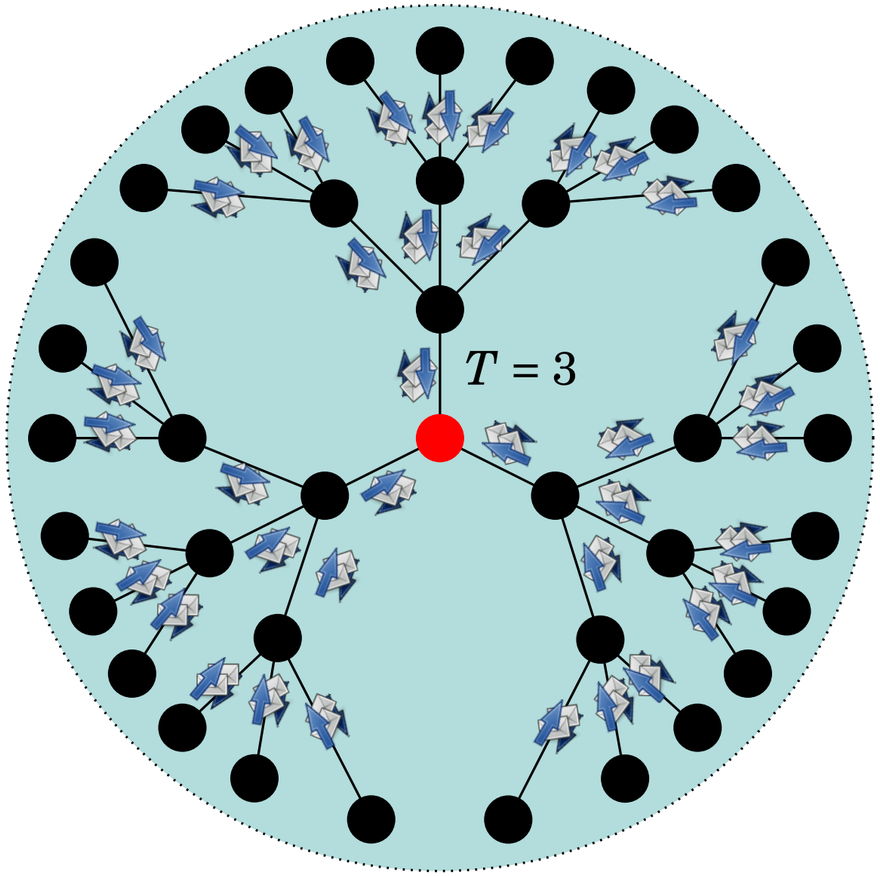


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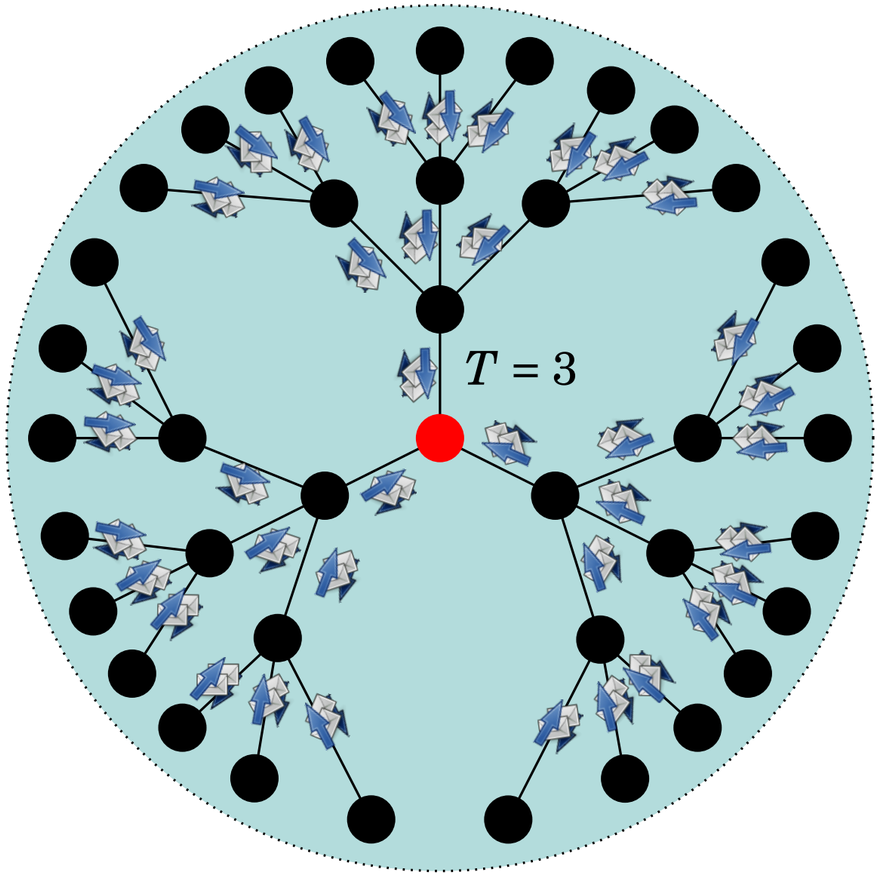
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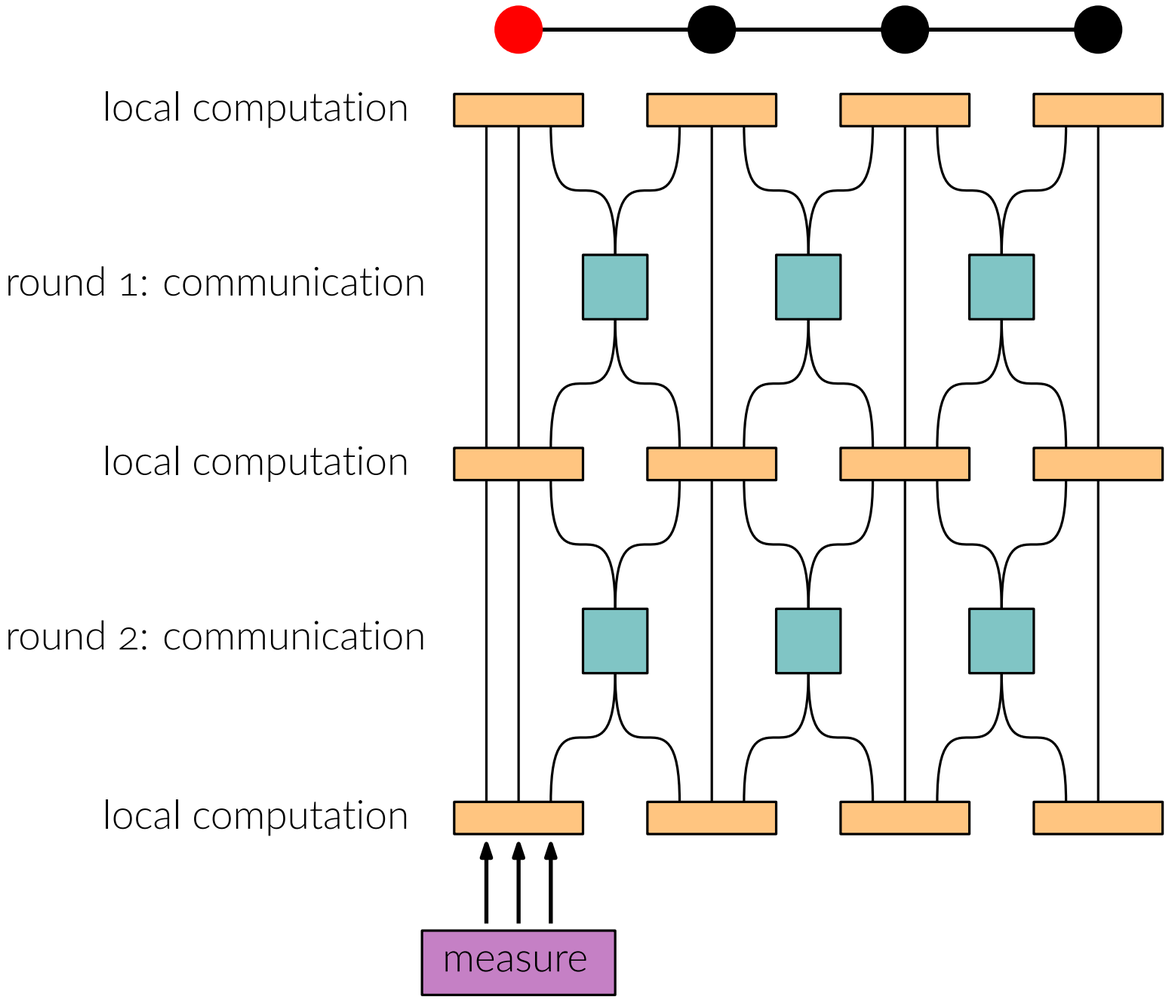
knowledge after T rounds of communication

- **Equivalence**: T-round algorithm ≈ funtion mapping radius-T neighborhoods to local outputs
- Locality $T = n + 1 \geq \mathbf{diam}(G) + 1$ is always sufficient to solve any problem: gathering algorithm

Quantum-LOCAL

[Gavoille, Kosowski, and Markiewicz, DISC '09]

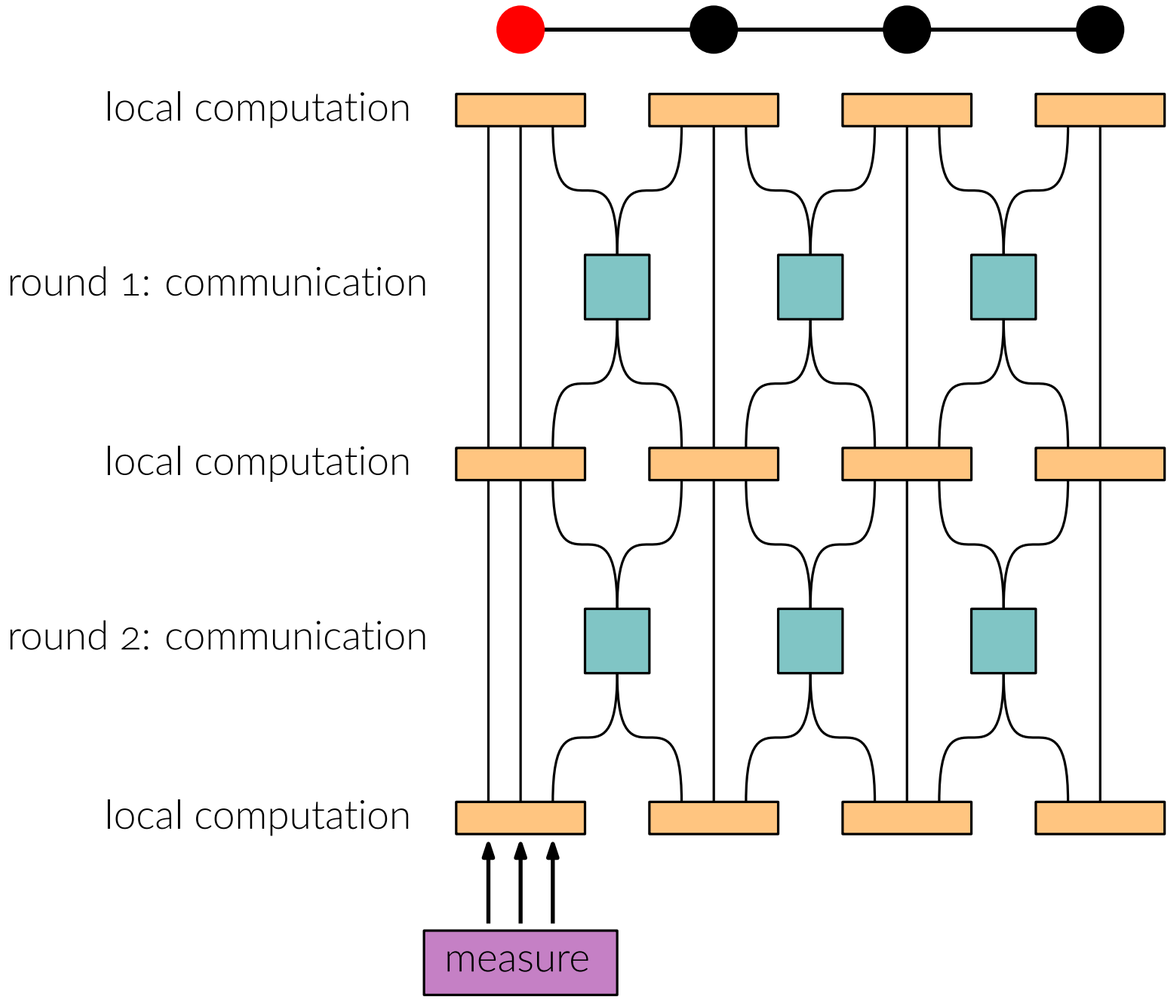
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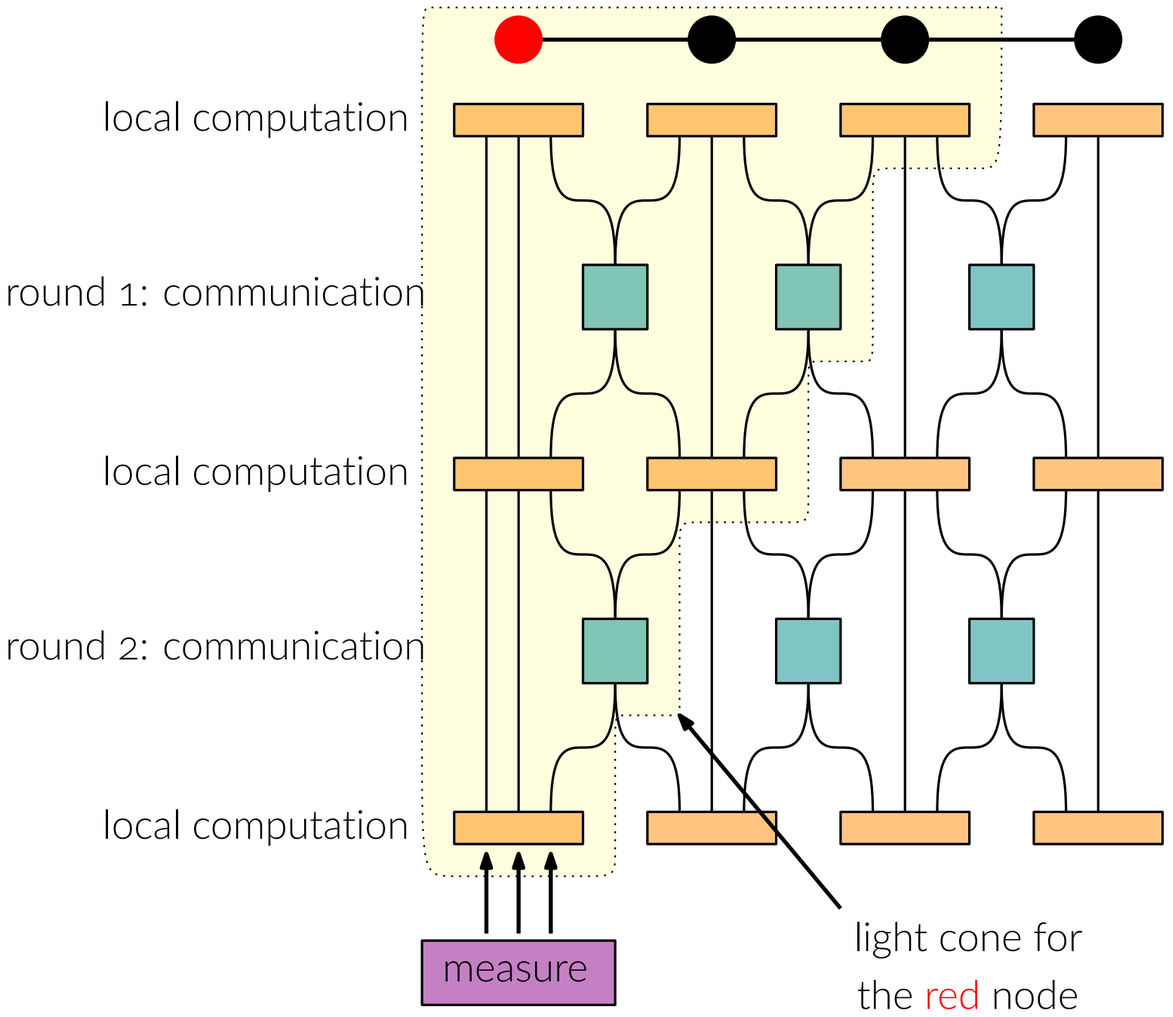
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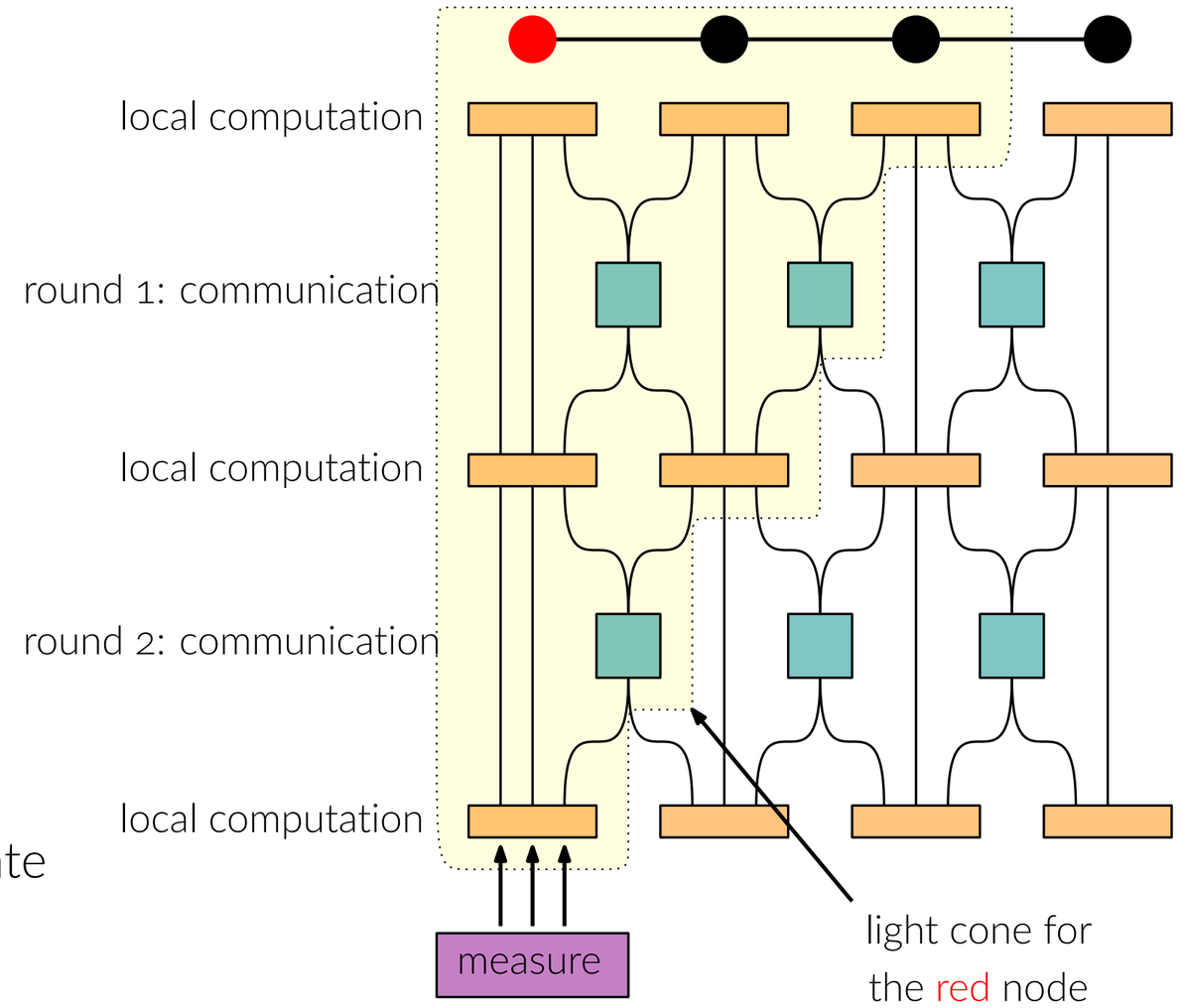
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  - quantum states cannot be *cloned* (no-cloning theorem)



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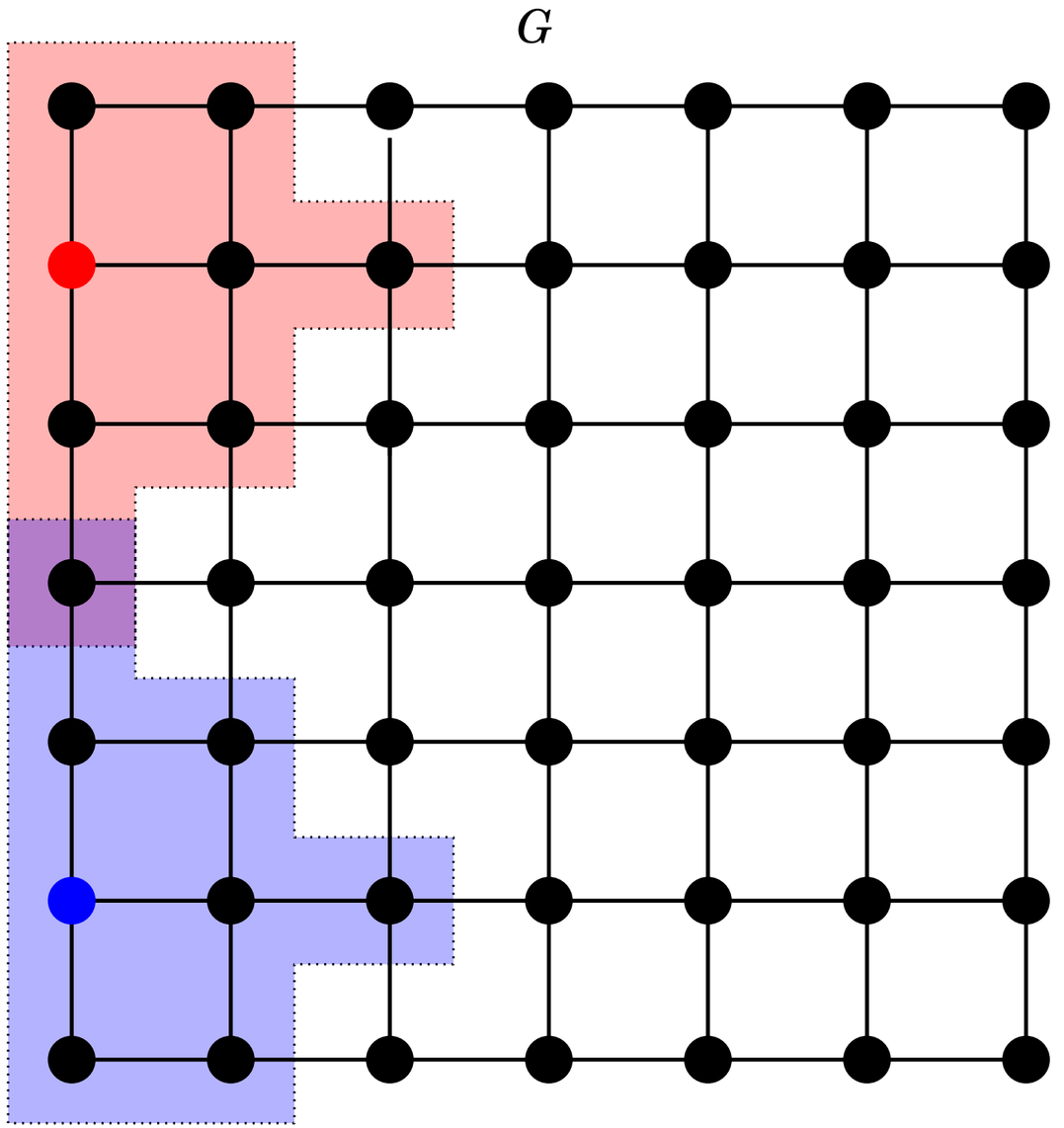
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  - quantum states cannot be *cloned* (no-cloning theorem)
- Still, locality identifies how *far* nodes need to communicate



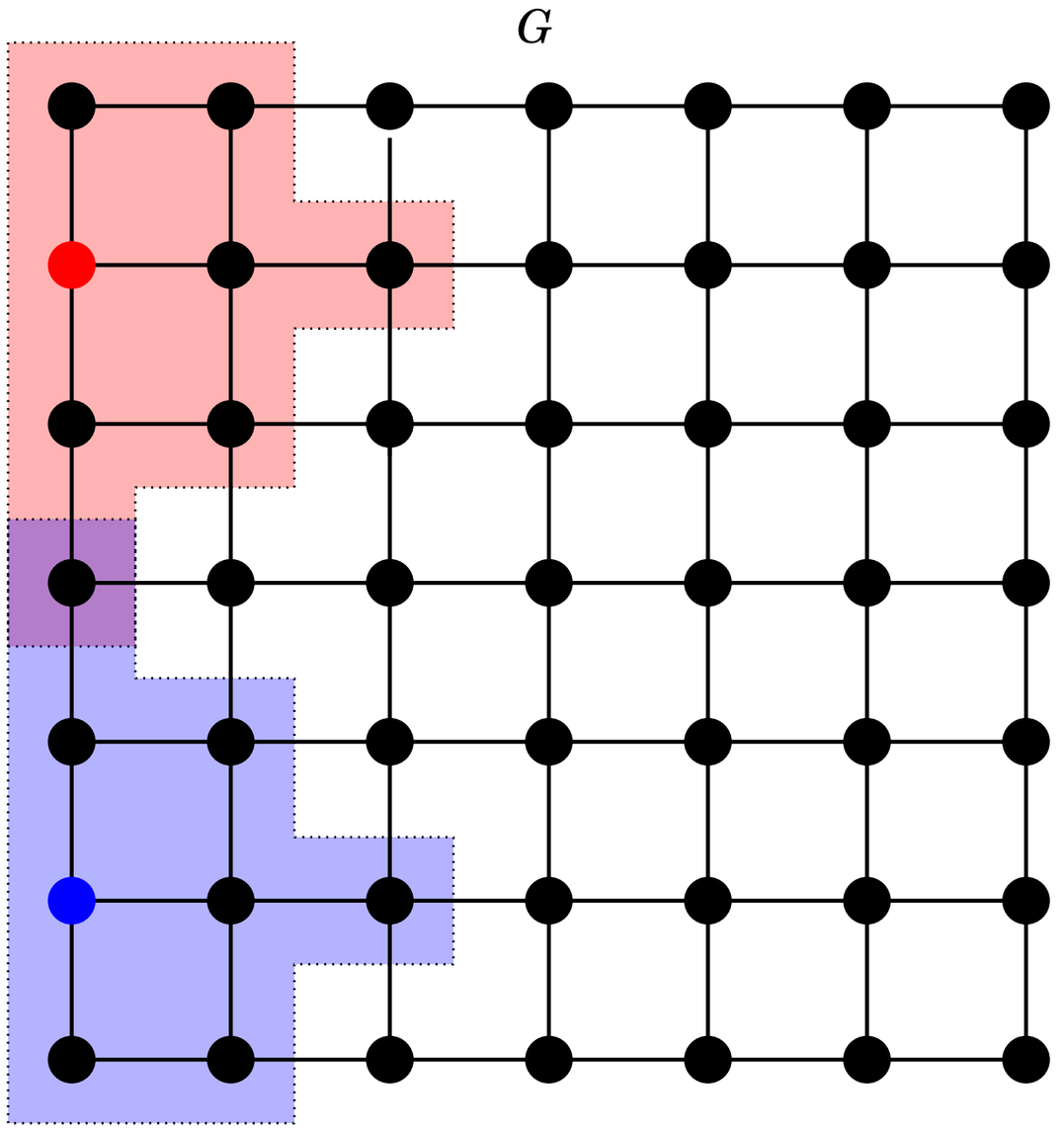
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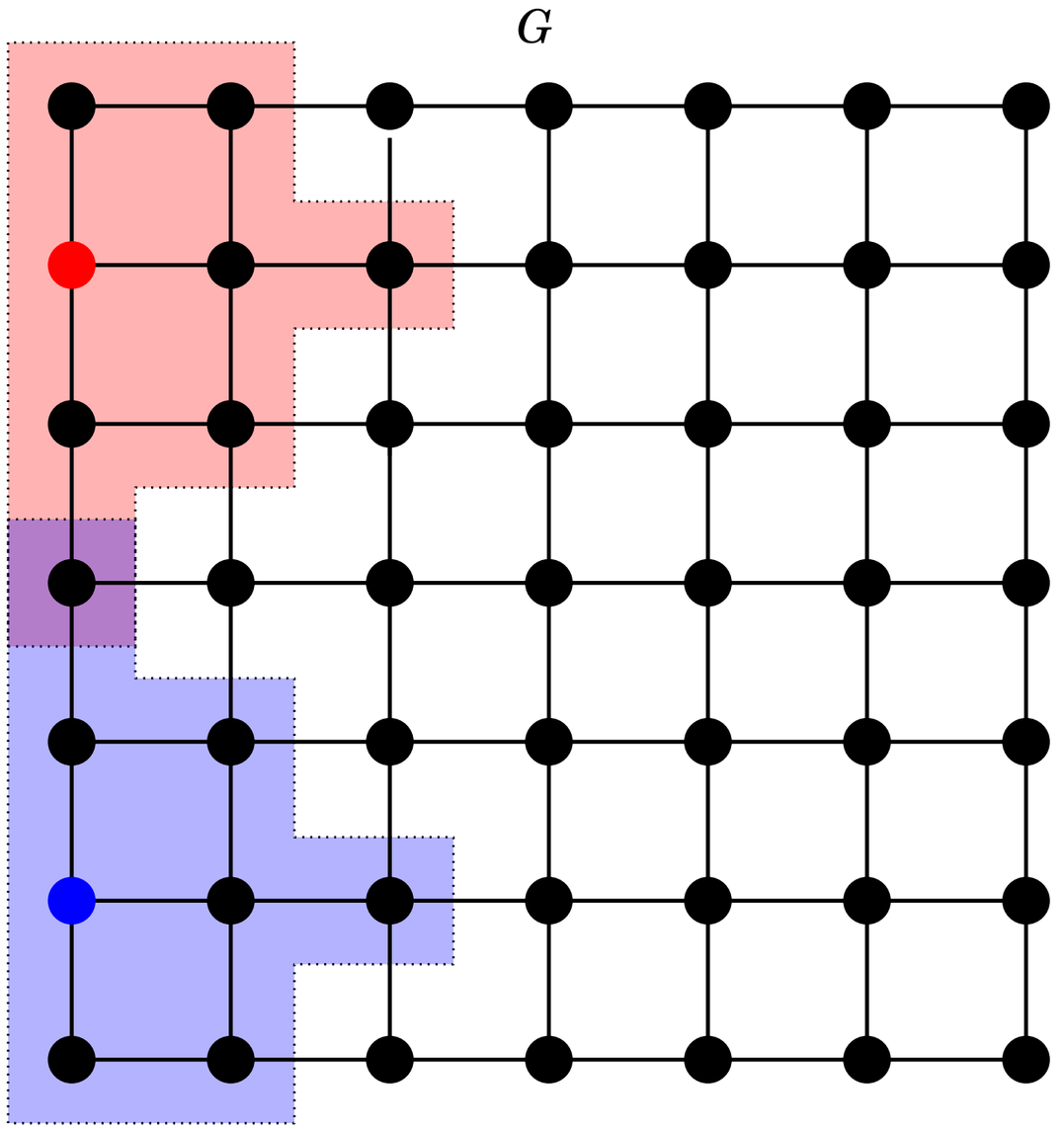
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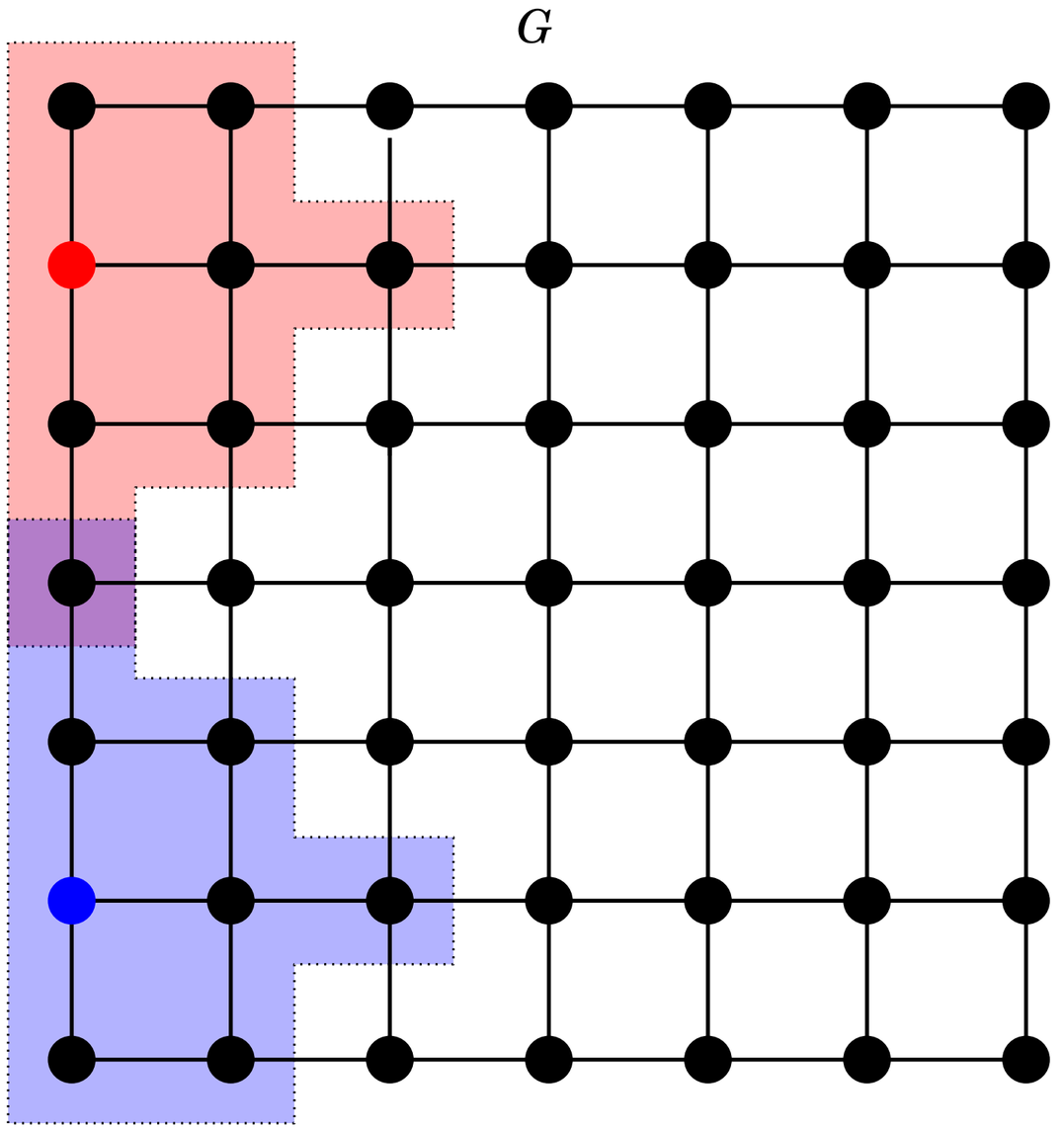
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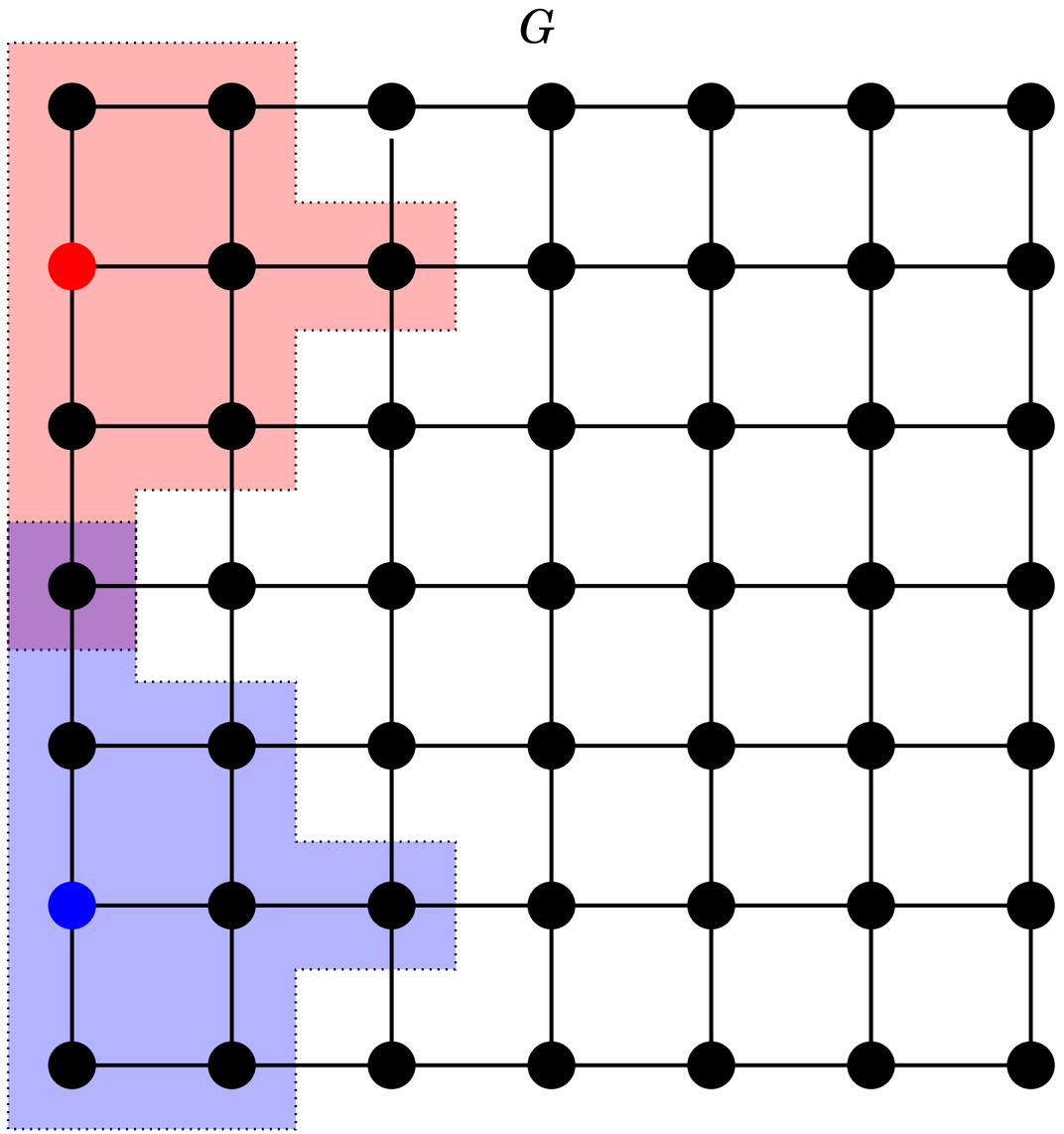


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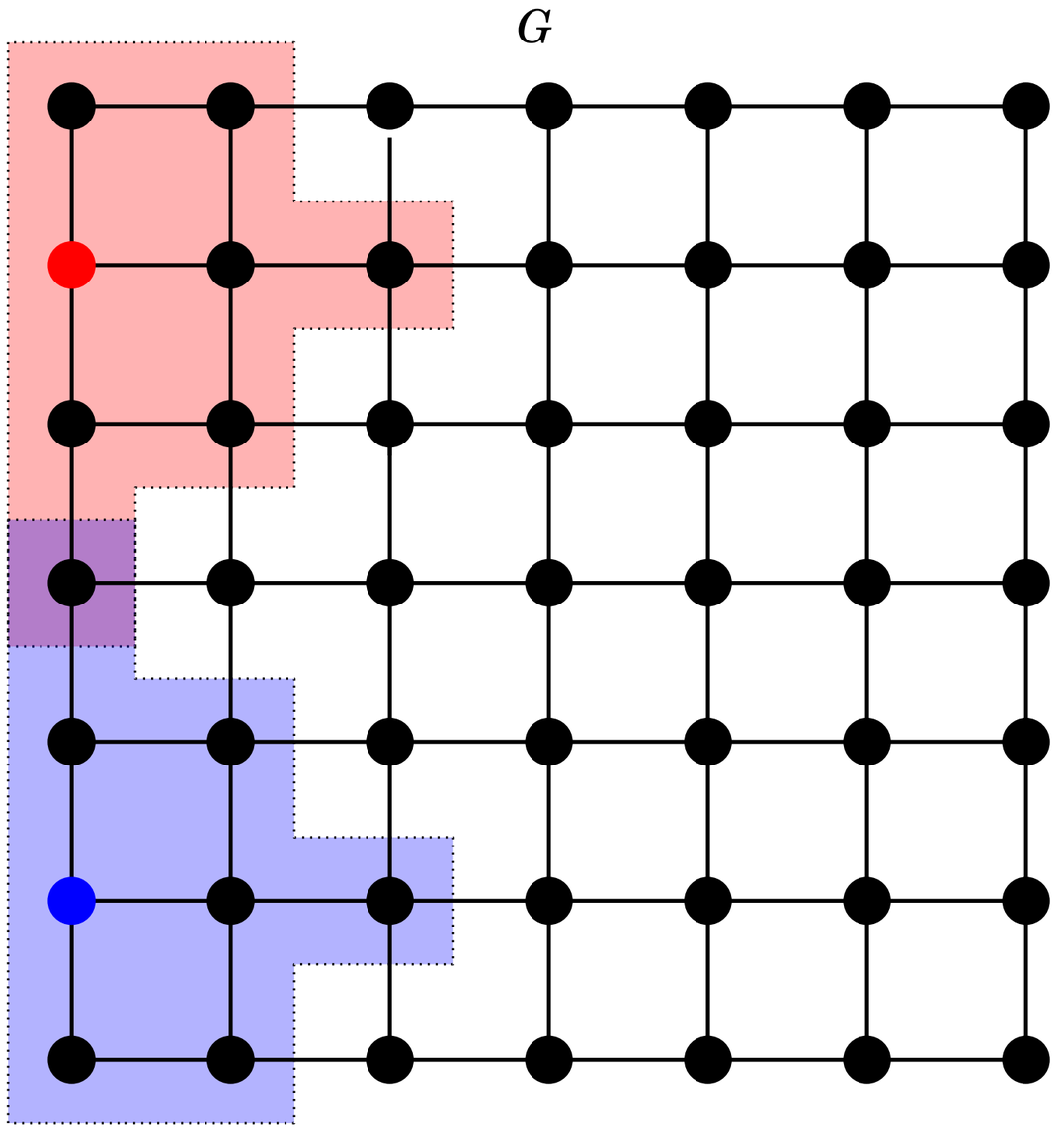
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- **No-cloning**
- Still, we can describe *output distributions*
- **Question**: *is there any graph problem that admits quantum advantage?*

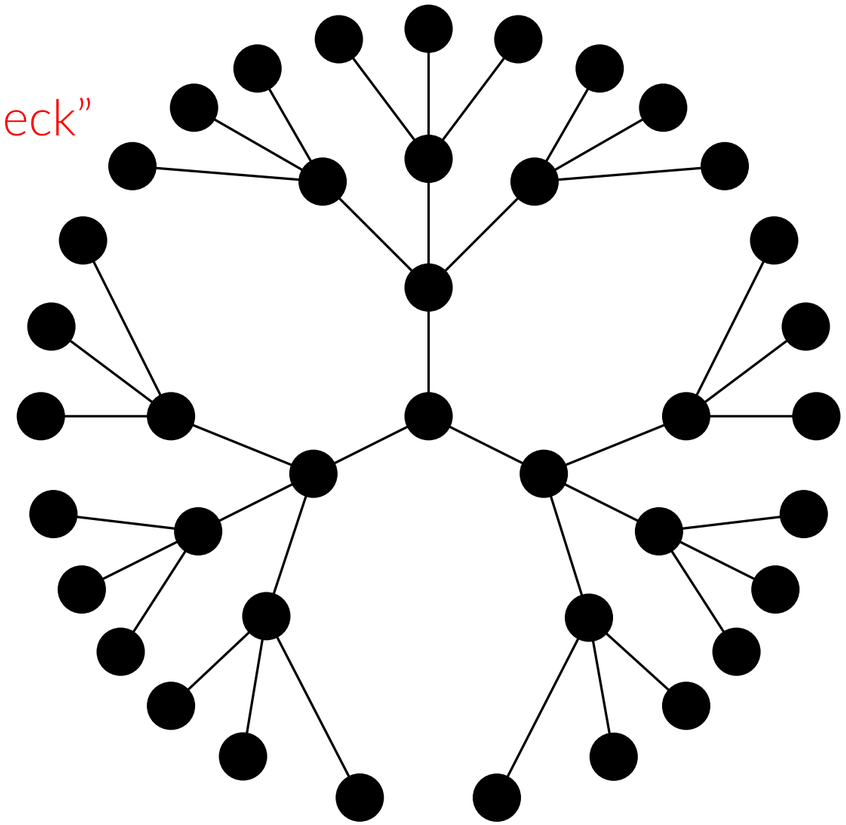


Locally checkable labeling (LCL) problems

[Naor and Stockmeyer, STOC ’93 & SICOMP ’95]

- **Problems** whose solutions might be “hard to find” but are “easy to check”

  - “analogue” of NP in the distributed setting
  - coloring, maximal independent set, maximal matching, etc.

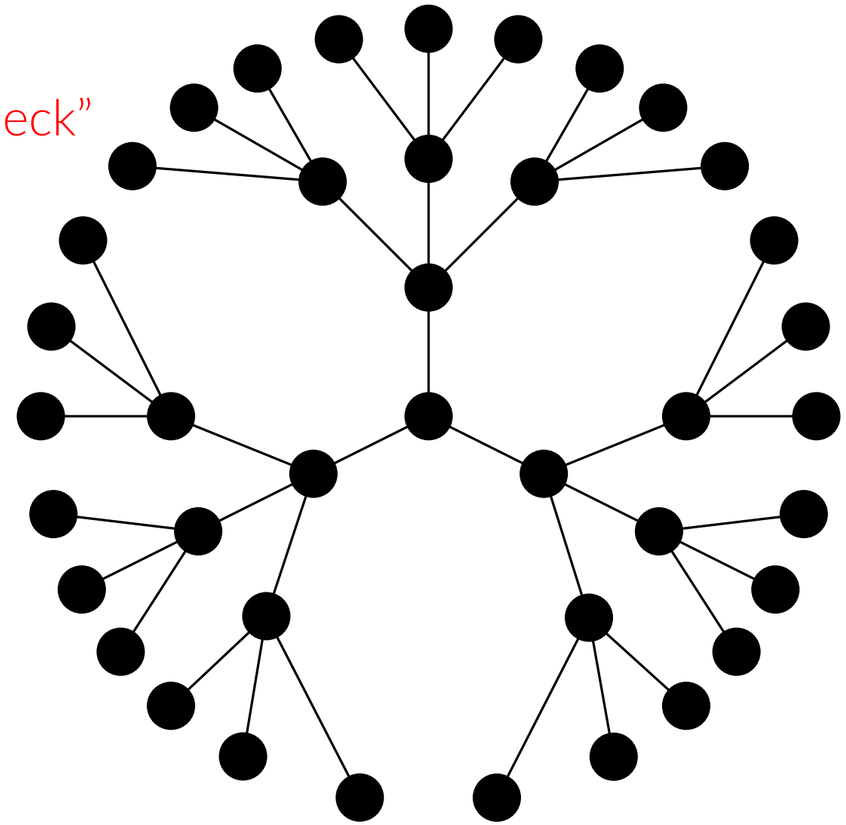


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- **Problems** whose solutions might be “hard to find” but are “easy to check”
  - “analogue” of NP in the distributed setting
  - coloring, maximal independent set, maximal matching, etc.

- **“Easy to check”**
  - radius $r = \Theta(1)$
  - each node can check its solution within its radius-r neighborhood
  - a globally valid iff each node is locally happy

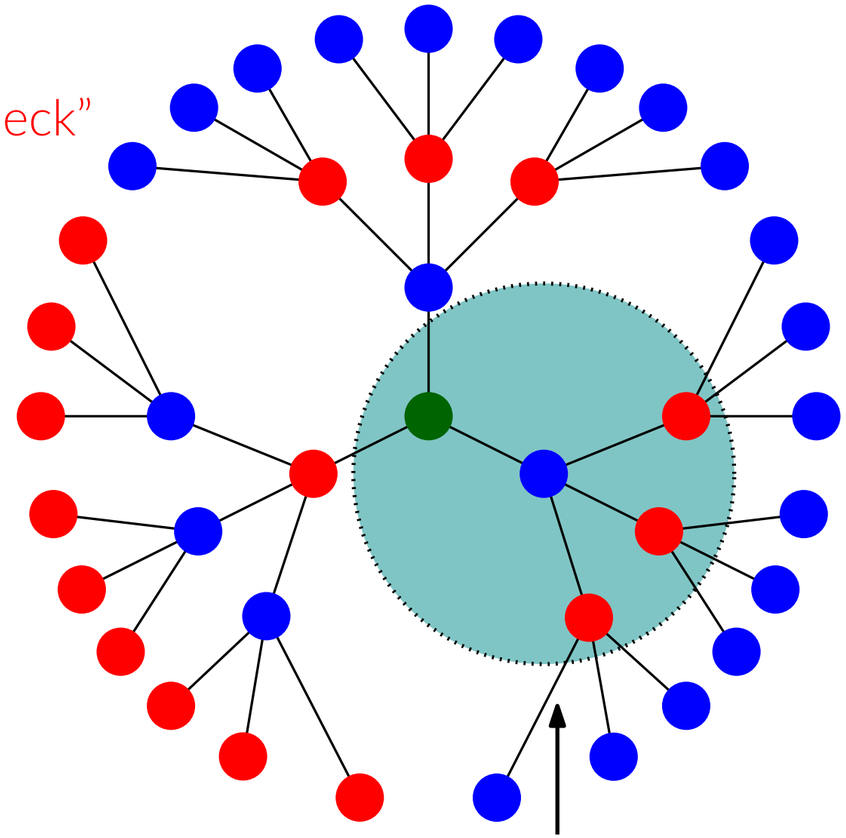


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3-coloring: the blue node checks if its color is different from those of its neighbors

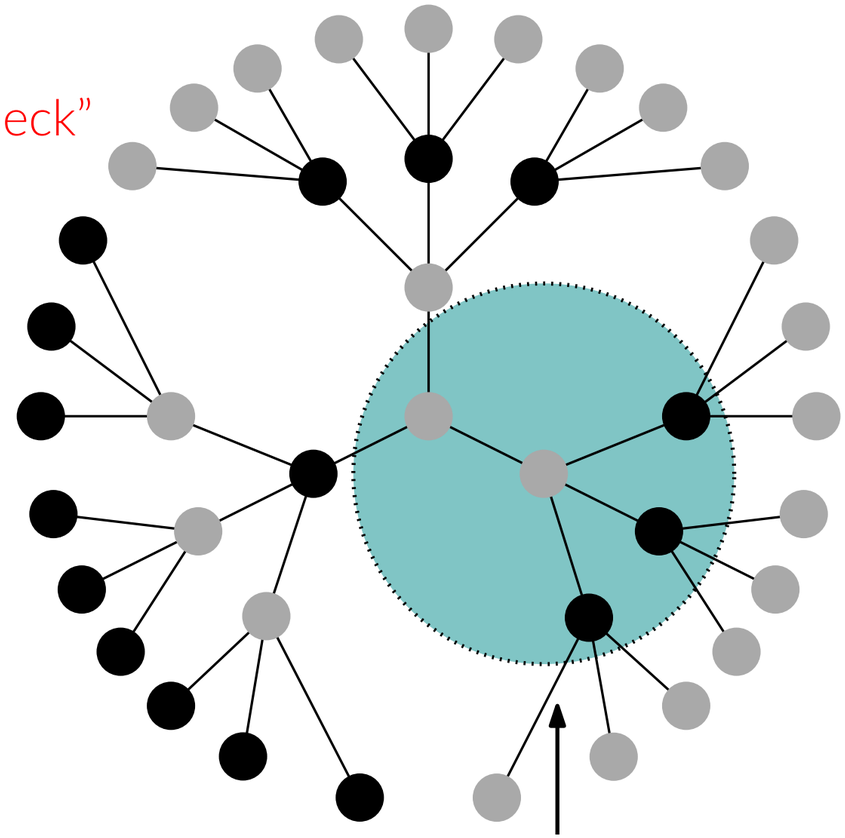
valid LCL

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MIS: each node checks if it is in the IS or if it has a neighbor in the IS

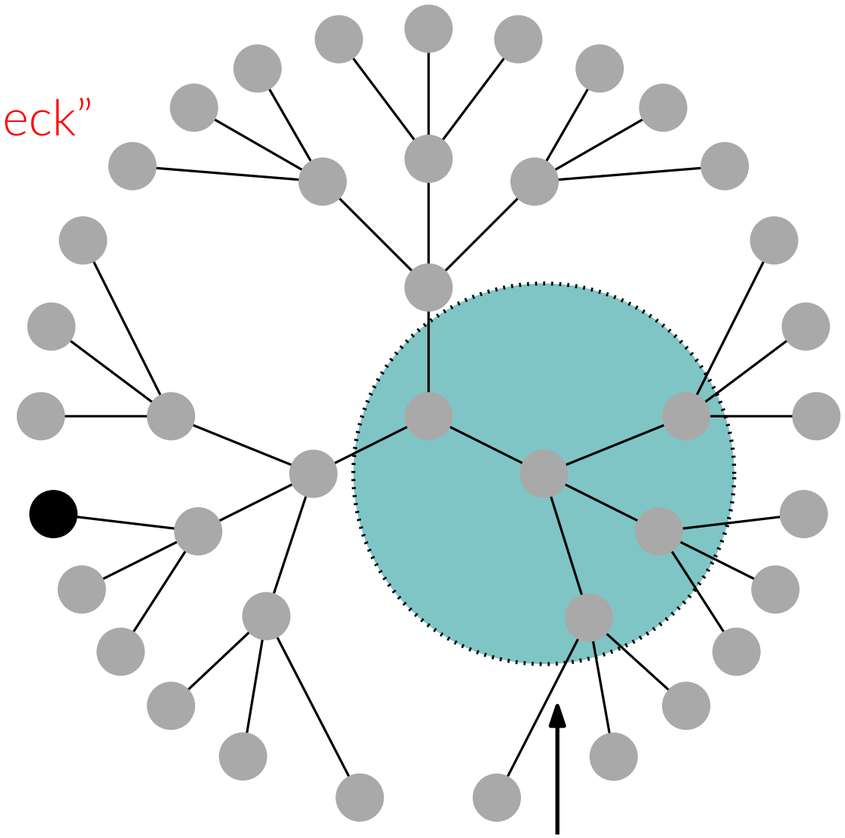
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Leader election: the checking radius should be $r = \text{diam}(G)$

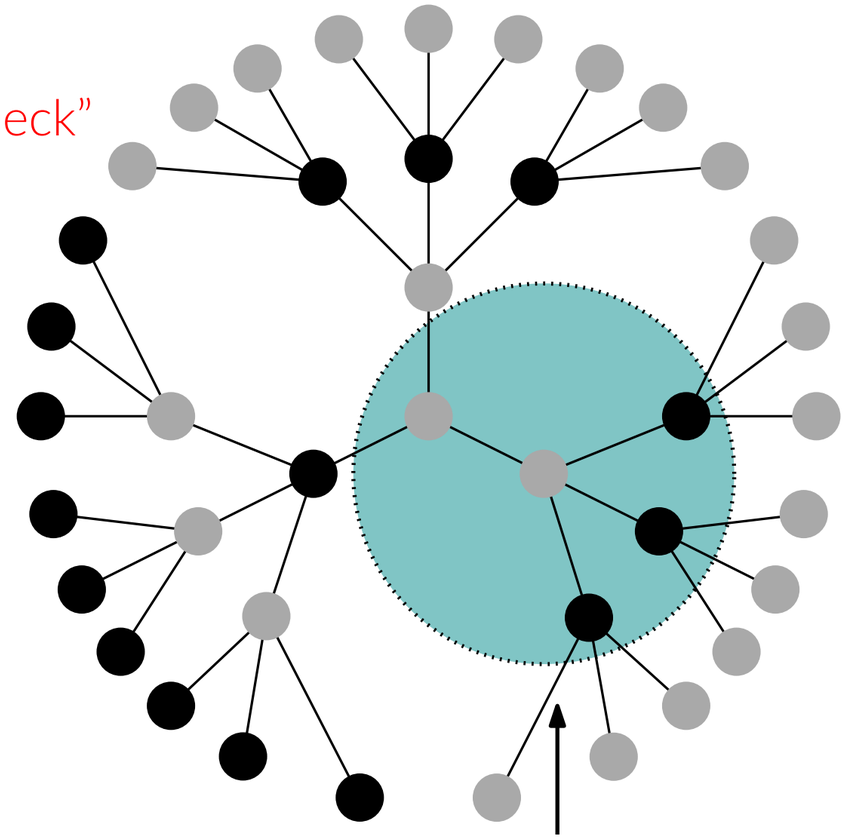
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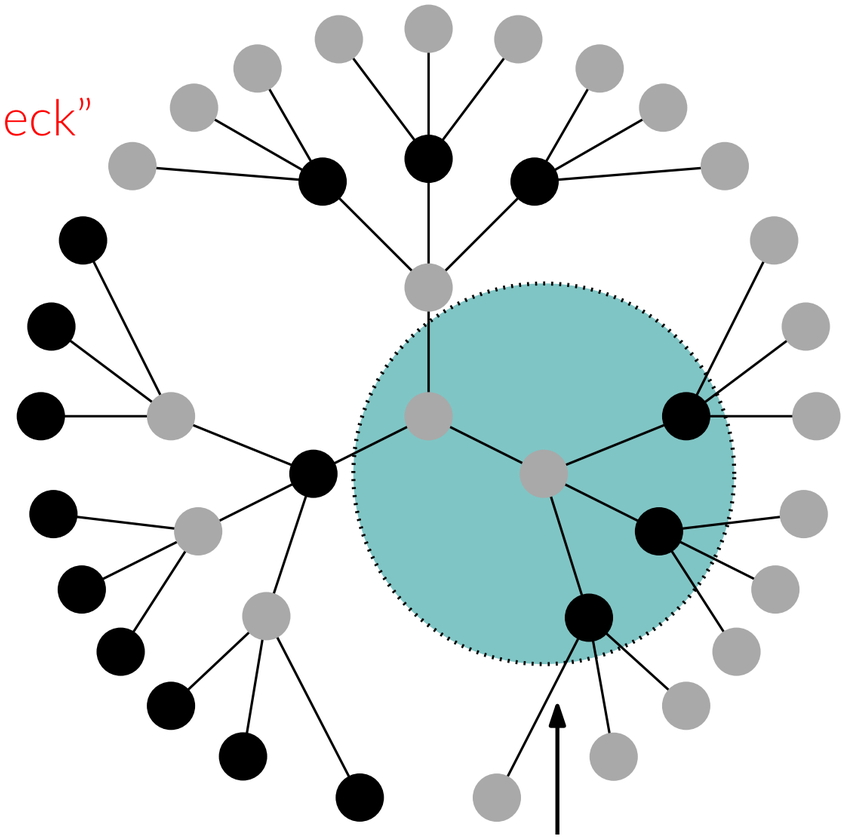
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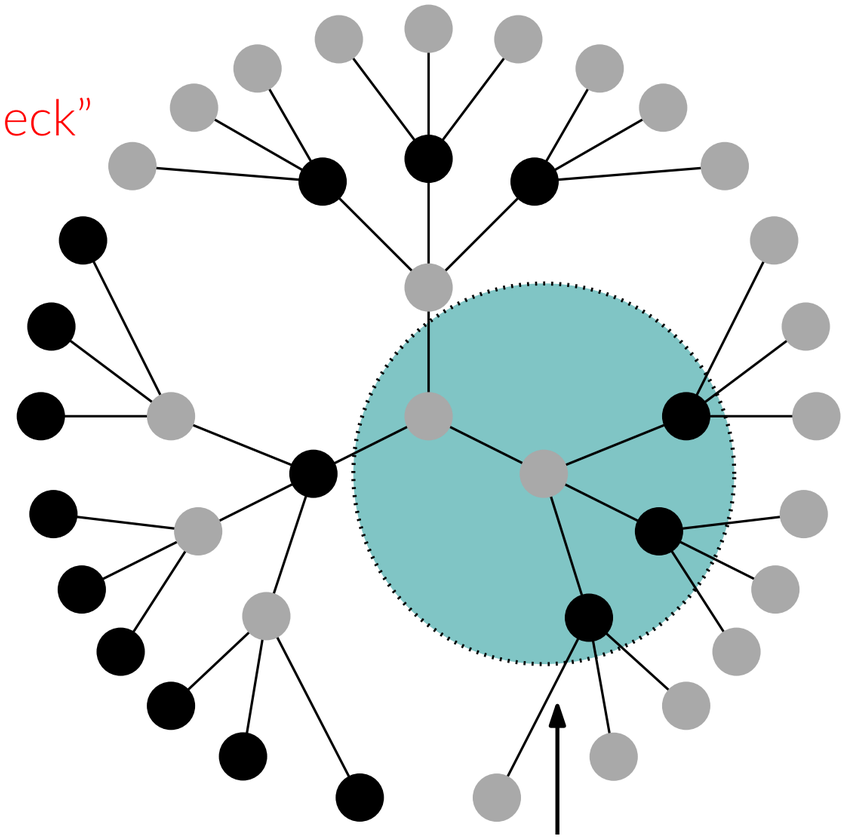
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- A **lot of literature** studying LCLs:
  - classification of LCLs based on complexity (locality)
  - e.g.: complexity $T(n)$ in randomized-LOCAL $\implies$ $O(T(2^{n^2}))$ in deterministic-LOCAL [Chang et al., SICOMP '19]



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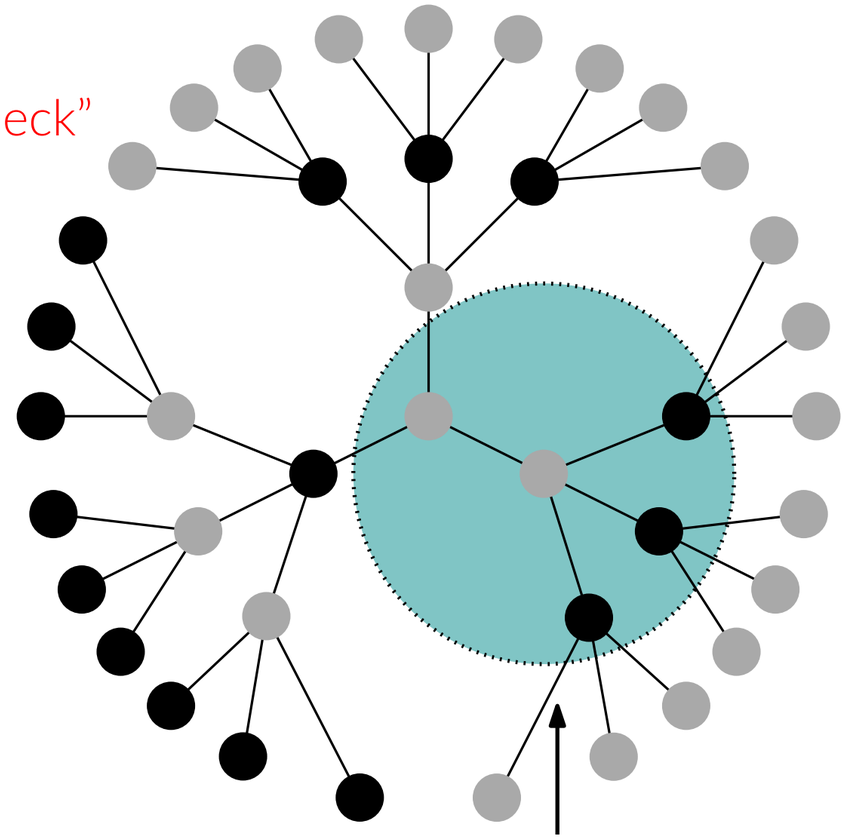
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  - [BFHKLRSU STOC '16; BHKLOPRSU PODC'17; GKM STOC '17; GHK FOCS '18; CP SICOMP '19; BHKLOS STOC '18; BBCORS PODC '19; BBOS PODC '20; BBHORS JACM '21; BBCOSS DISC '22; AELMSS ICALP '23; etc.]



MIS: each node checks if it is in the IS or if it has a neighbor in the IS

Complexity landscape of LCL problems

- **Paths and cycles**

det-LOCAL	$O(1)$	$\Theta(\log^\star n)$	$\Theta(n)$
rand-LOCAL	$O(1)$	$\Theta(\log^\star n)$	$\Theta(n)$

Complexity landscape of LCL problems

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- **Balanced d-dimensional toroidal grids**

det-LOCAL	$O(1)$	$\Theta(\log^\star n)$	$\Theta(n^{1/d})$
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det-LOCAL	$O(1)$	$\Theta(\log^* n)$	$\Theta(\log n)$		$\Theta(n^{1/k})$	$\Theta(n)$
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- **General graphs**

det-LOCAL						
rand-LOCAL						

Complexity landscape of LCL problems

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det-LOCAL	$O(1)$	$\Omega(\log\log^\star n)\dots O(\log^\star n)$				$\Theta(n)$
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Complexity landscape of LCL problems

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Complexity landscape of LCL problems

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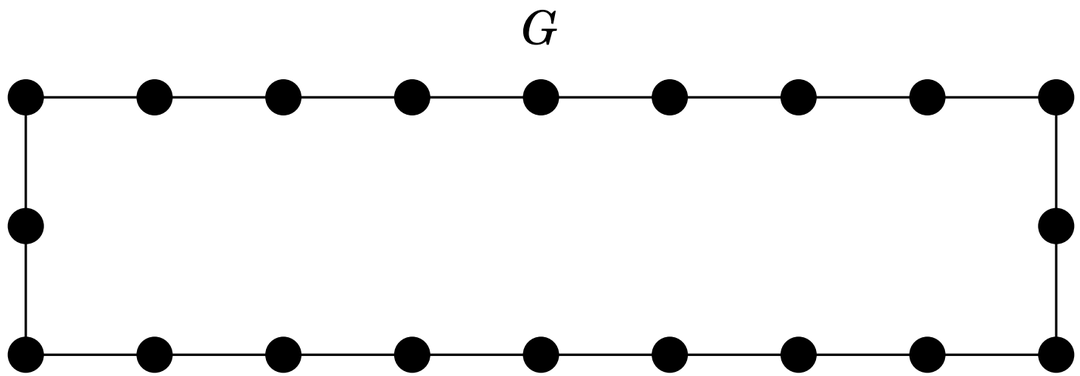
  - role of quantum??

Table of content

1. **Intro**: distributed algorithms, the LOCAL model, the quantum-LOCAL model, locally checkable labeling problems
2. **Classical lower bounds**: the indistinguishability argument
3. **Properties of distributed algorithms**: independence and non-signaling
4. **Super-quantum models**: bounded-dependence and non-signaling model
5. **State of the art results**
6. **Quantum advantage**

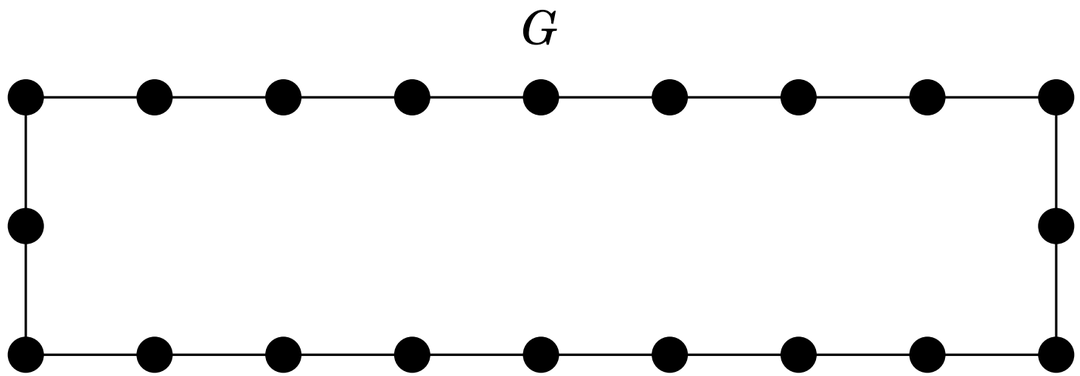
Propagation argument

- **Problem**: 2-coloring paths & even cycles



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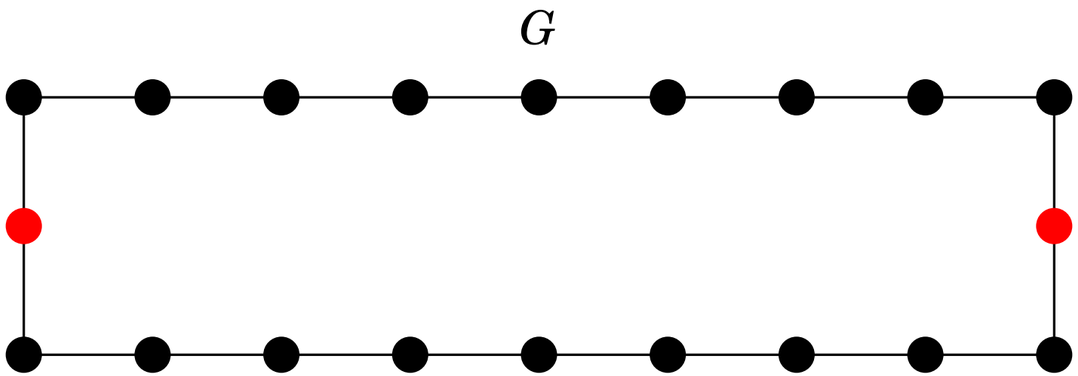
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- Suppose algorithm $\mathcal{A}$ with running time $T \leq n/5$

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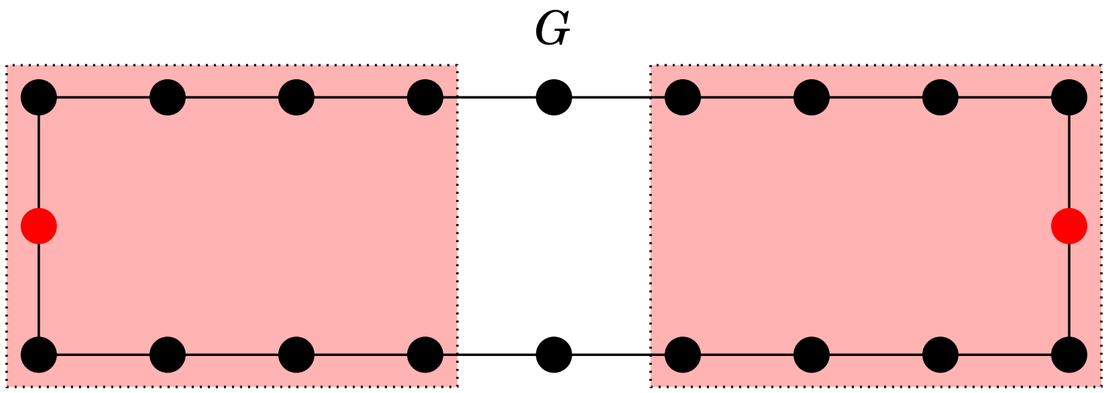
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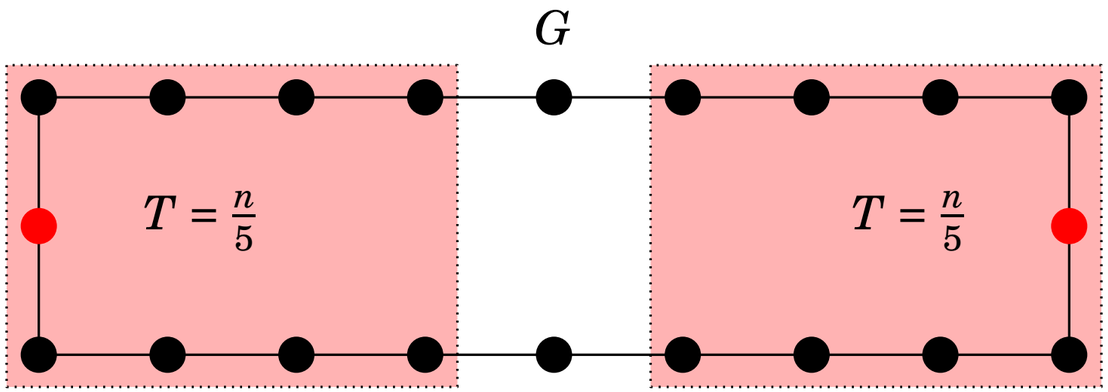
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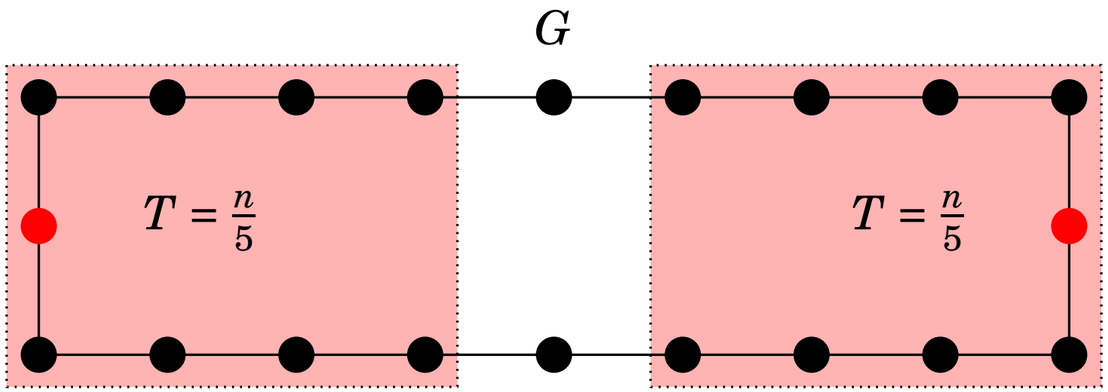
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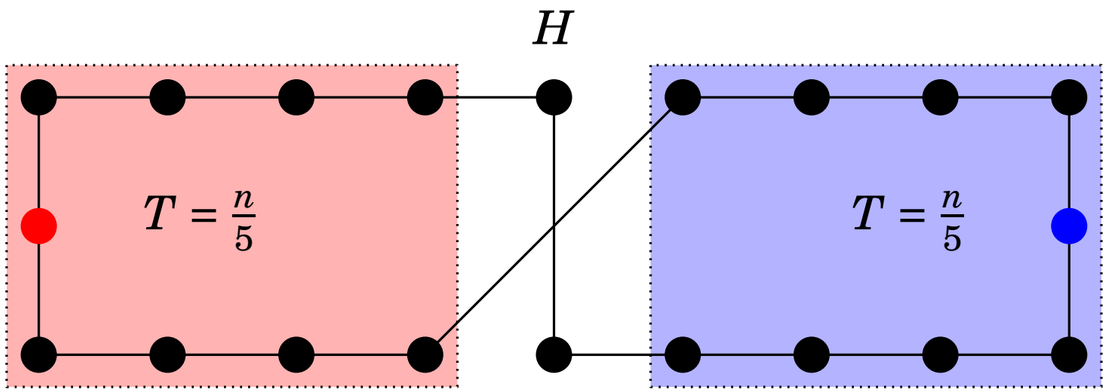


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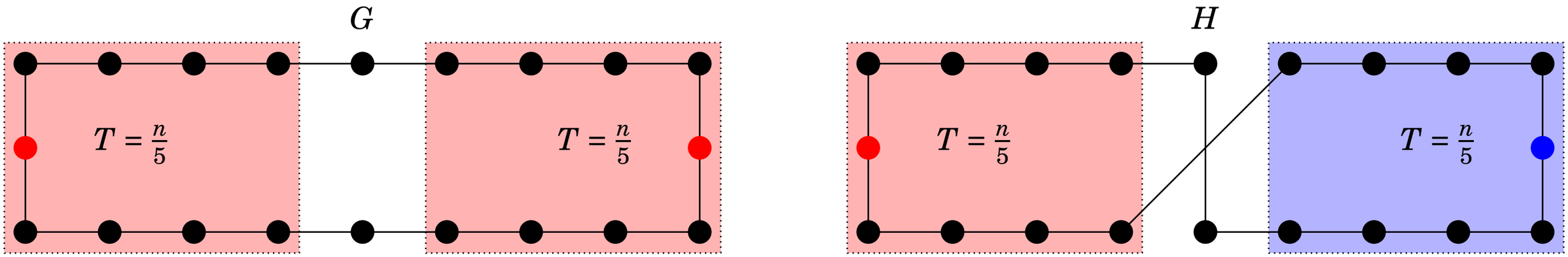




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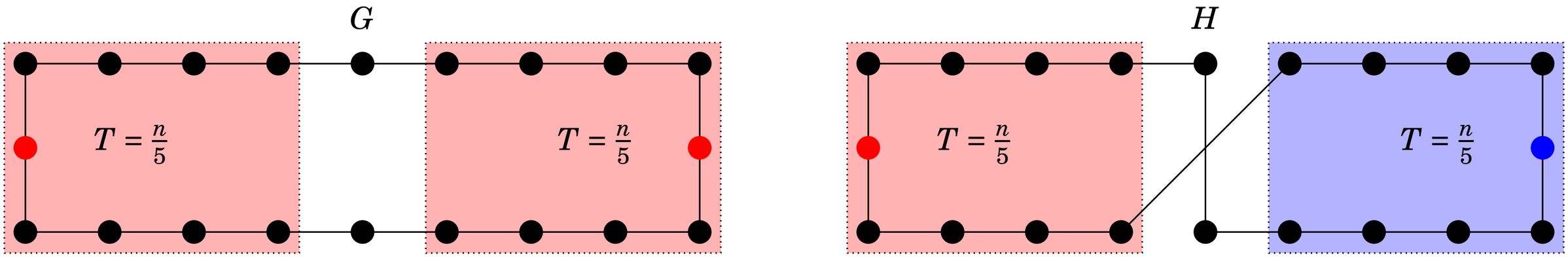


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Propagation argument

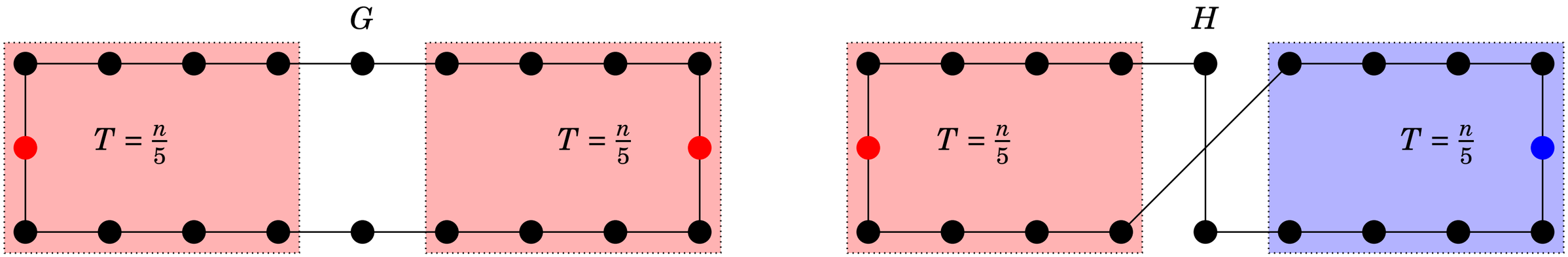
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Propagation argument

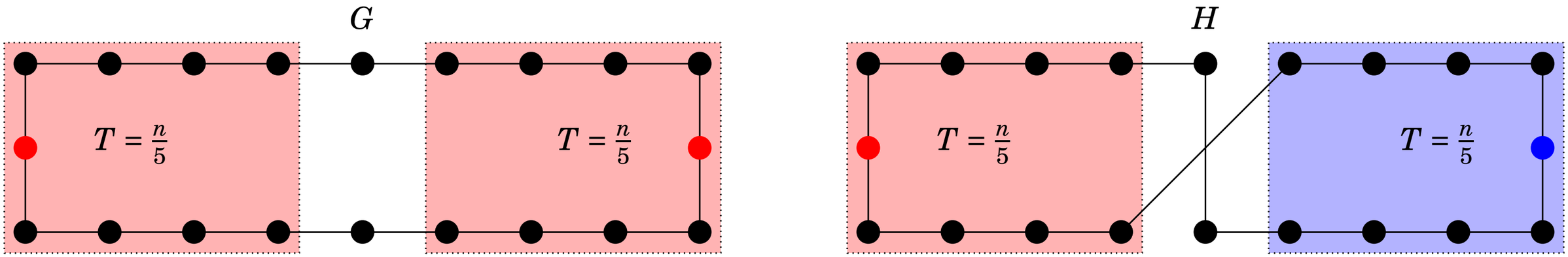
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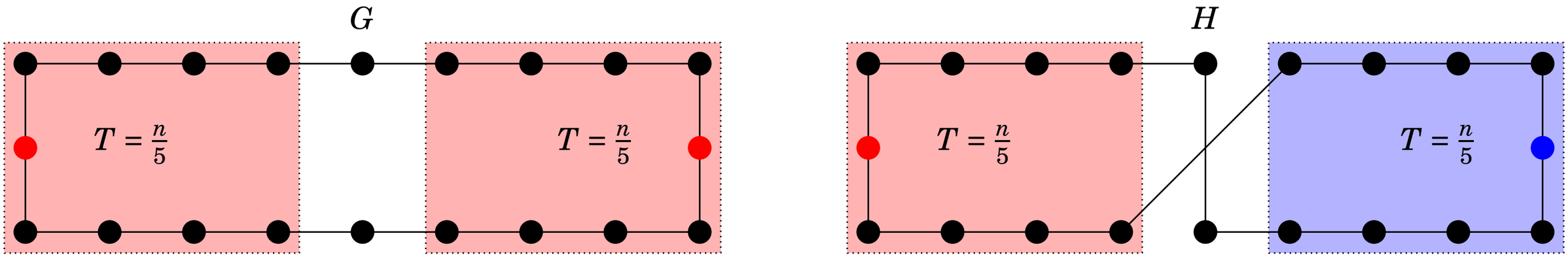


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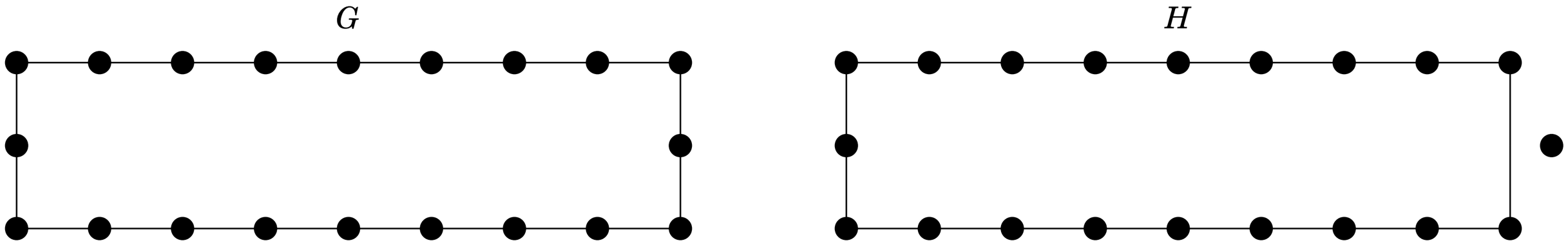
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propagation argument
exploiting
independence

- **Global problem**: complexity $\Theta(n)$

Graph-existential indistinguishability (randomized)

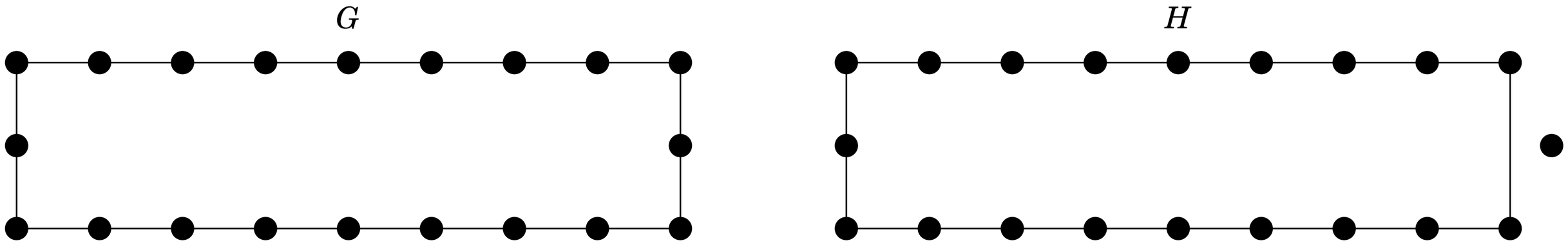
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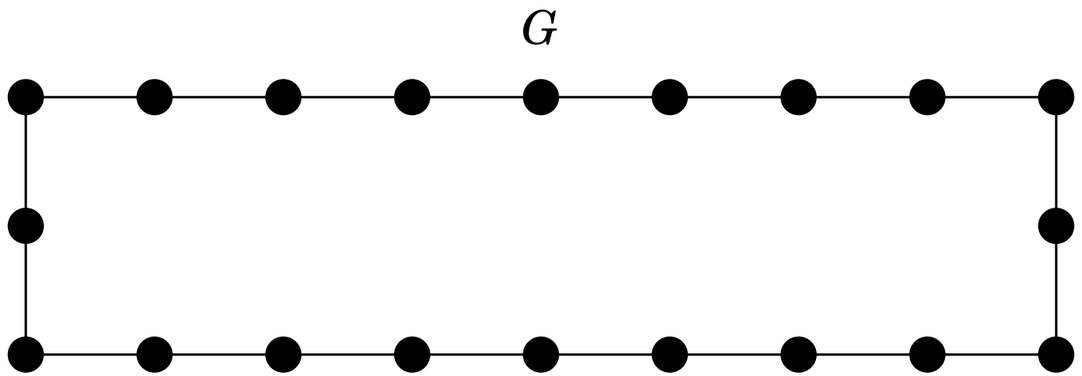


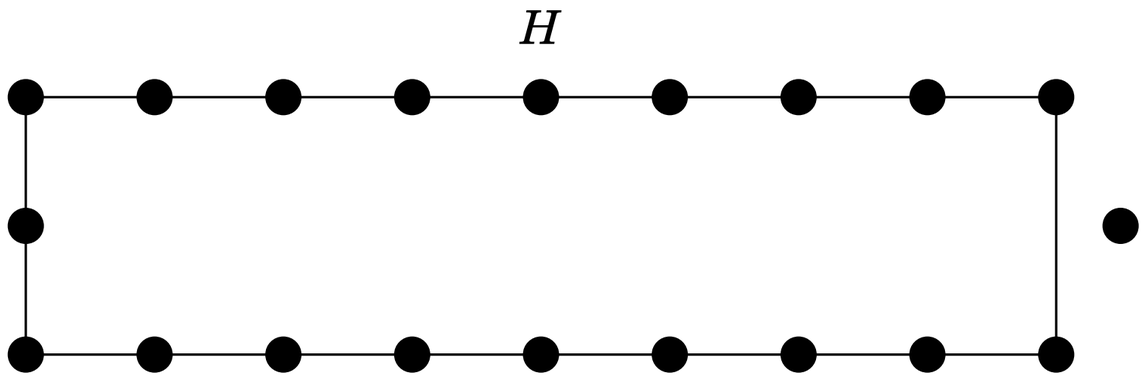
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- Run $\mathcal{A}$ both in G and H

Graph-existential indistinguishability (randomized)

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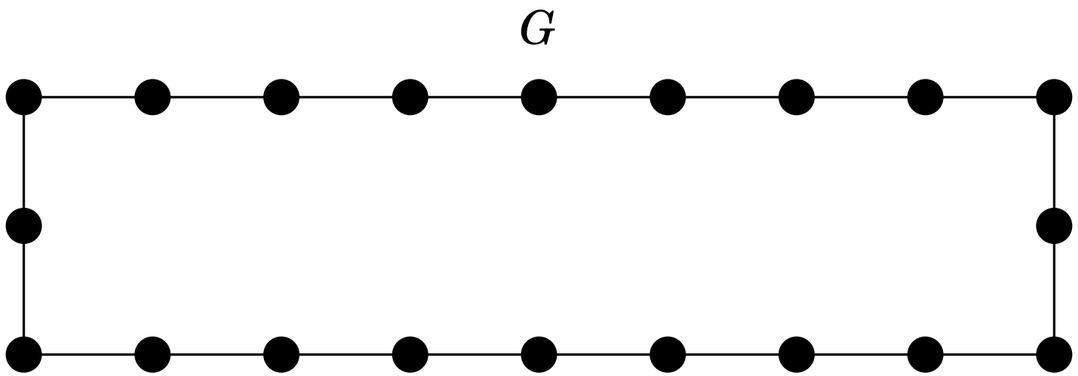


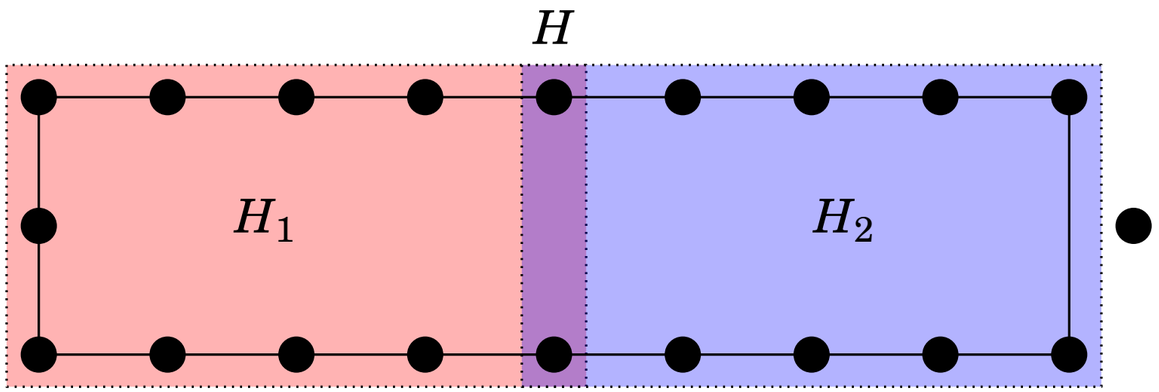


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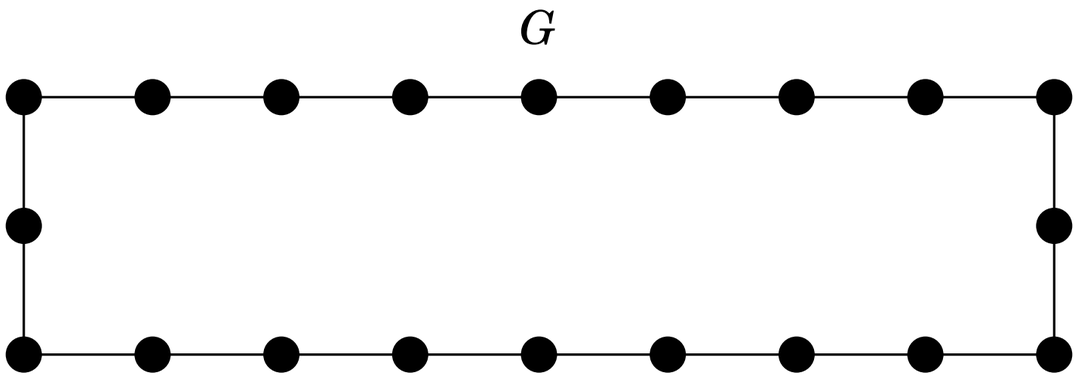


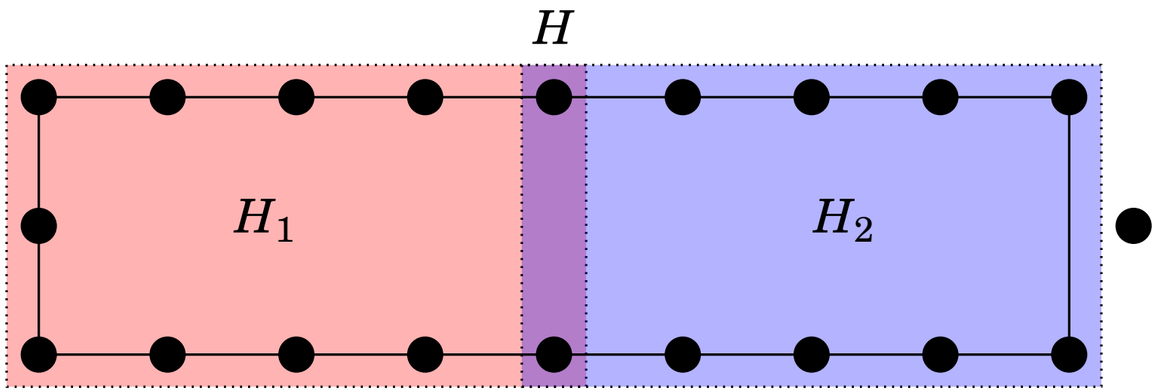


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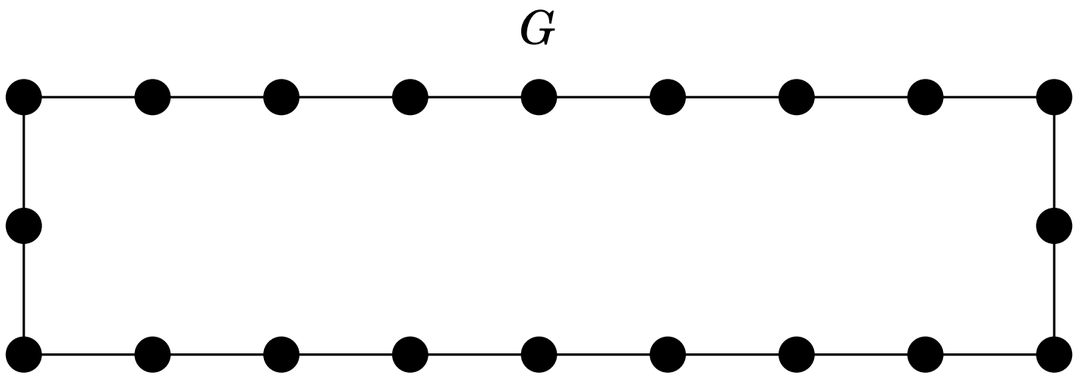


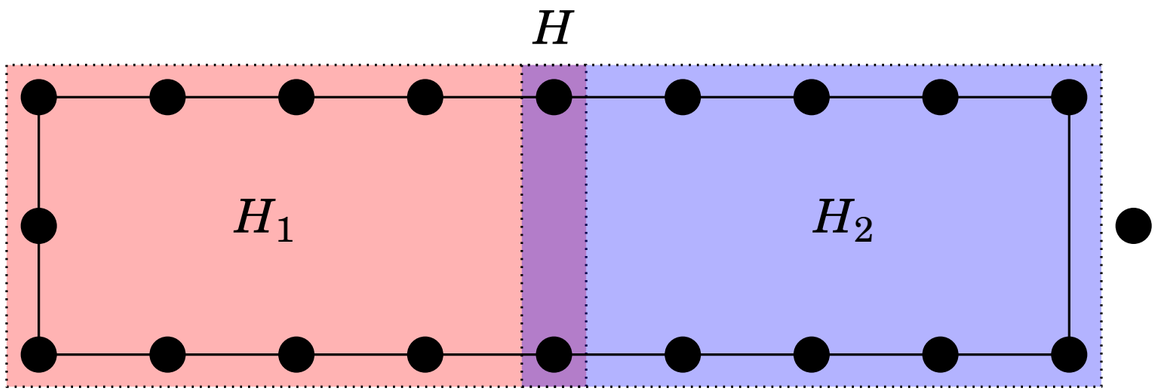


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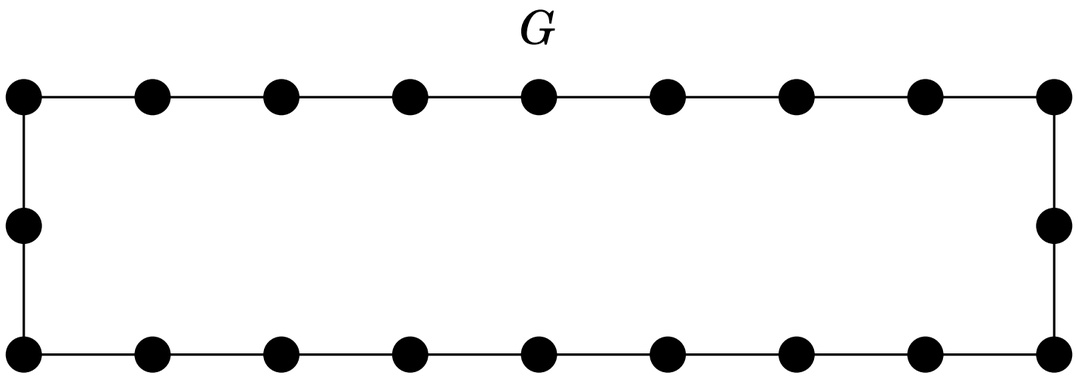


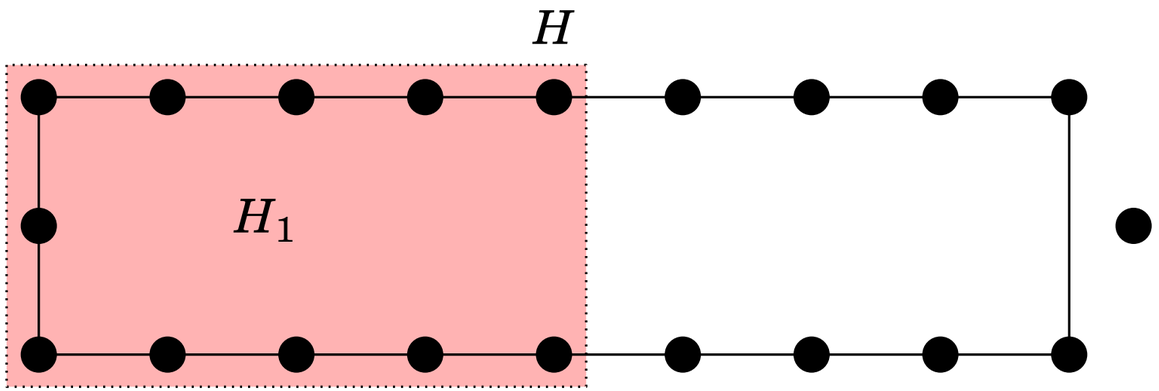


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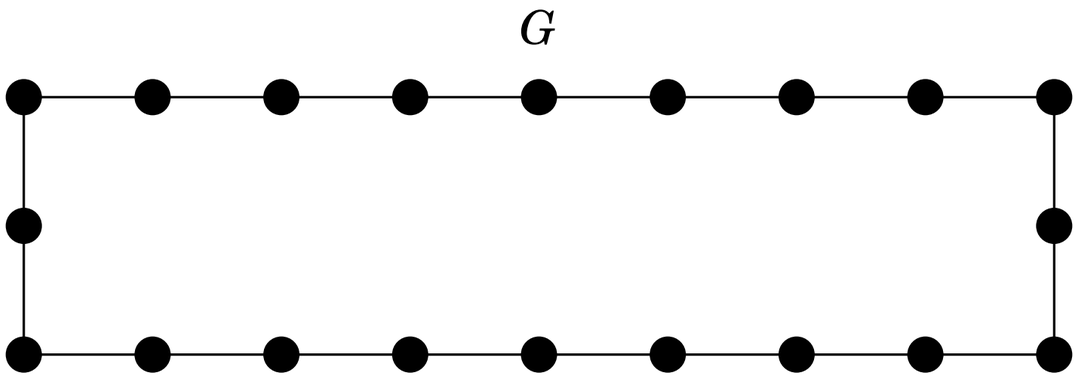


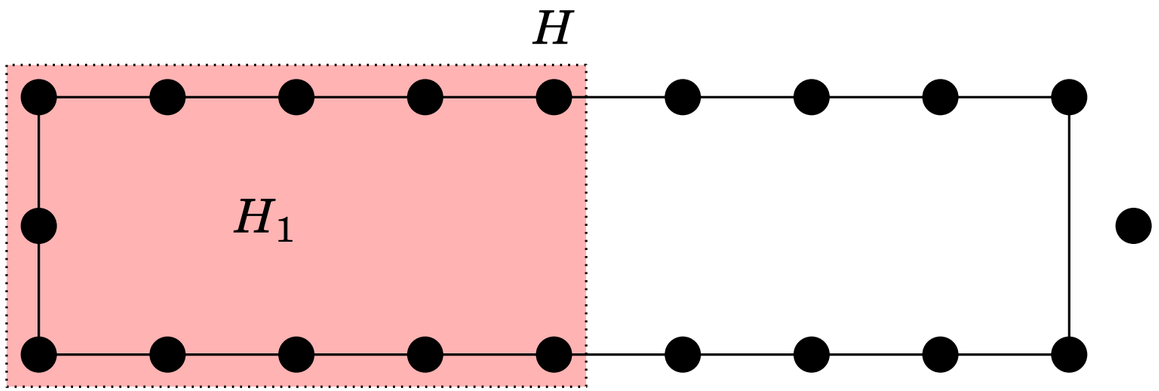


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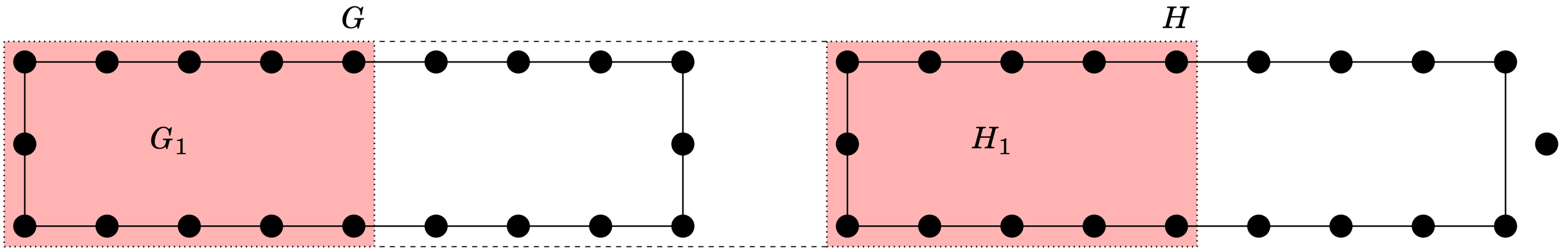




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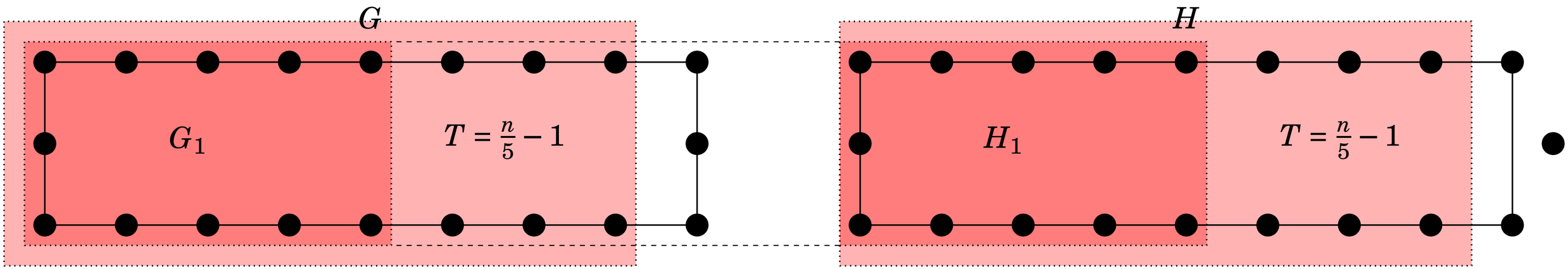
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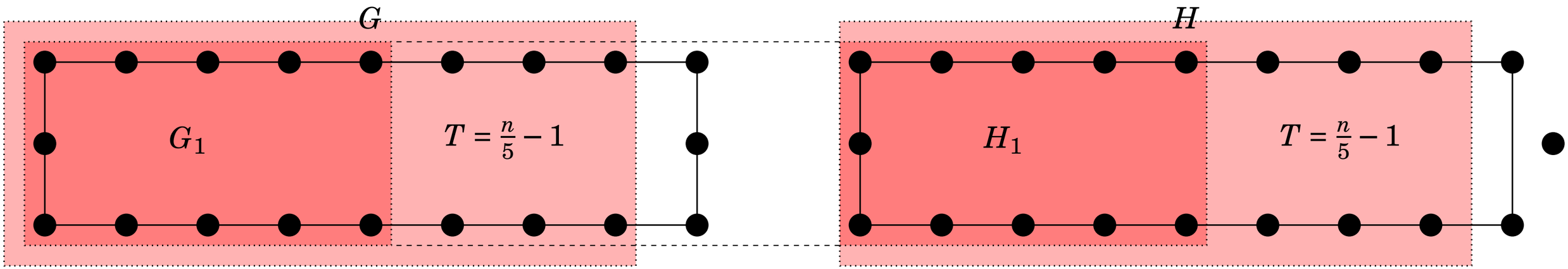
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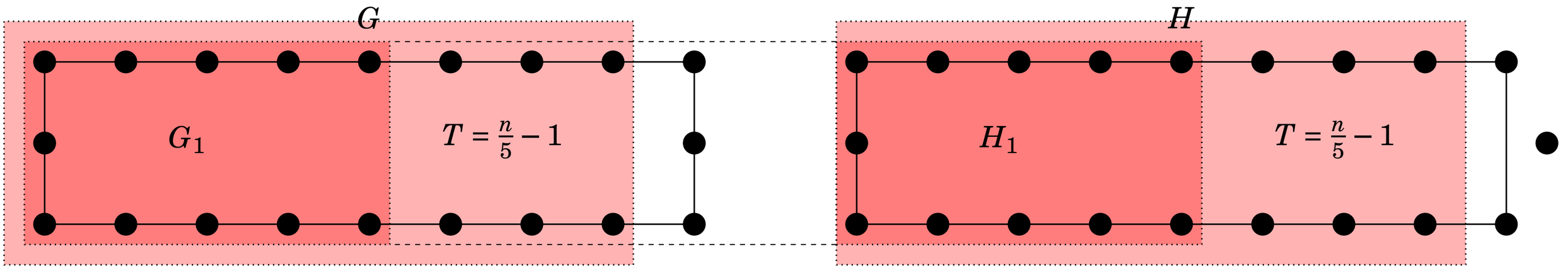
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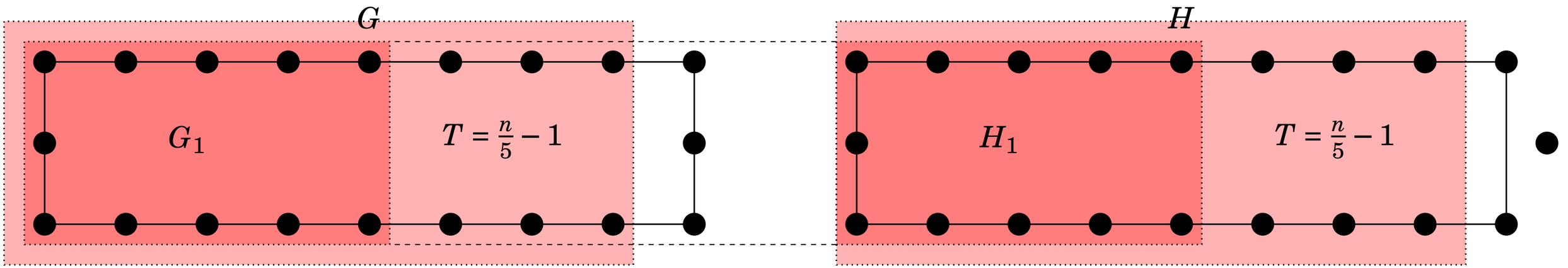


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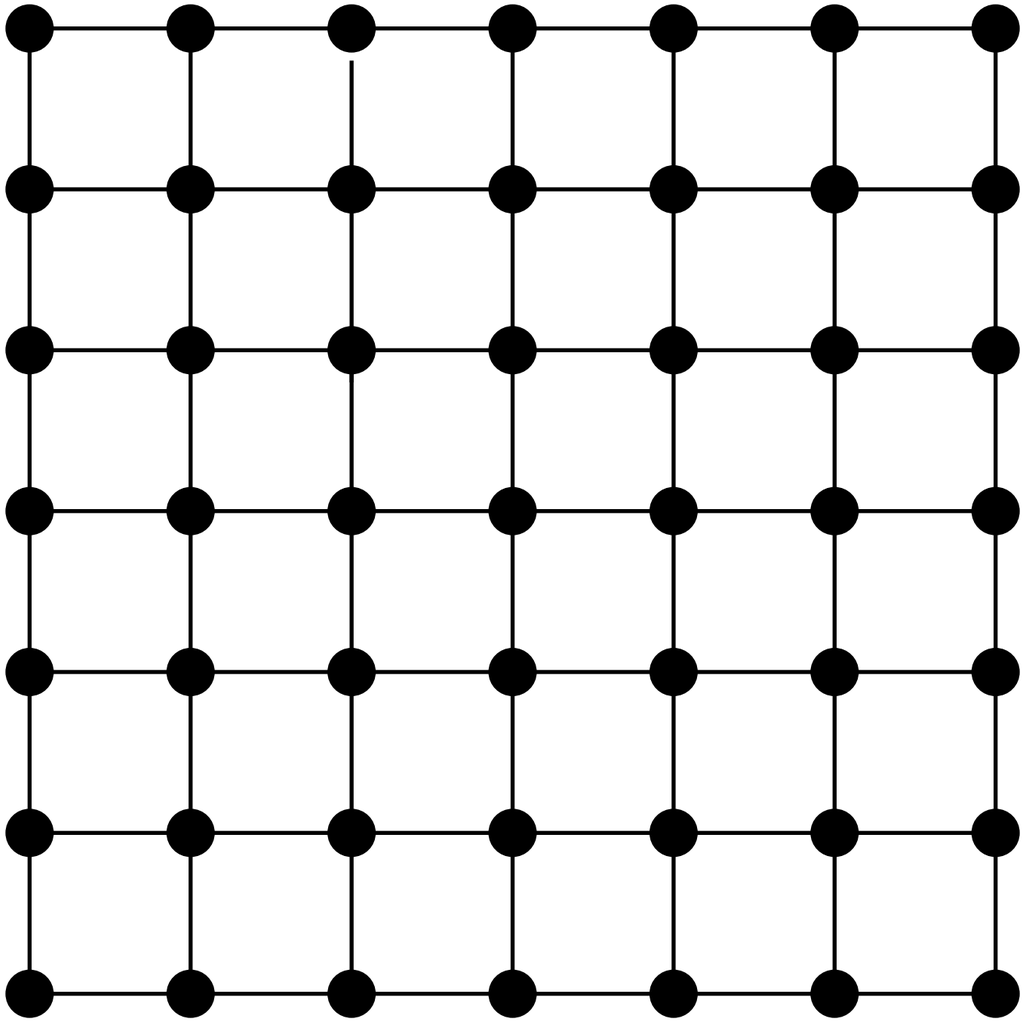
indistinguishability argument exploiting **non-signaling** properties

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4. **Super-quantum models**: bounded-dependence and non-signaling model
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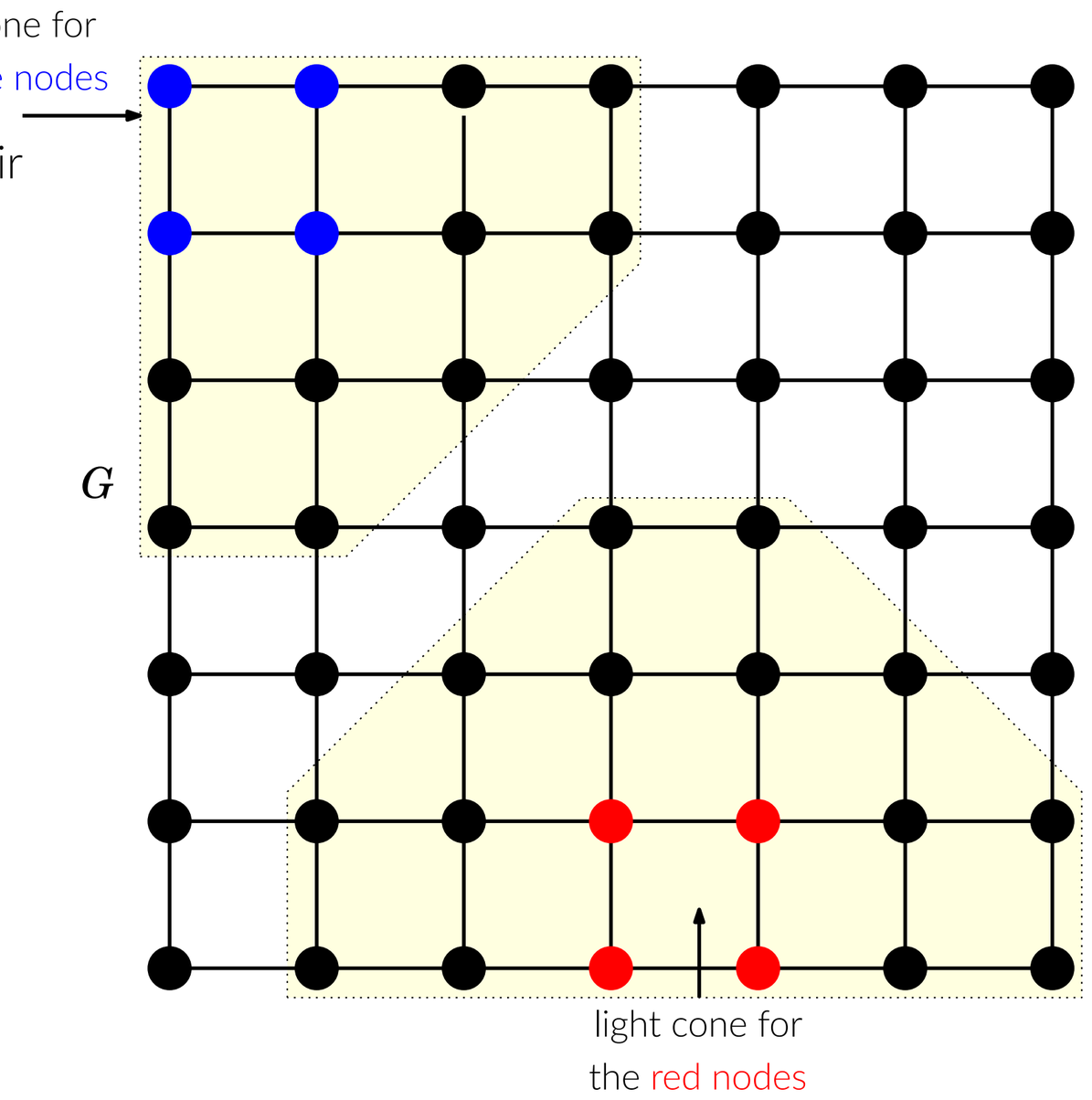
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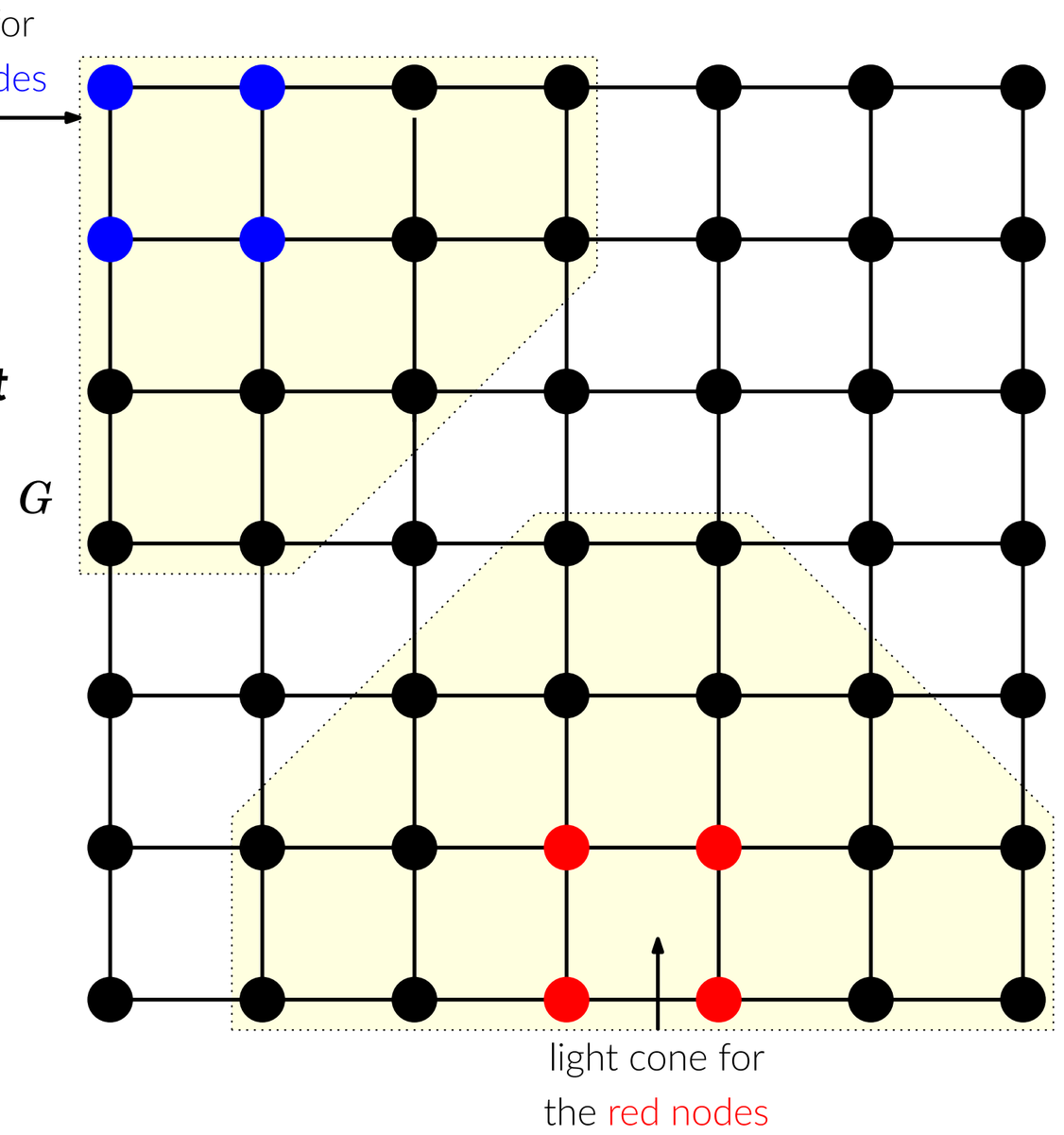
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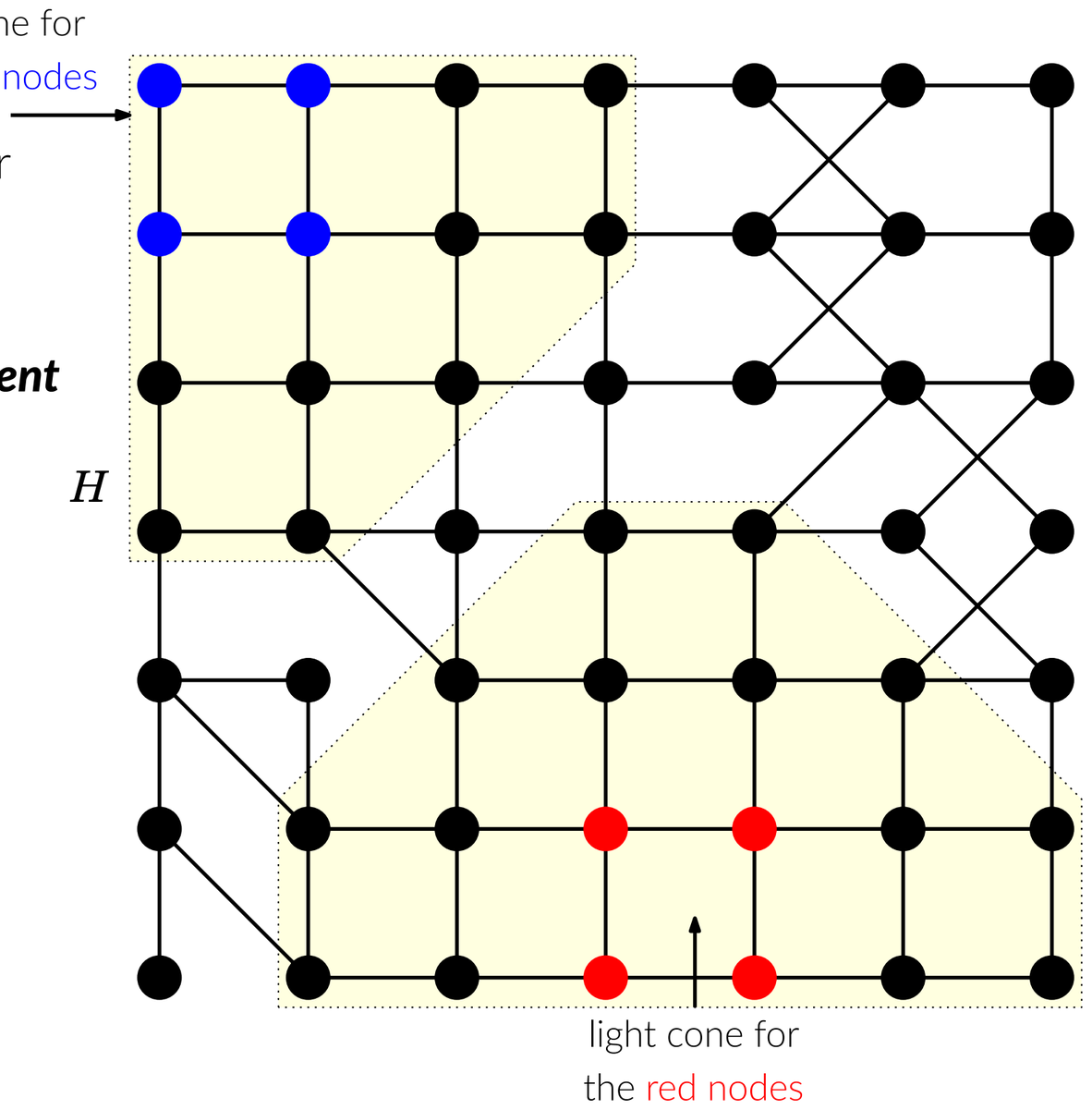
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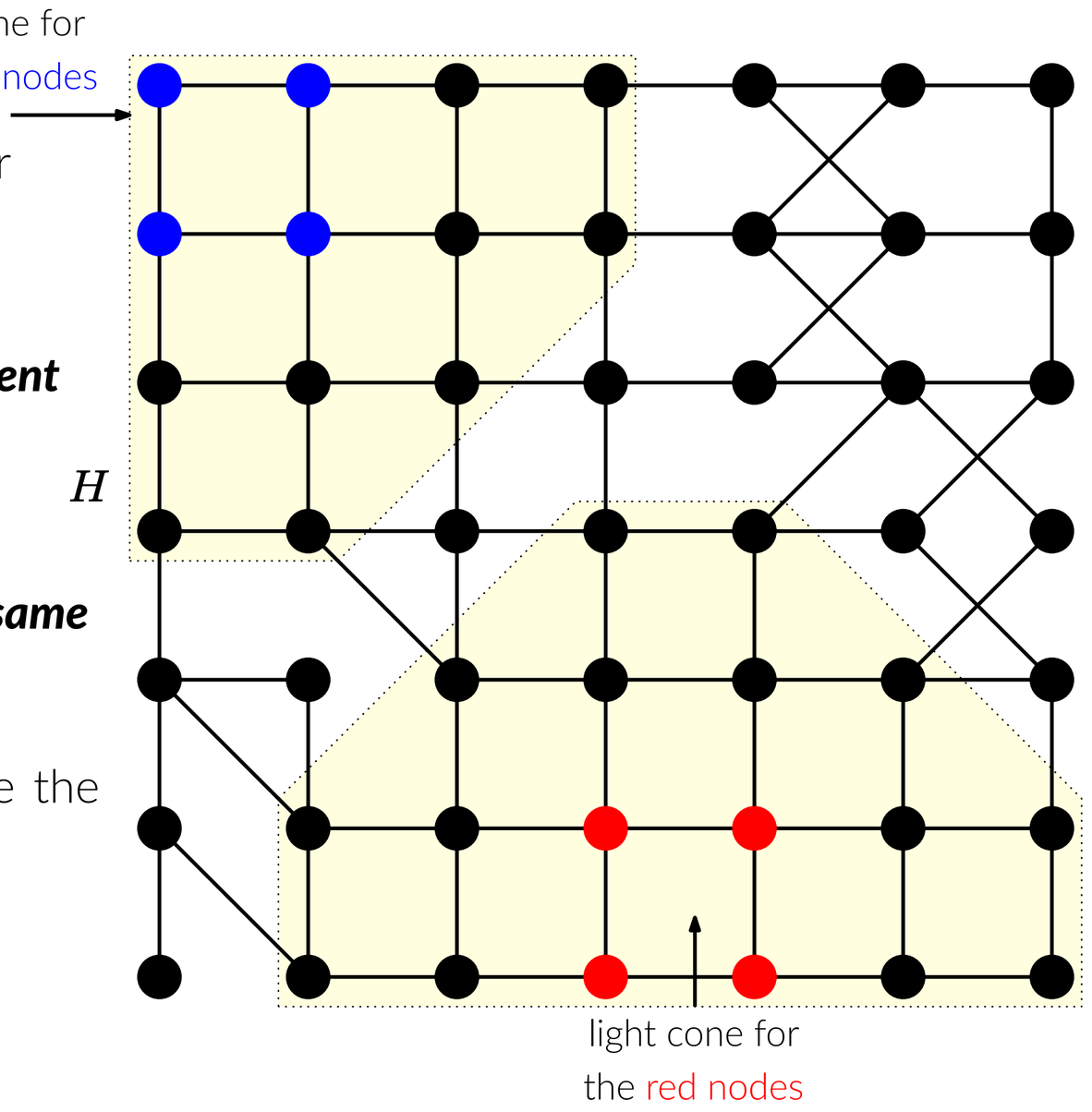


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- **Output distributions** remains ***the same*** if ***light cone is the same***
  - non-signaling property
  - changes that are beyond 2-hops away do not influence the output distribution
  - also known as causality



Abstracting output distributions

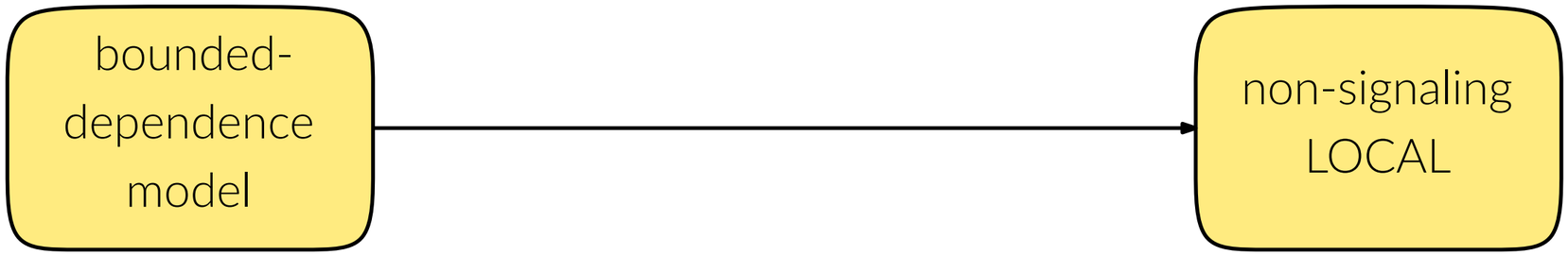
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locality T =
independence at distance $2T+1$ plus
non-signaling beyond distance T

[Holroyd and Liggett, Forum of Mathematics, Pi '14]
[STOC '25a]

* finitely-dependent distributions if $T = O(1)$

locality T =
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[Gavoille, Kosowski, and Markiewicz, DISC '09]
[Arfaoui and Fraigniaud, PODC '12 & SIGACT News '14]

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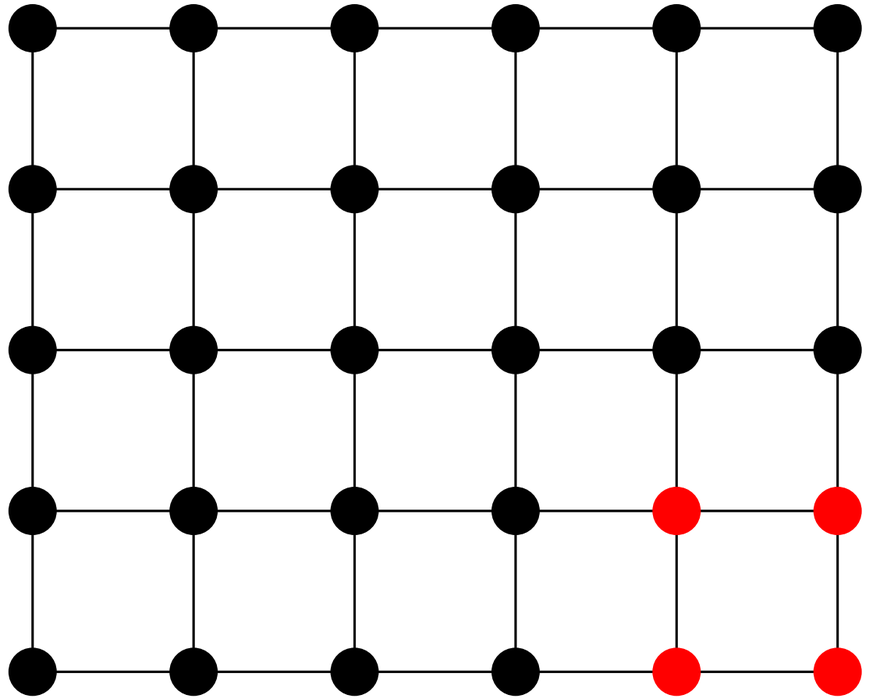
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The non-signaling model

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  - always contains garbage output $\perp$

[STOC '24]



G

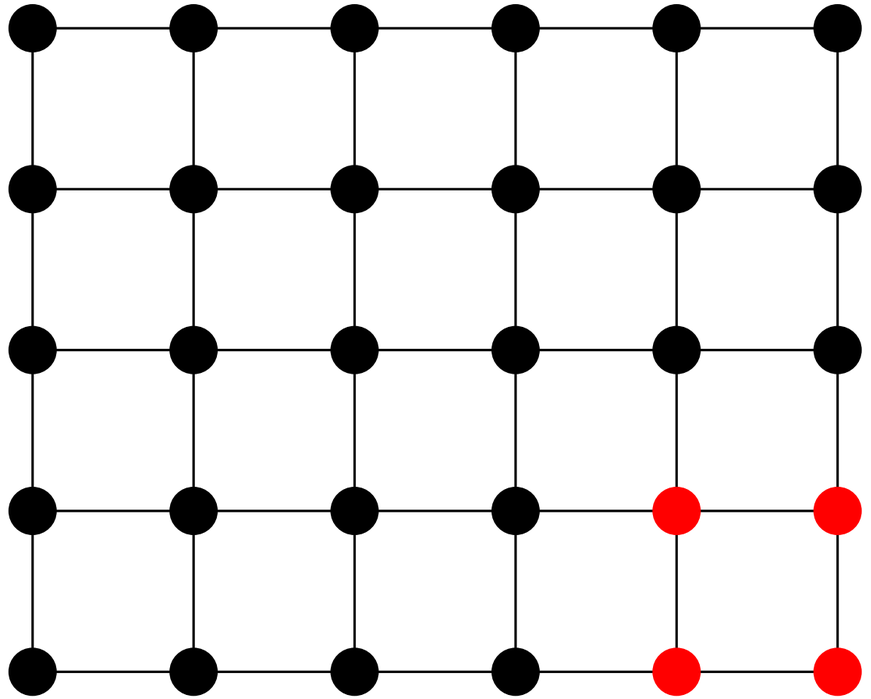
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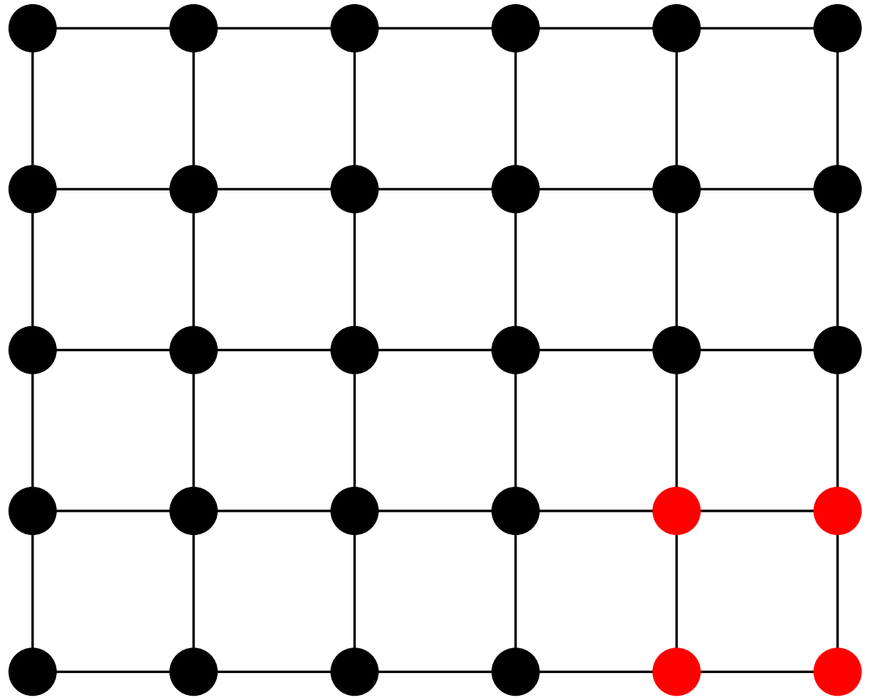
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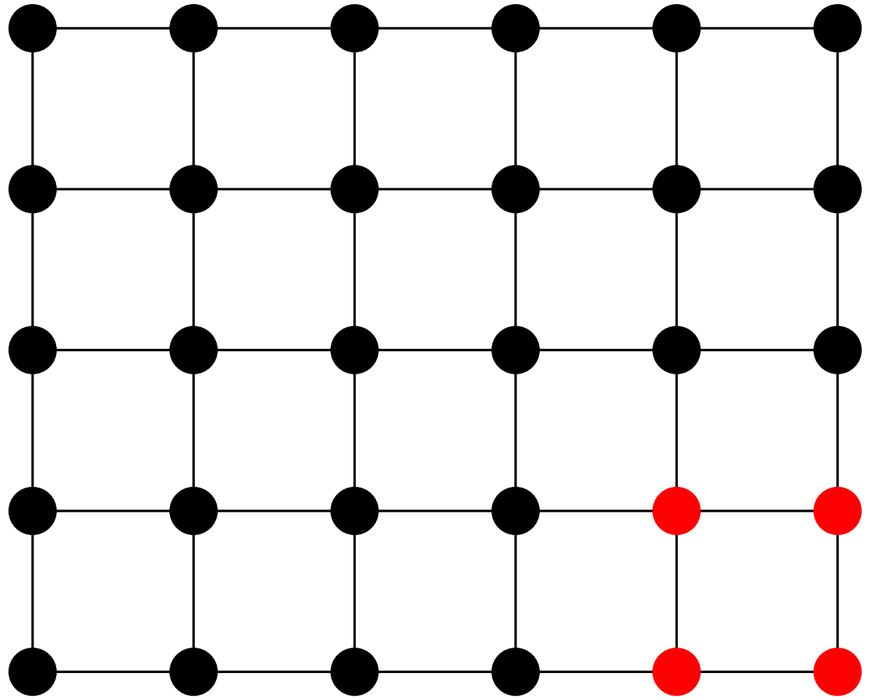


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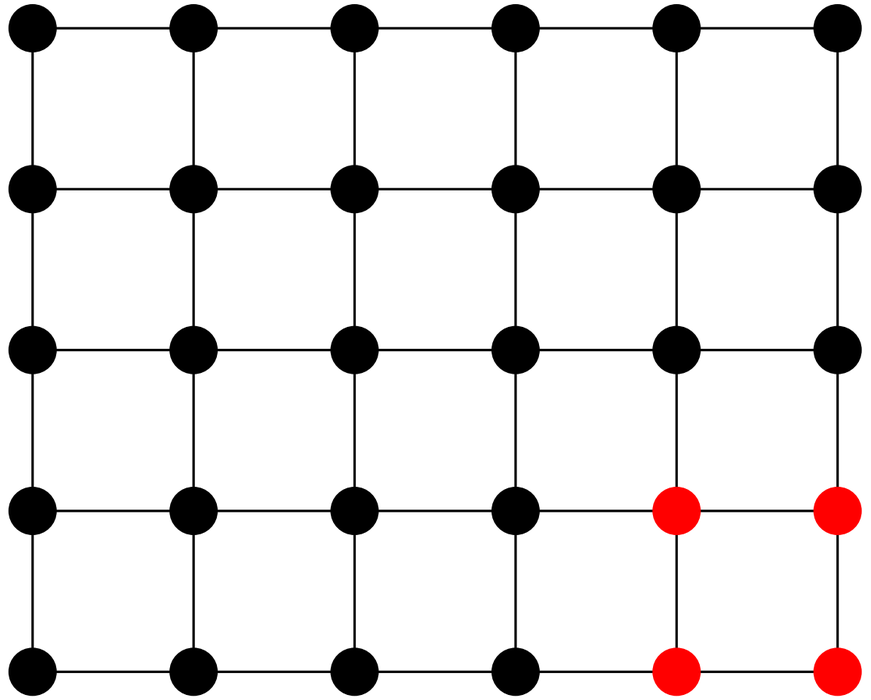


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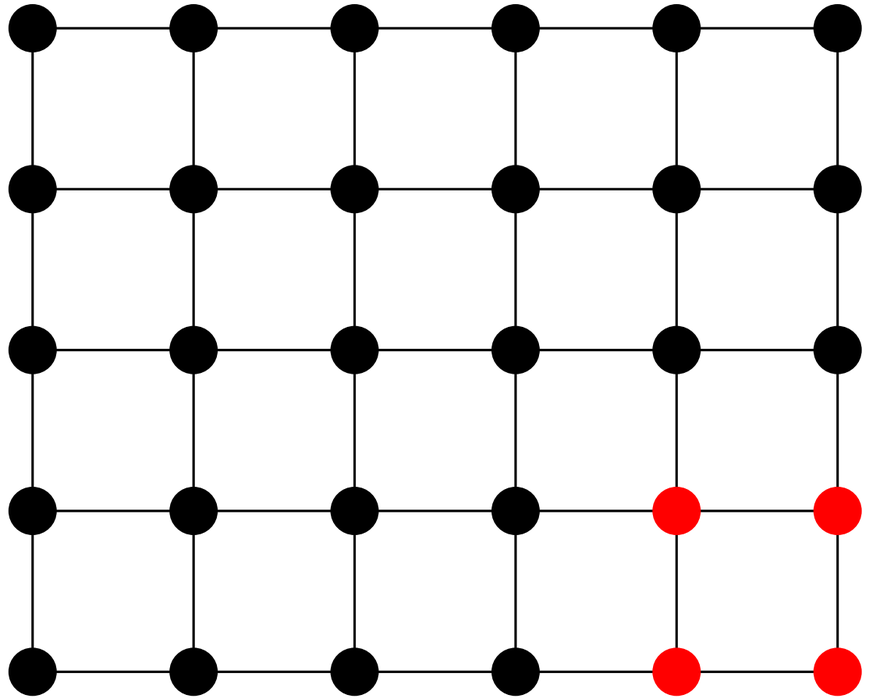


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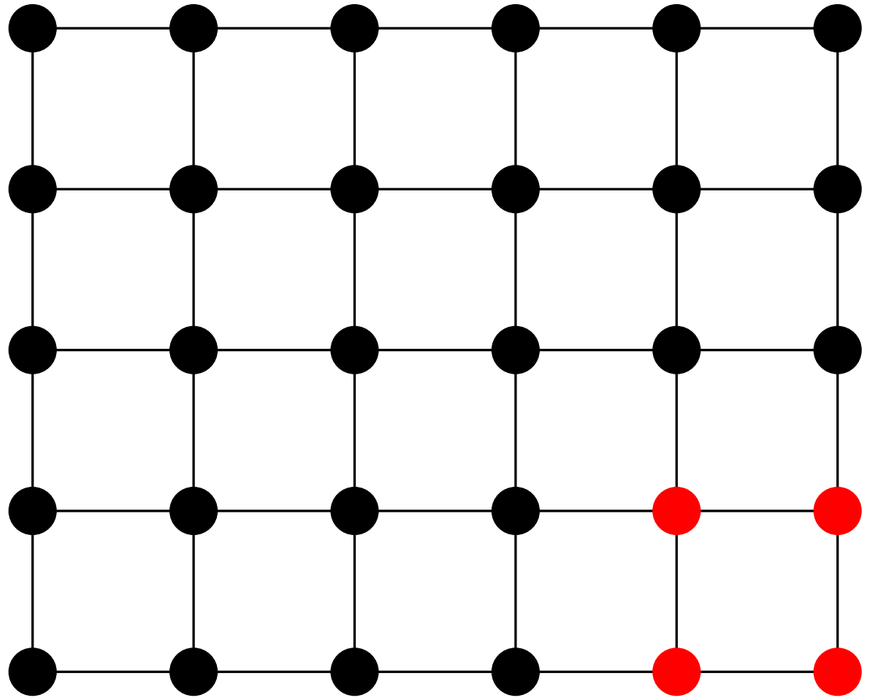


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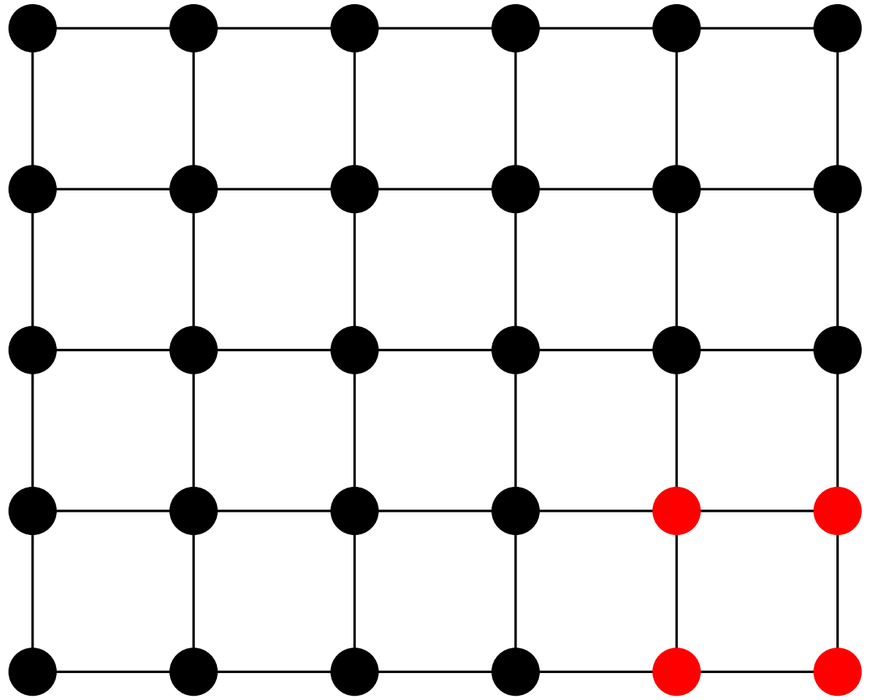


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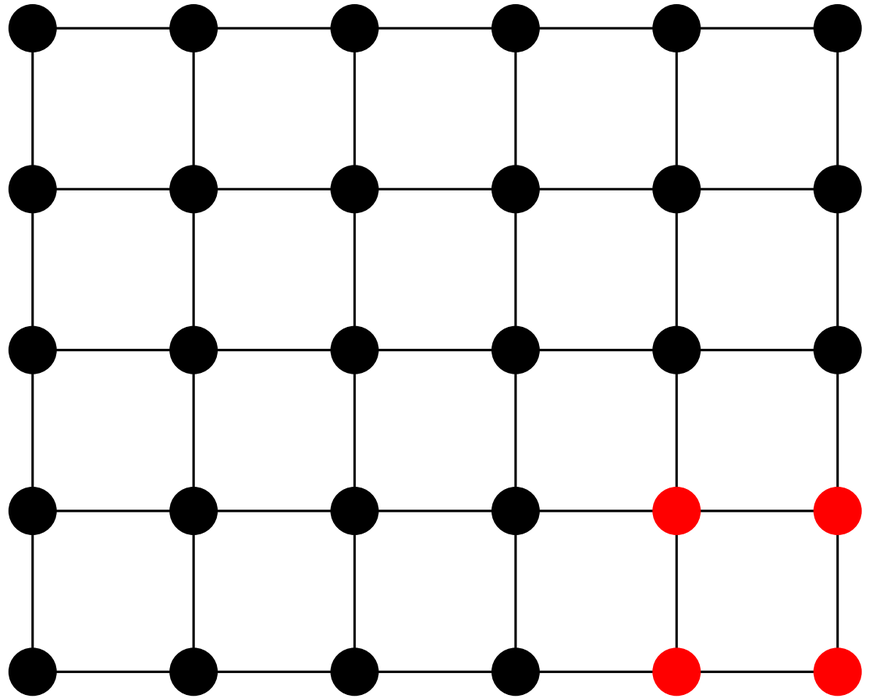


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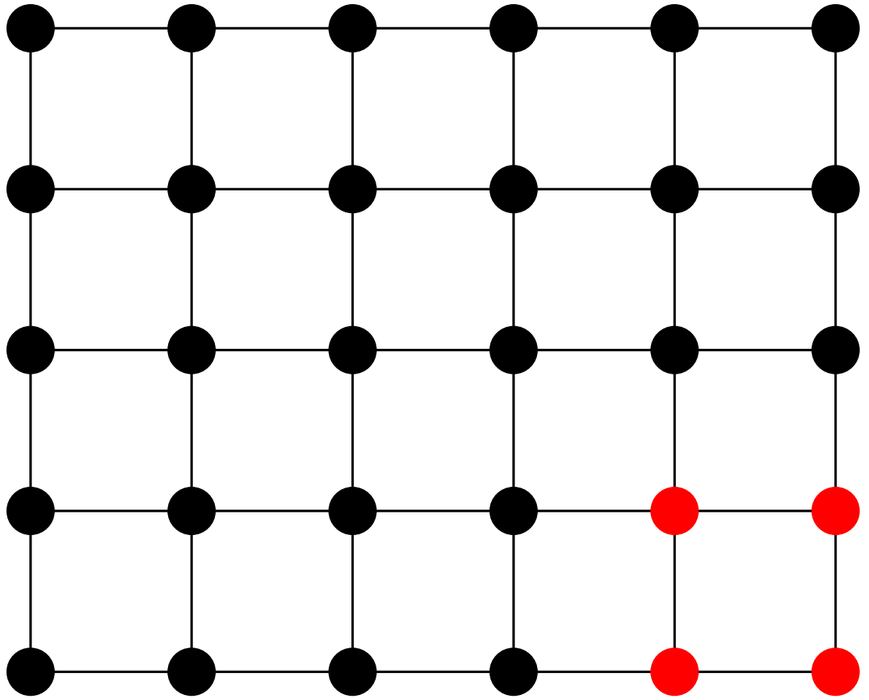


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G

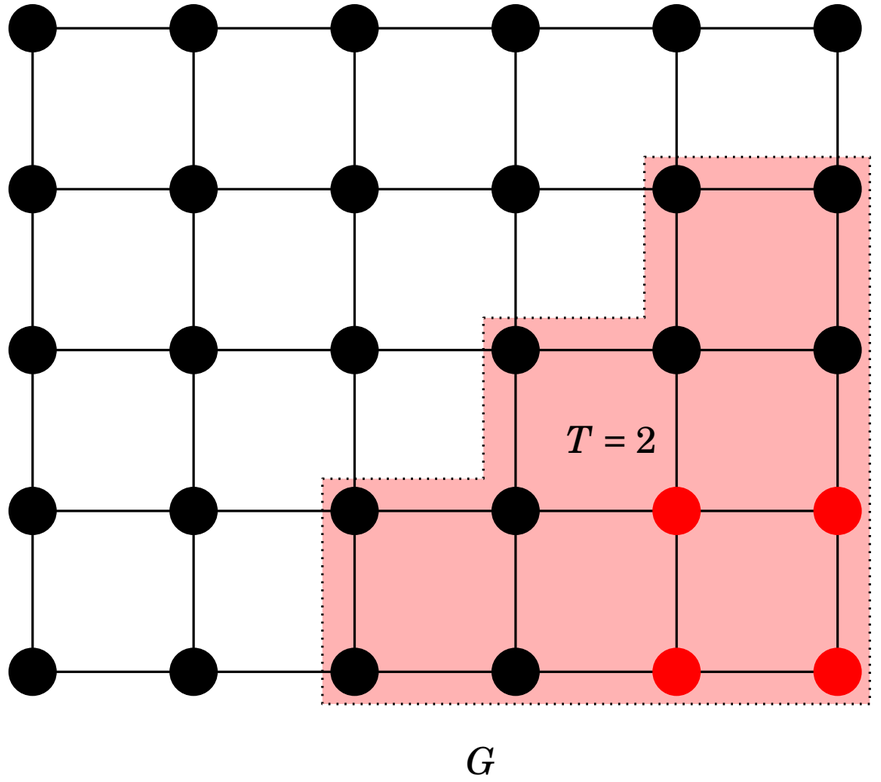
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The non-signaling model

[STOC '24]

- Σ finite set of labels
  - always contains garbage output $\perp$

- **Outcome**: function $\mathrm{O}\colon (G,x) \mapsto \{(y_i, p_i)\}_{i\in I}$
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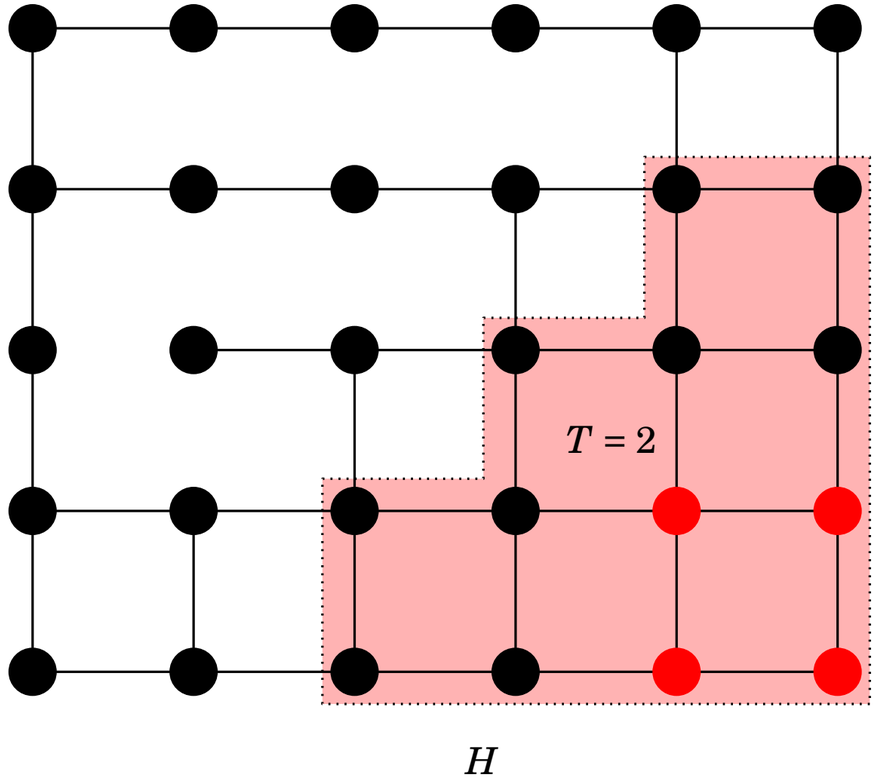
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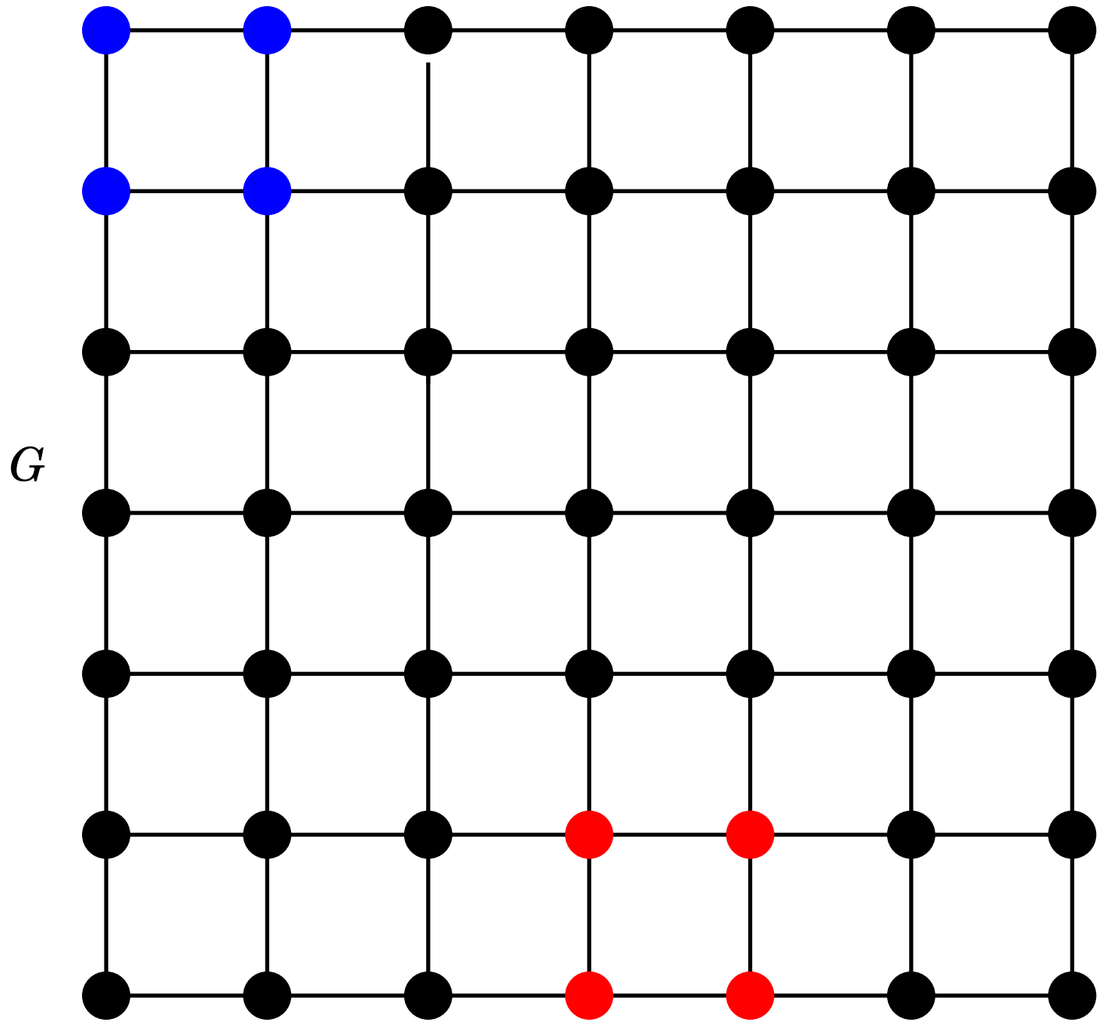


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The bounded-dependence model

[STOC '25a]

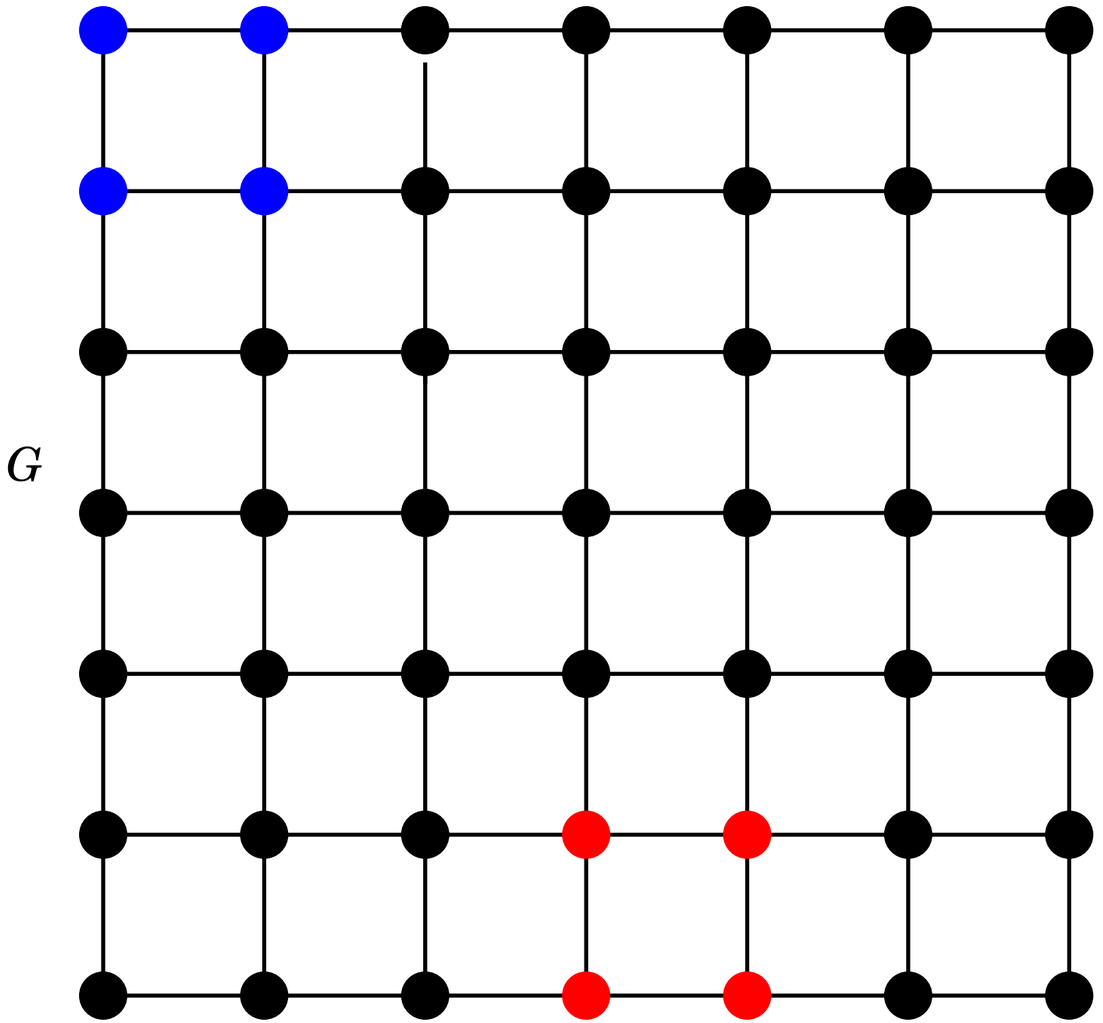
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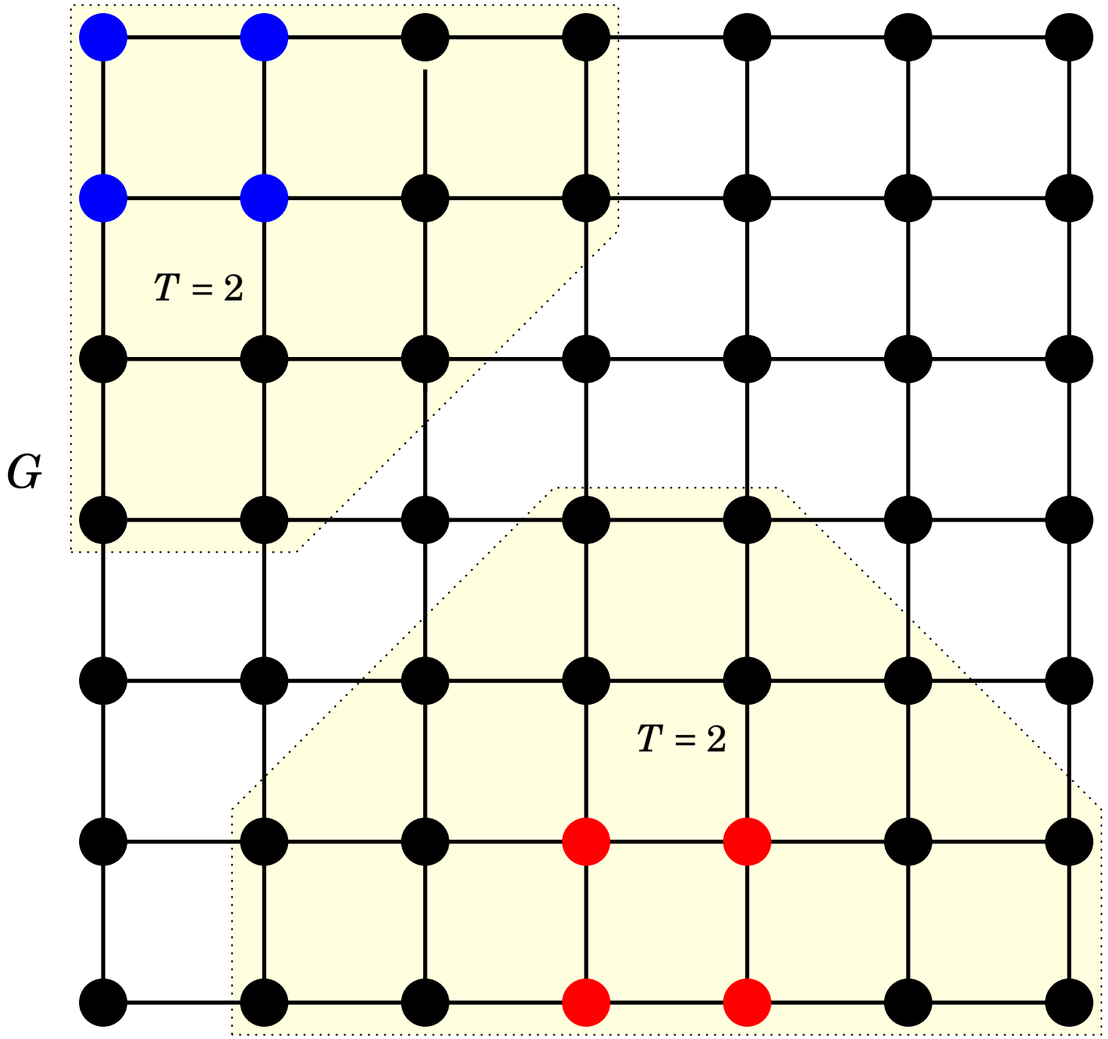
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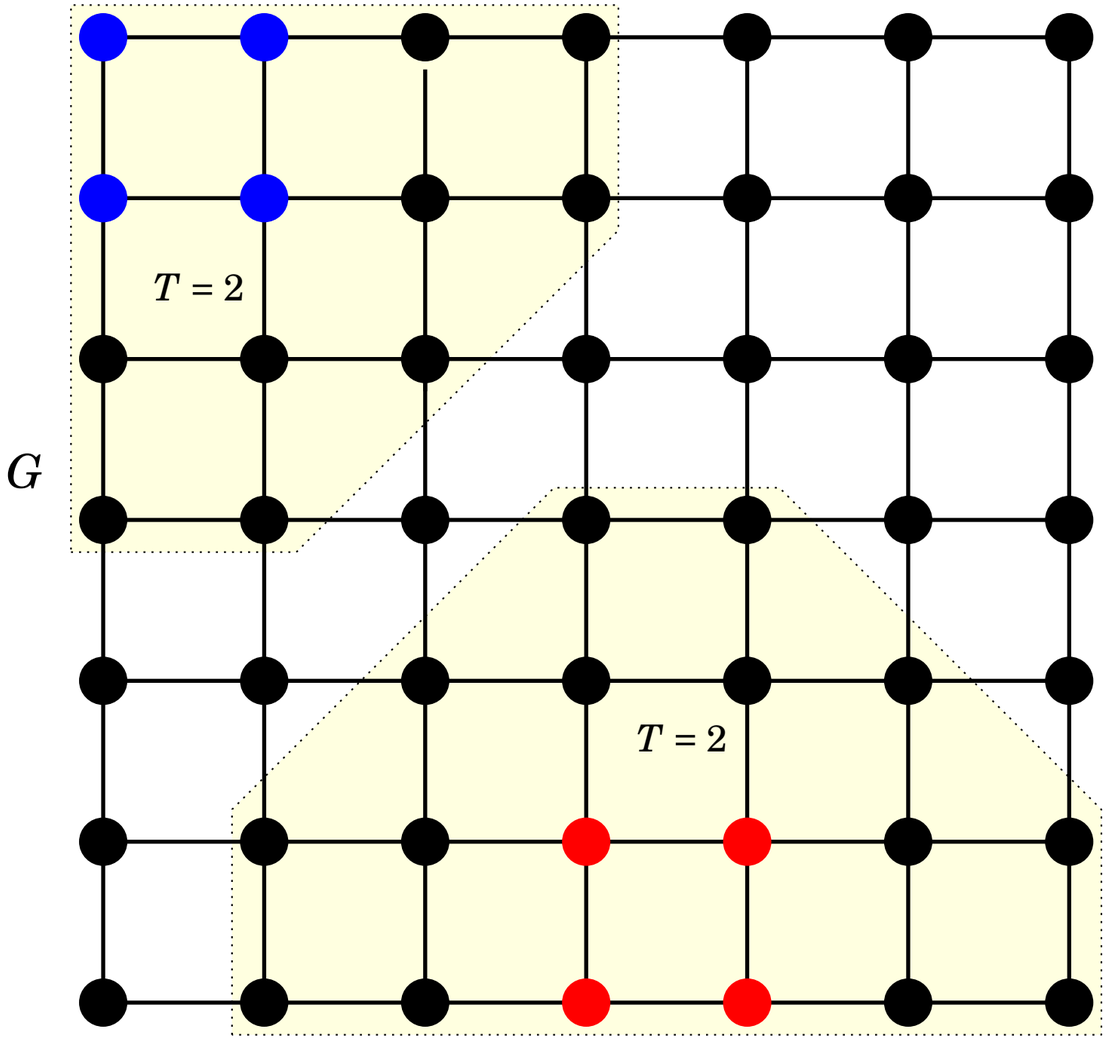
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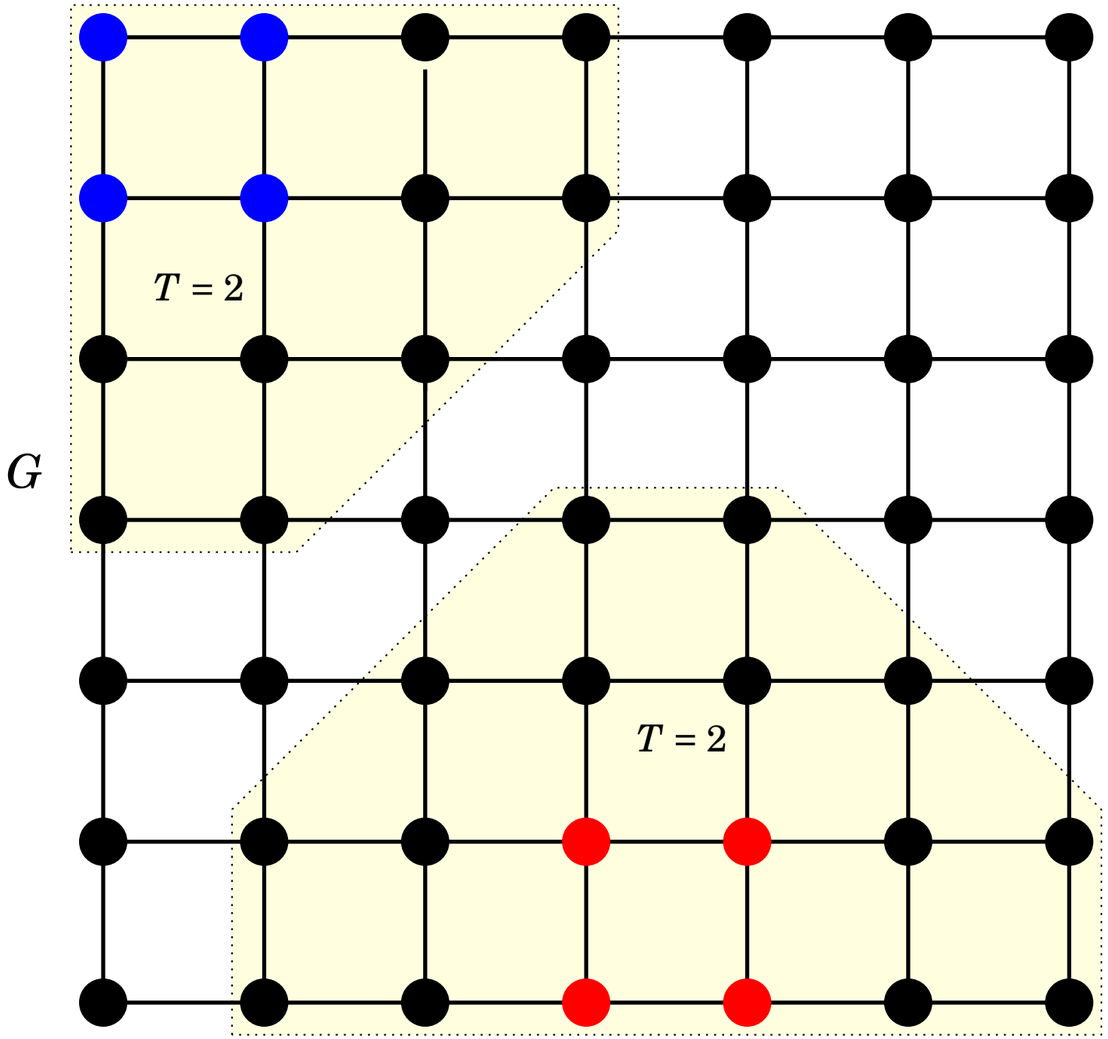
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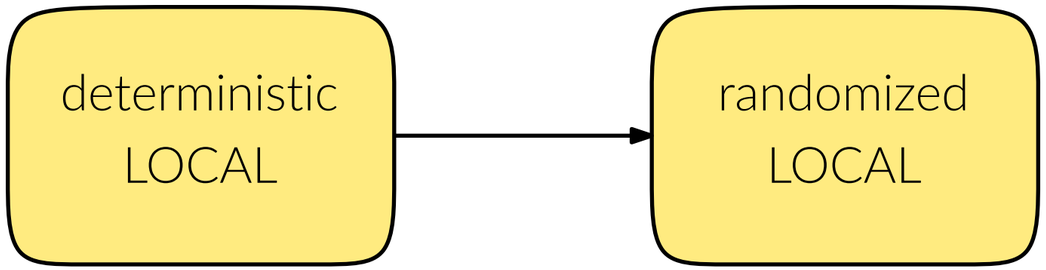
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  - if $T = O(1)$, *finitely-dependent* distribution



Relations among models

- $X \to Y$ means that locality T in X becomes locality $O(T)$ in Y



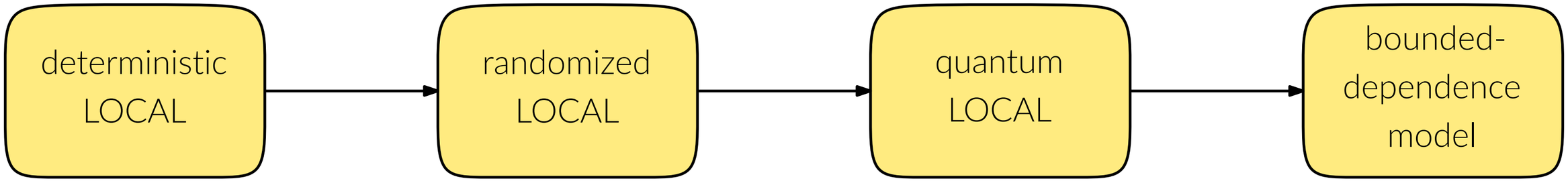
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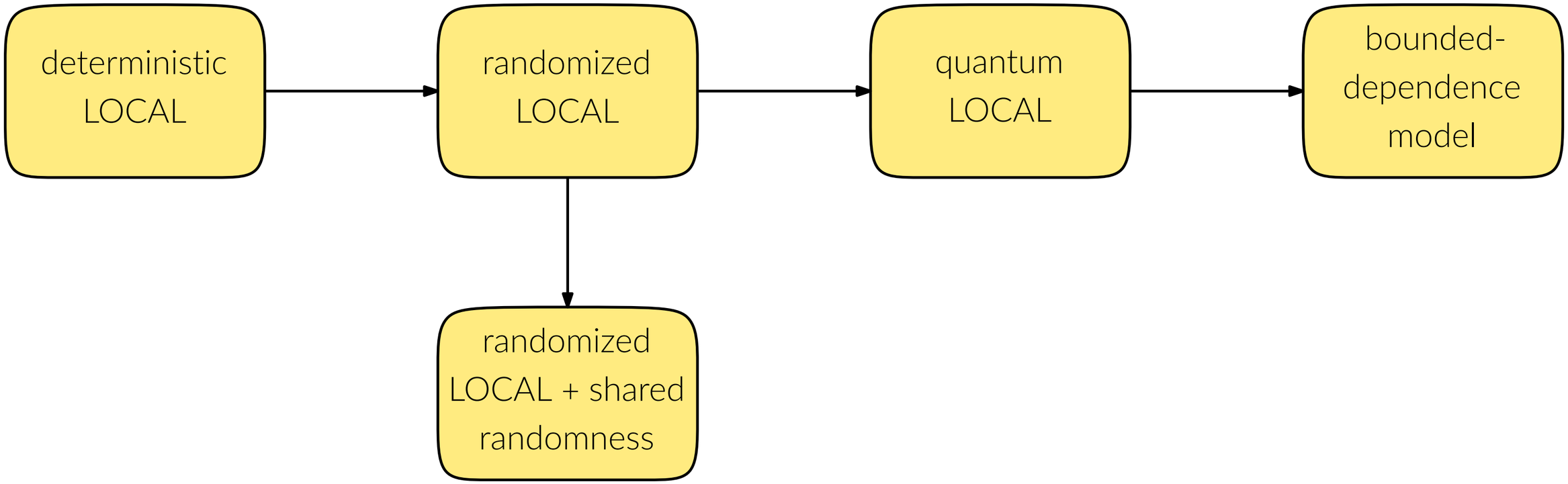
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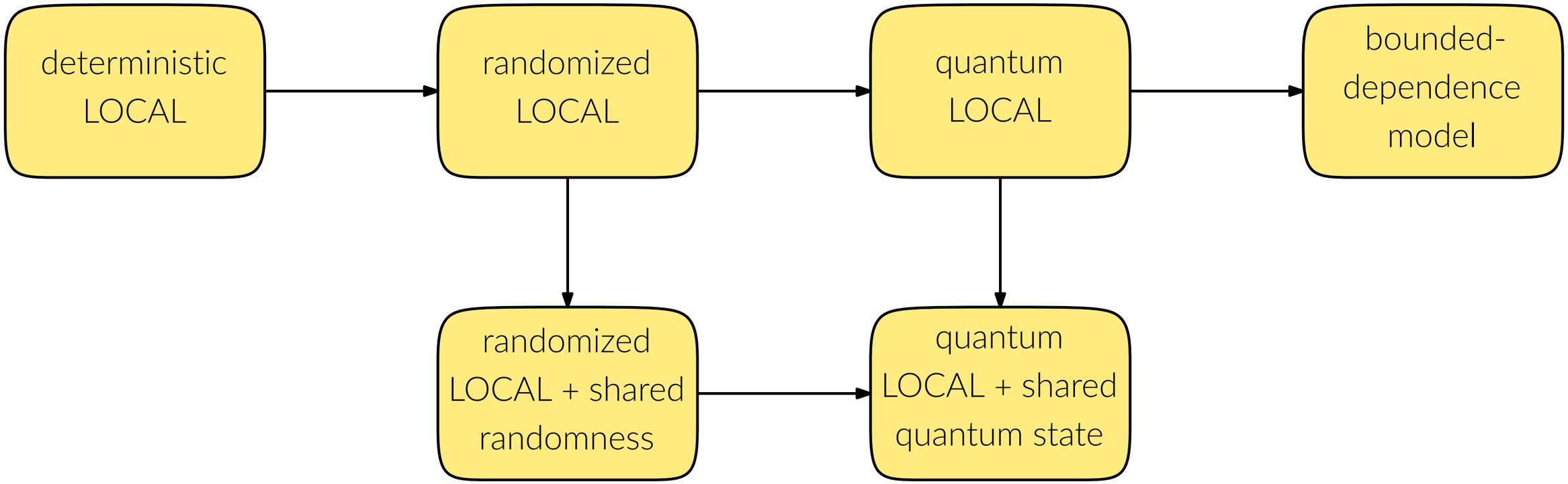
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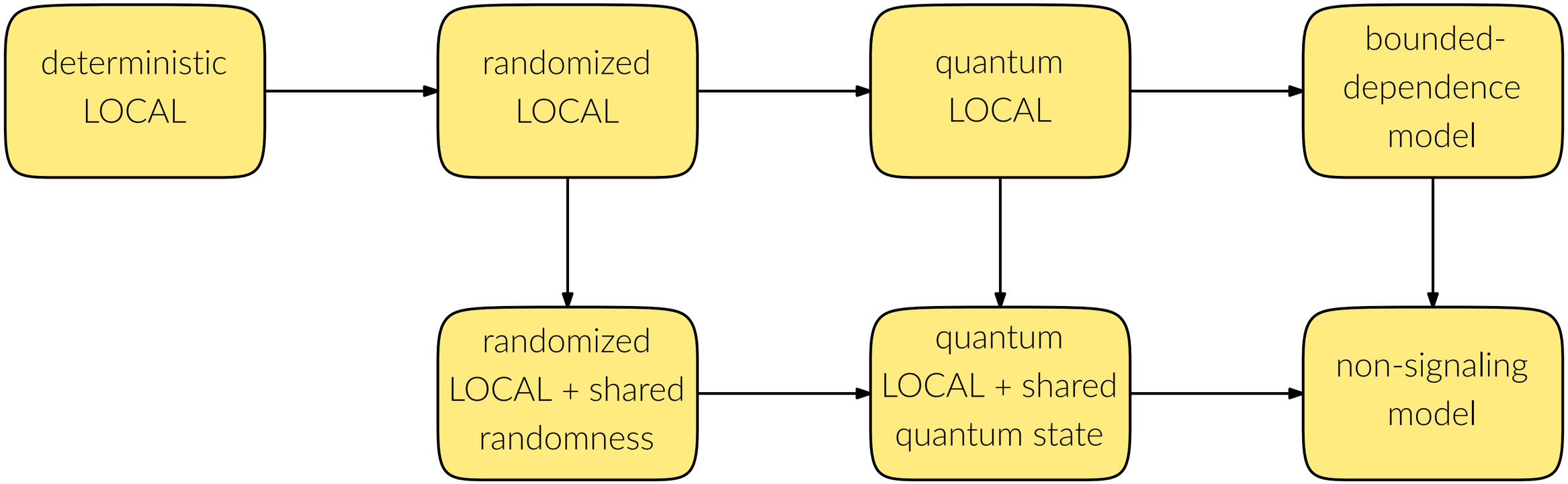
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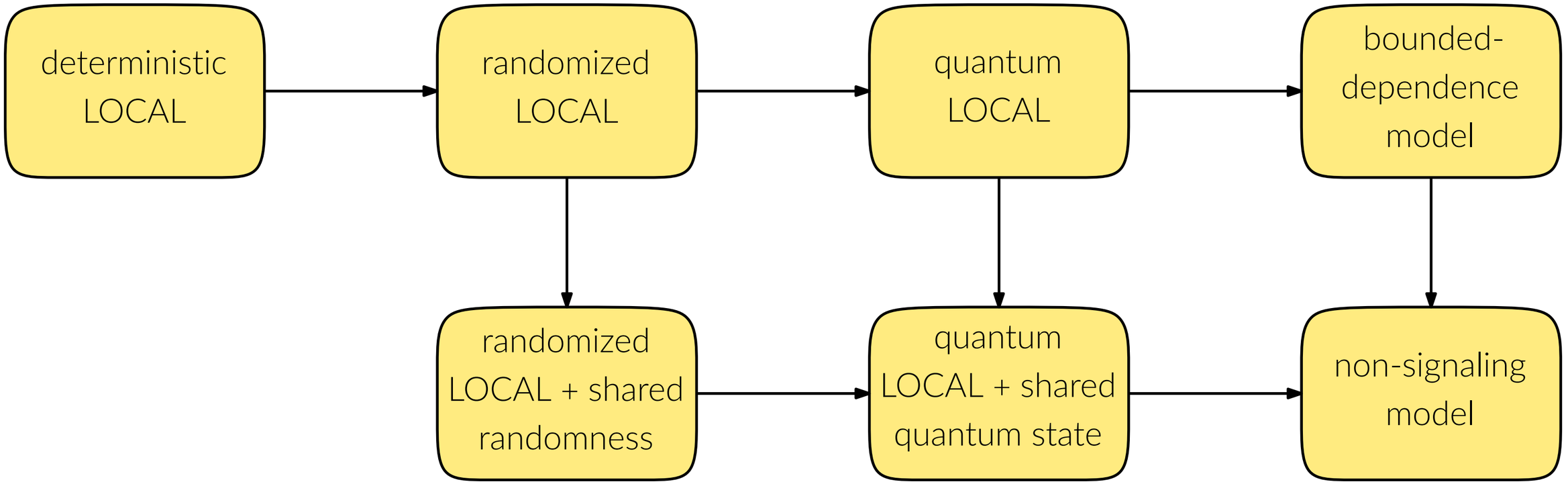
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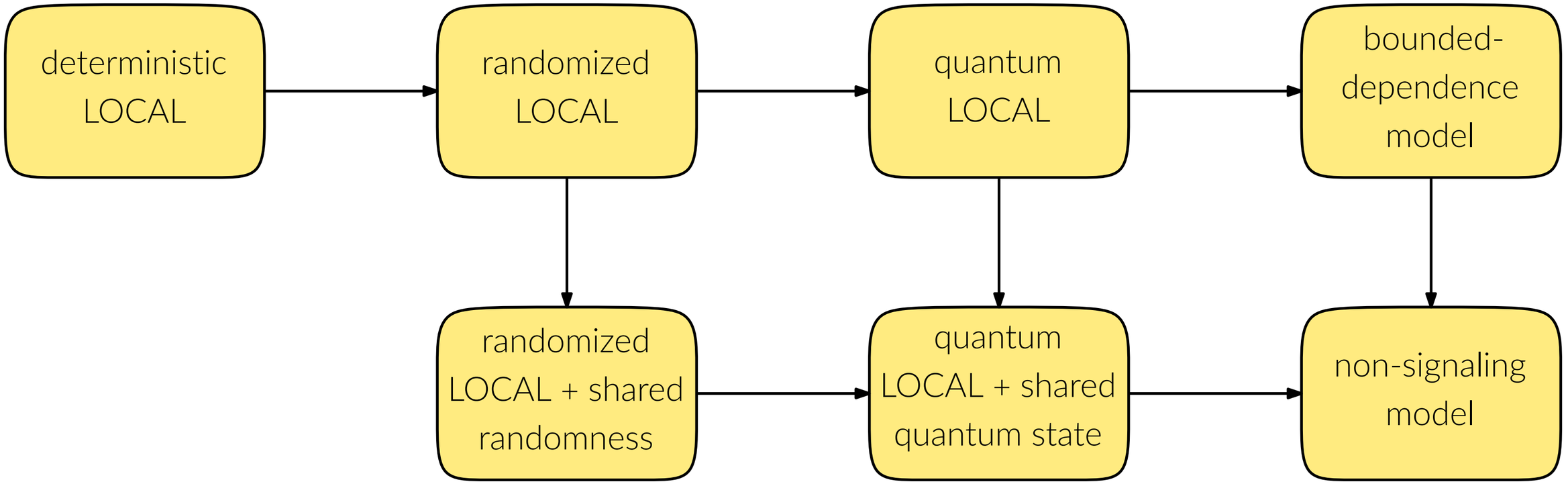
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  - yes!

Table of content

1. **Intro**: distributed algorithms, the LOCAL model, the quantum-LOCAL model, locally checkable labeling problems
2. **Classical lower bounds**: the indistinguishability argument
3. **Properties of distributed algorithms**: independence and non-signaling
4. **Super-quantum models**: bounded-dependence and non-signaling model
5. **State of the art results**
6. **Quantum advantage**

Complexity of approximate graph coloring

Problem: c-coloring χ-chromatic graphs [STOC '24]

		upper bound		lower bound		
χ	c	old	new	old	new	ref
2	2	$O(n)$	$O(n)$	$\Omega(n)$	$\Omega(n)$	trivial
2	3					
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Some results: non-signaling model

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What about other known lower bounds? E.g., 3-coloring cycles has complexity $\Theta(\log^\star n)$ [Linial, FOCS '87]

Some results: bounded-dependence model

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  - no!
  - For any LCL Π on bounded degree graphs, *there is a finitely-dependent distribution ($T = O(1)$) solving Π*
  - [STOC '25a]

Bounding general quantum advantage

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  - T-dependent distribution $\implies$ $O(\sqrt{Tn}$ poly log$(n))$-time deterministic LOCAL algorithm

Relations among models

- $X \to Y$ means that locality T in X becomes locality $O(T)$ in Y

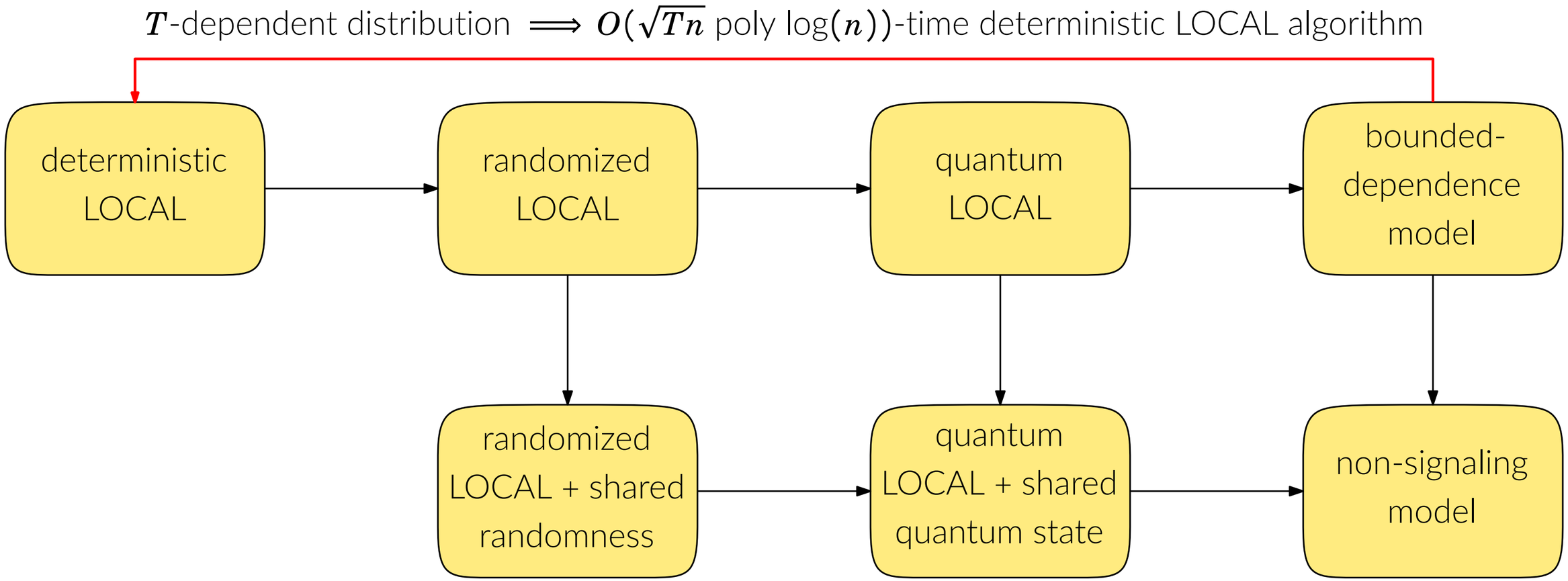


Table of content

1. **Intro**: distributed algorithms, the LOCAL model, the quantum-LOCAL model, locally checkable labeling problems
2. **Classical lower bounds**: the indistinguishability argument
3. **Properties of distributed algorithms**: independence and non-signaling
4. **Super-quantum models**: bounded-dependence and non-signaling model
5. **State of the art results**
6. **Quantum advantage**

Distributed quantum computing

- **Quantum-CONGEST**: computing the diameter of a network [Le Gall, Magniez, PODC '18]
  - **Classically**: $\Theta(n)$
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- **Quantum-LOCAL**: can we do something better? [STOC '25b, SODA '26]

Actual quantum advantage for LCLs

- Iterated GHZ (locally checkable)
  - **Classical**: $\Omega(\Delta)$ by round elimination [STOC '25b]
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- **Found** an LCL problem that admits quantum advantage
  - **useless** problem
  - what about problems that interest the community?
- At the moment, we have **no single example**
  - major open question

Non-signaling & quantum games

Alice



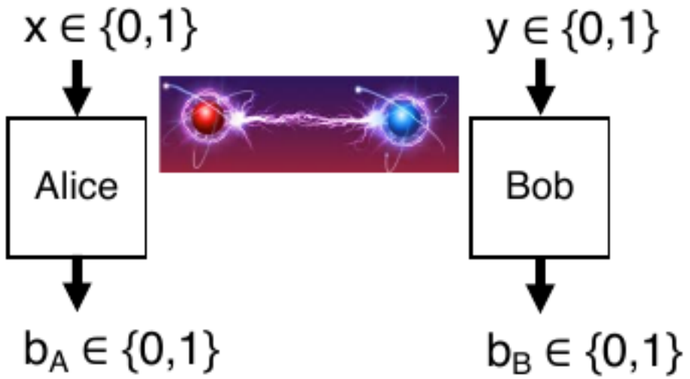
Bob



- Both Alice and Bob receive an input bit x and y in $\{0,1\}$
- They must output a bit each a and b according to some rule

CHSH game

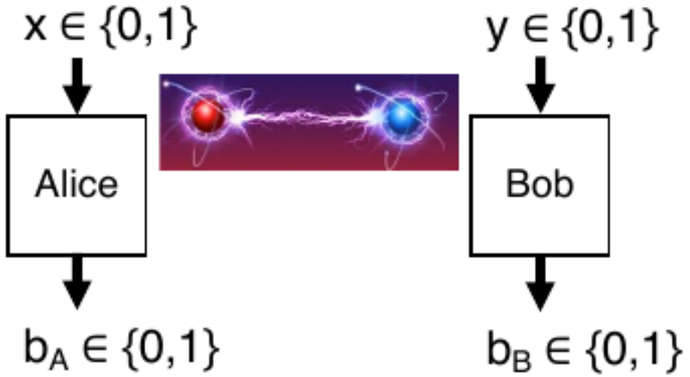
CHSH Game [Clauser, Horne, Shimony, Holt 1969]



winning condition: $b_A \oplus b_B = xy$

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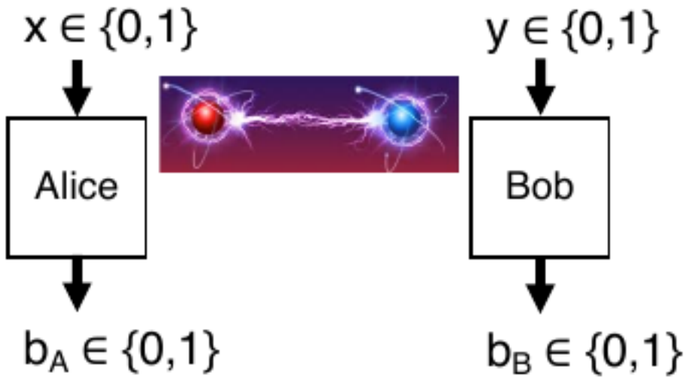


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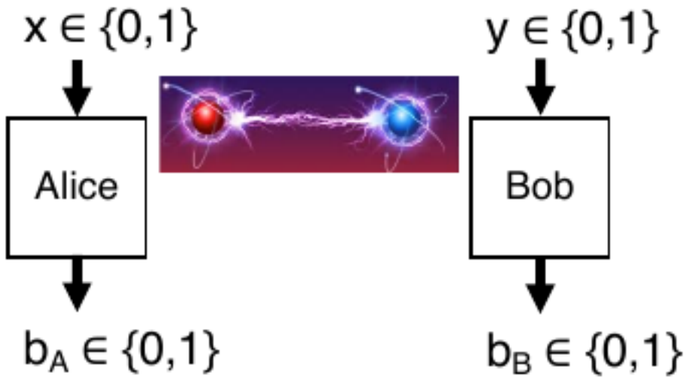
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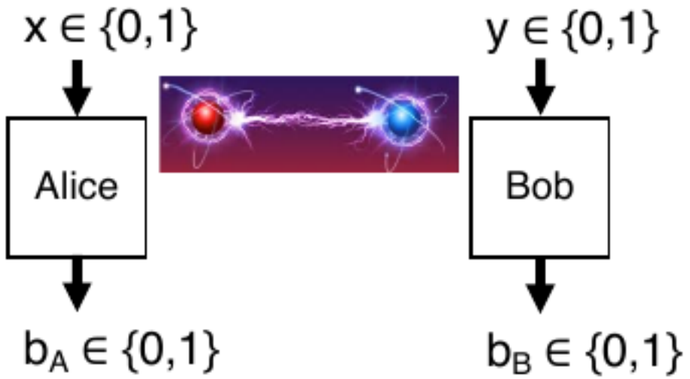
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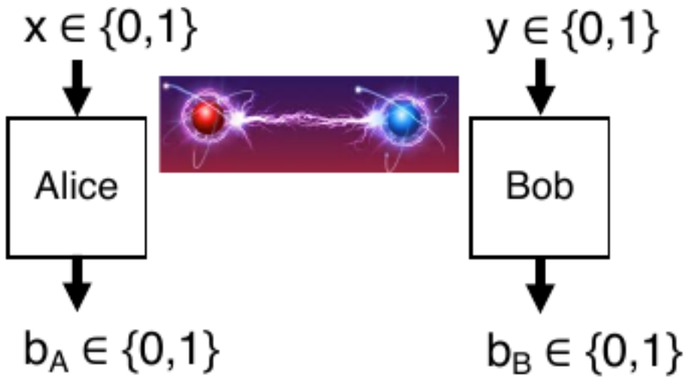
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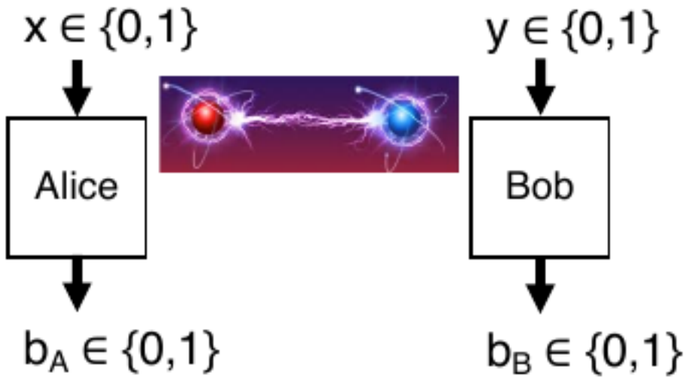
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  - *strategy*: sample u.a.r. among the correct solutions

Iterated CHSH problem

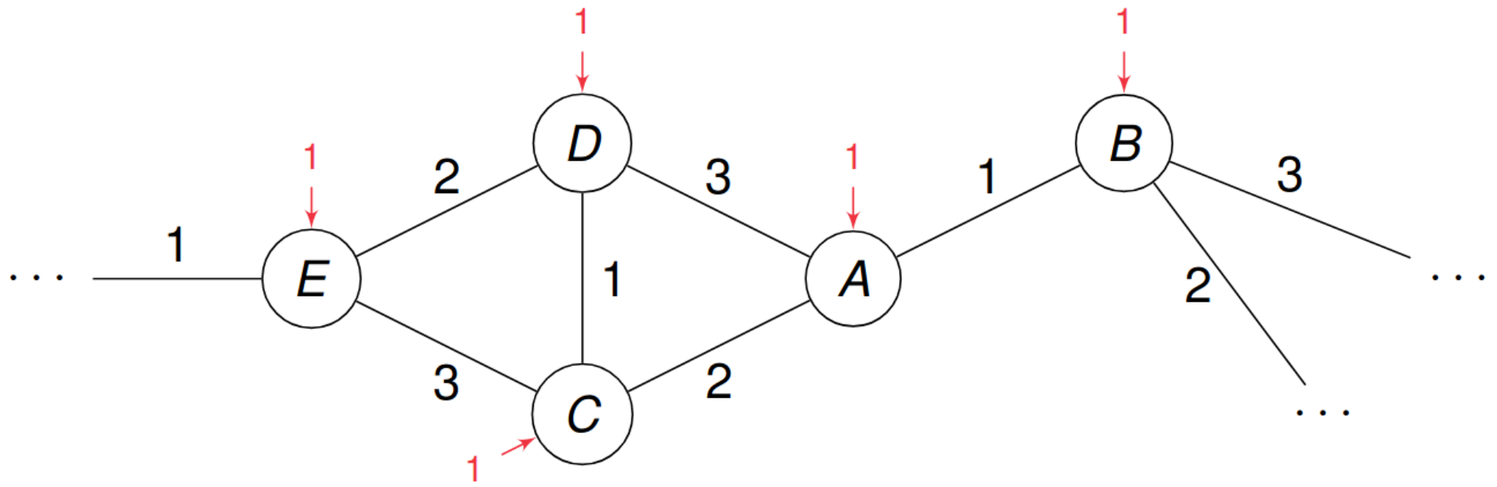
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Iterated CHSH problem

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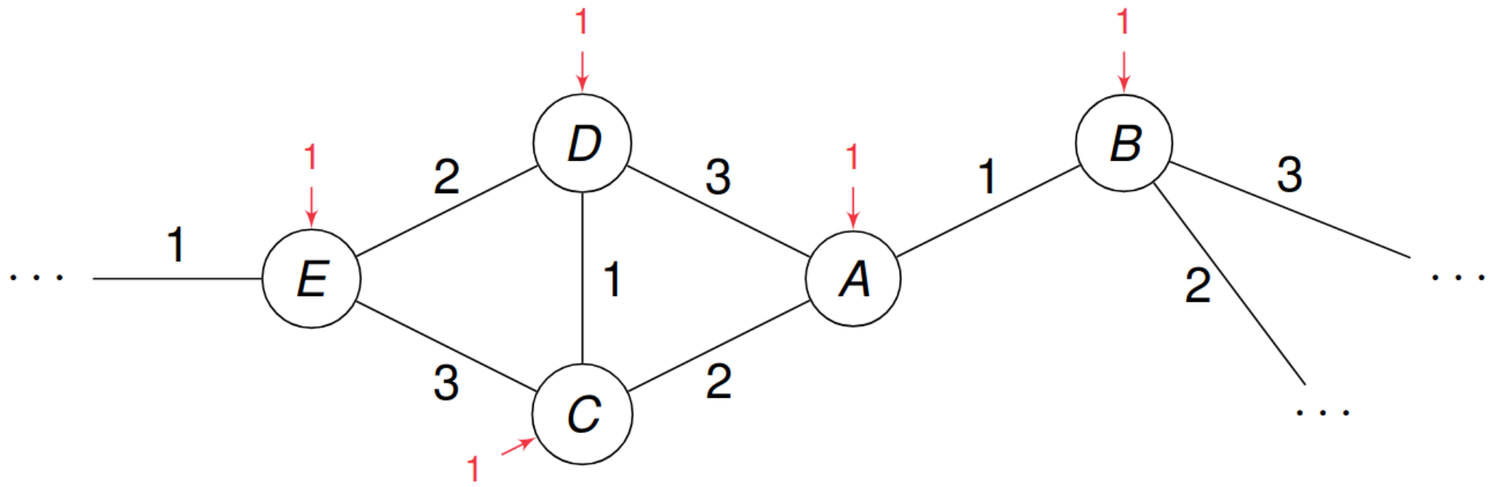
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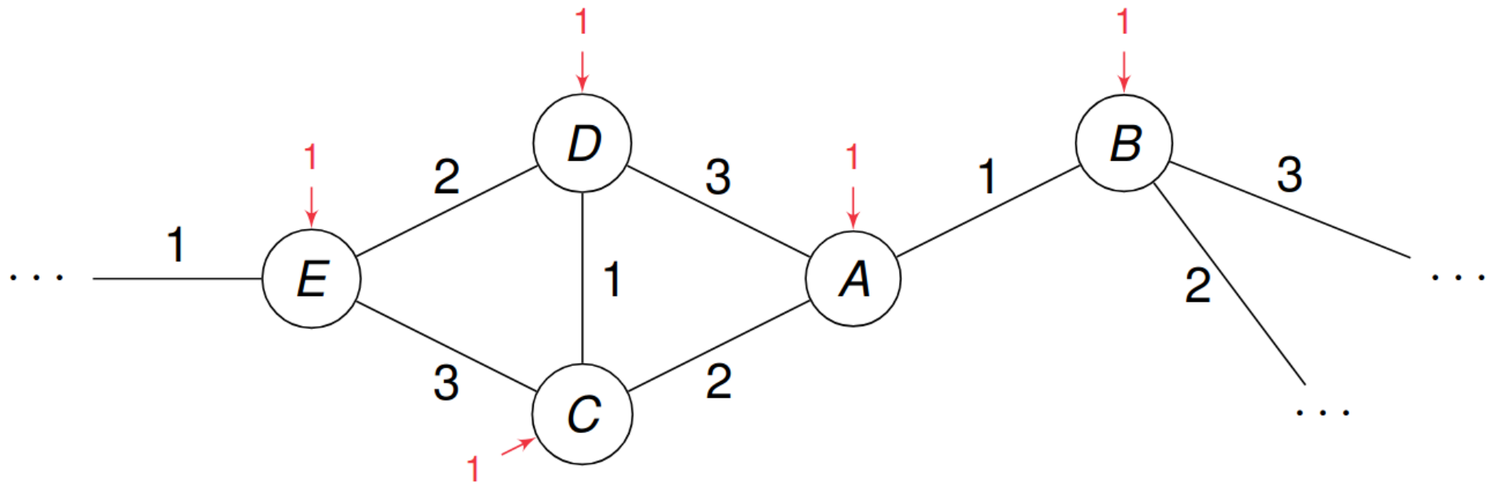
Iterated CHSH problem

- **Network of CHSH!**
- **How?**
- Δ-regular graph


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1

Iterated CHSH problem

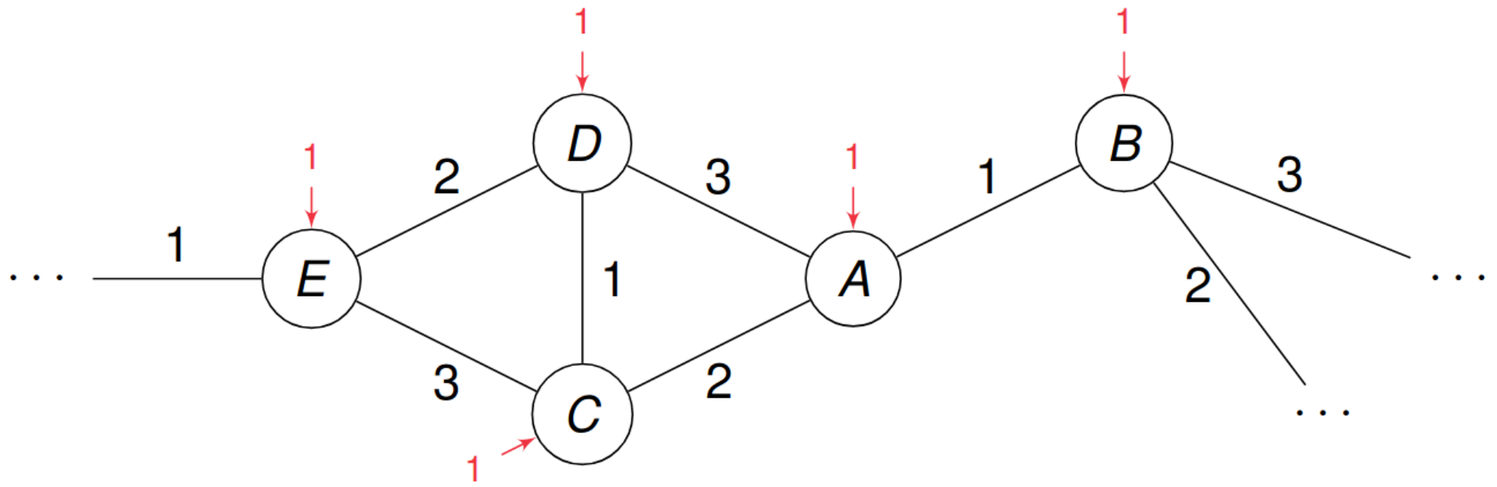
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- Input Δ-edge coloring



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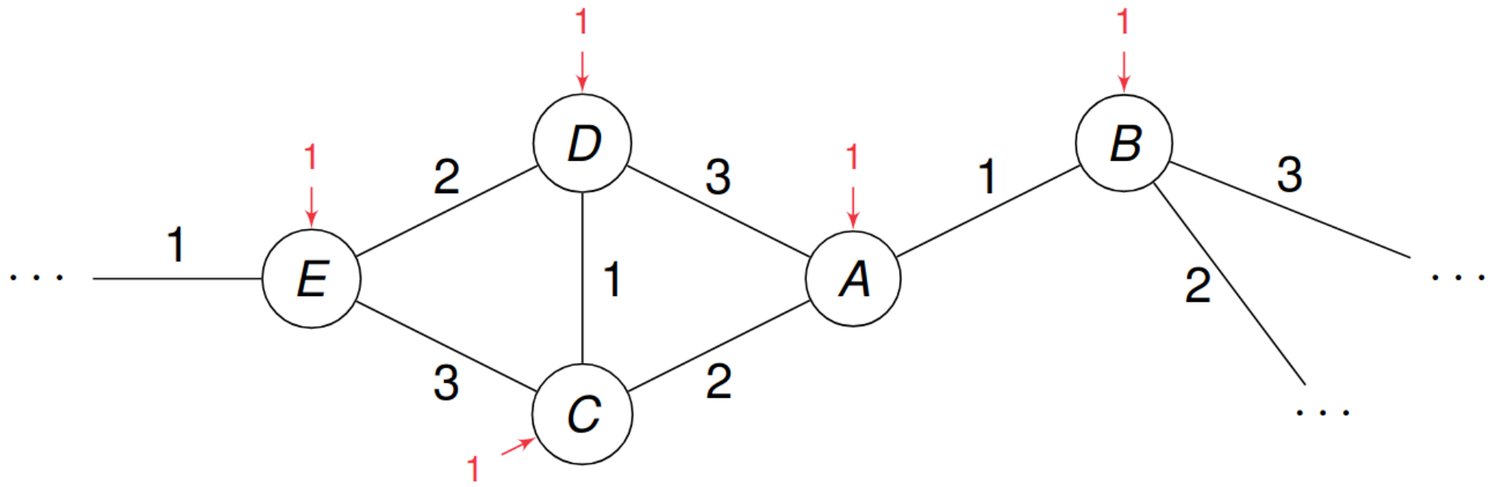
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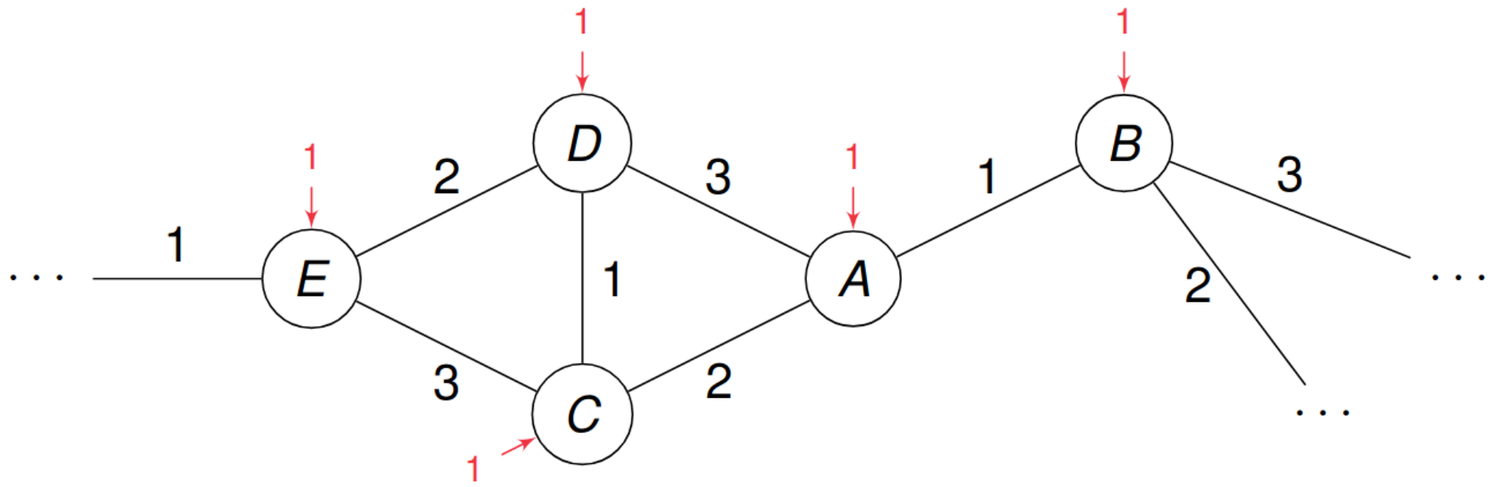
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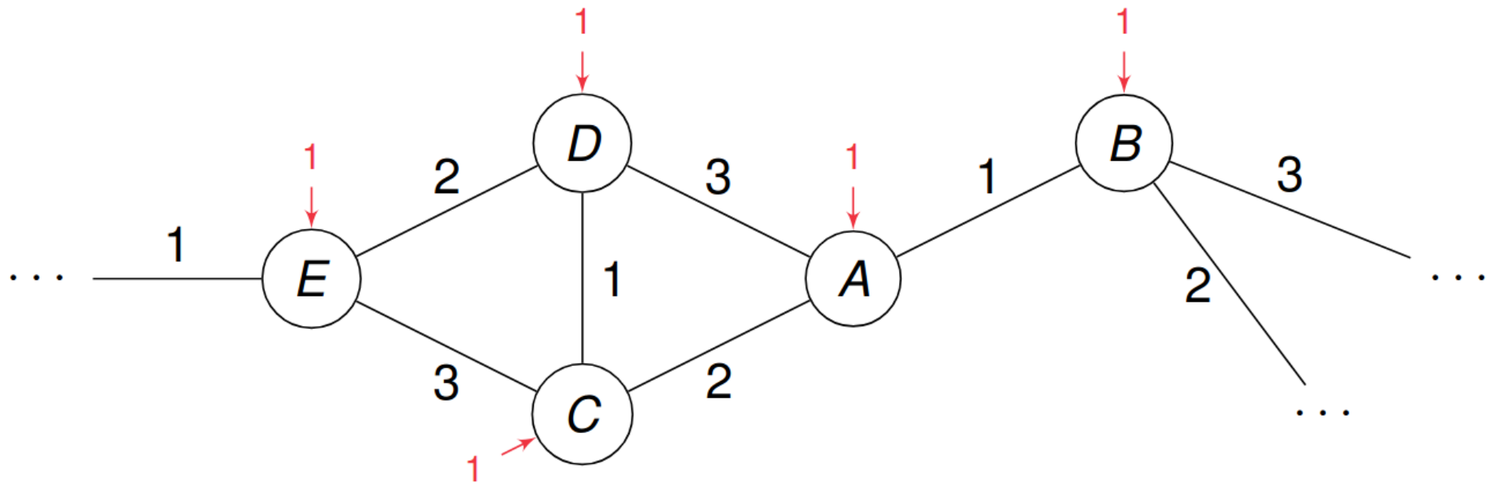
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- Input of game i is output of game $i-1$



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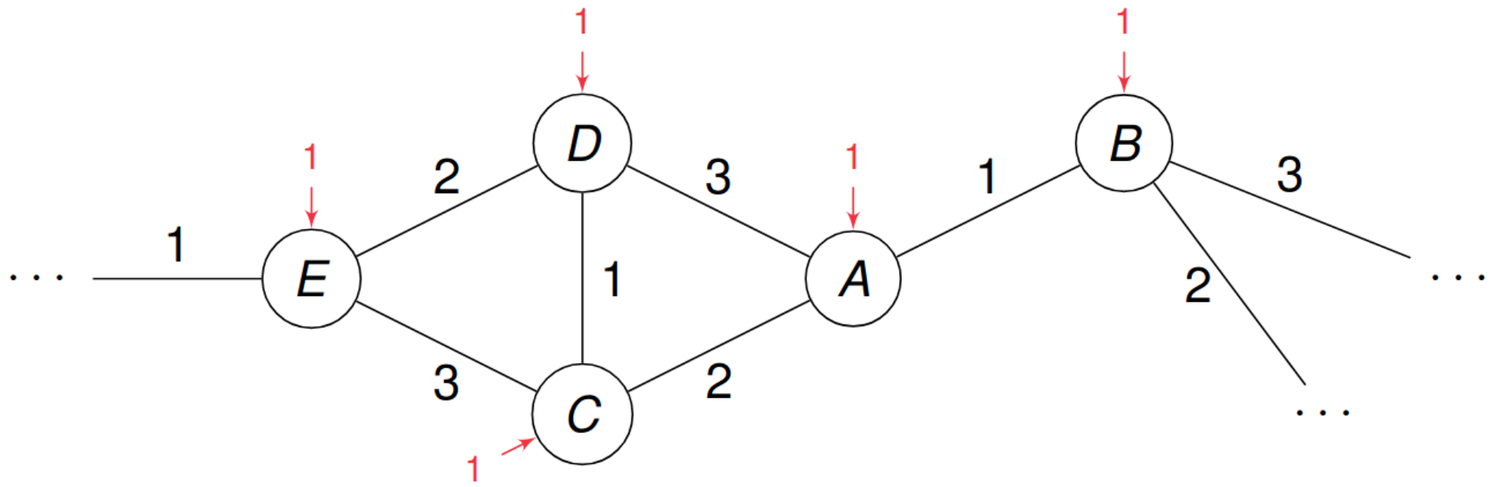
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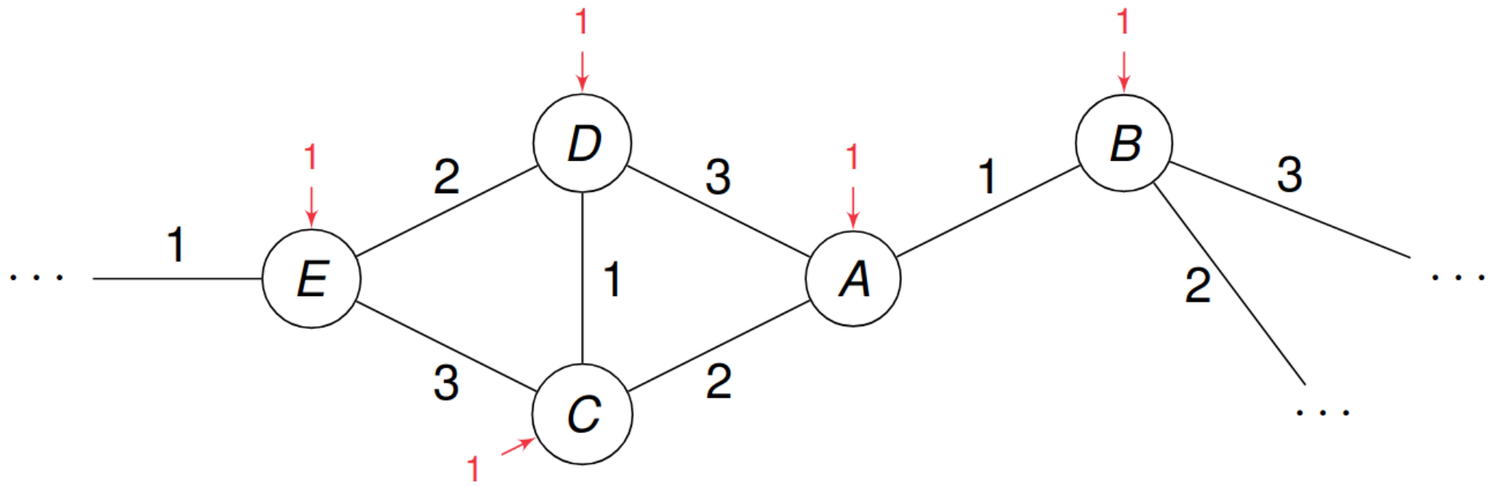
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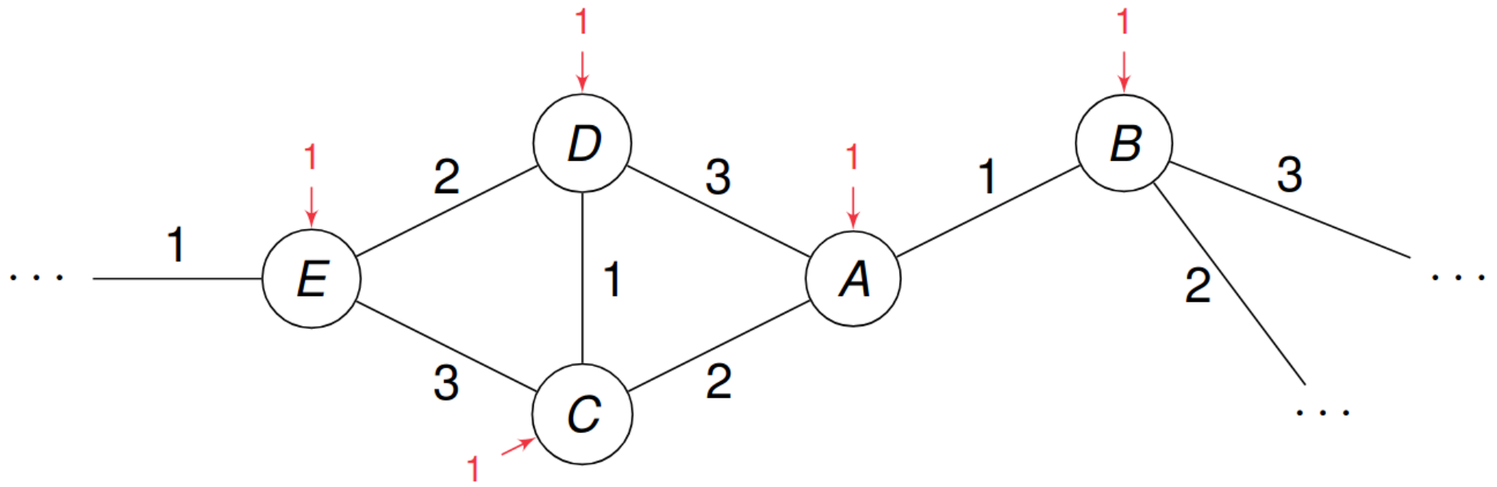


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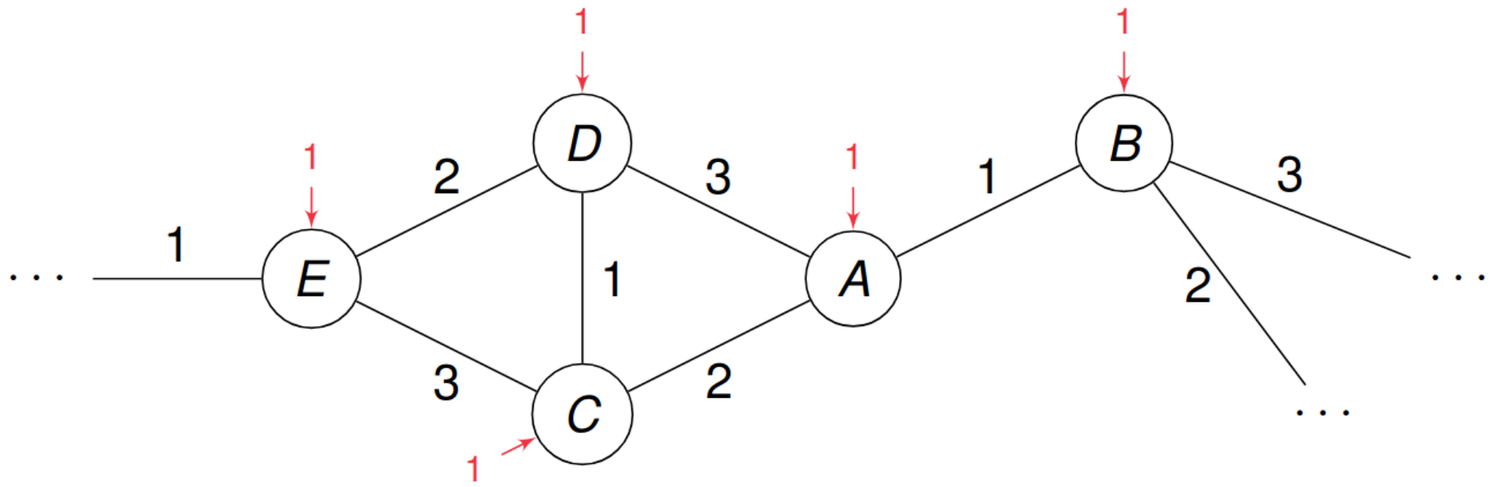


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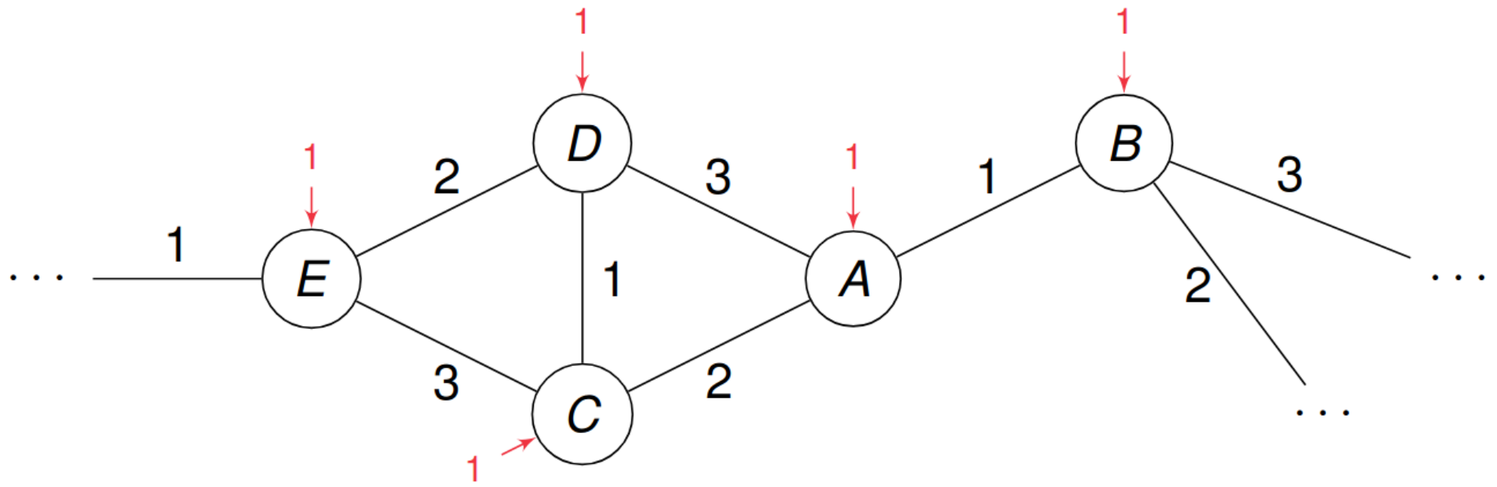
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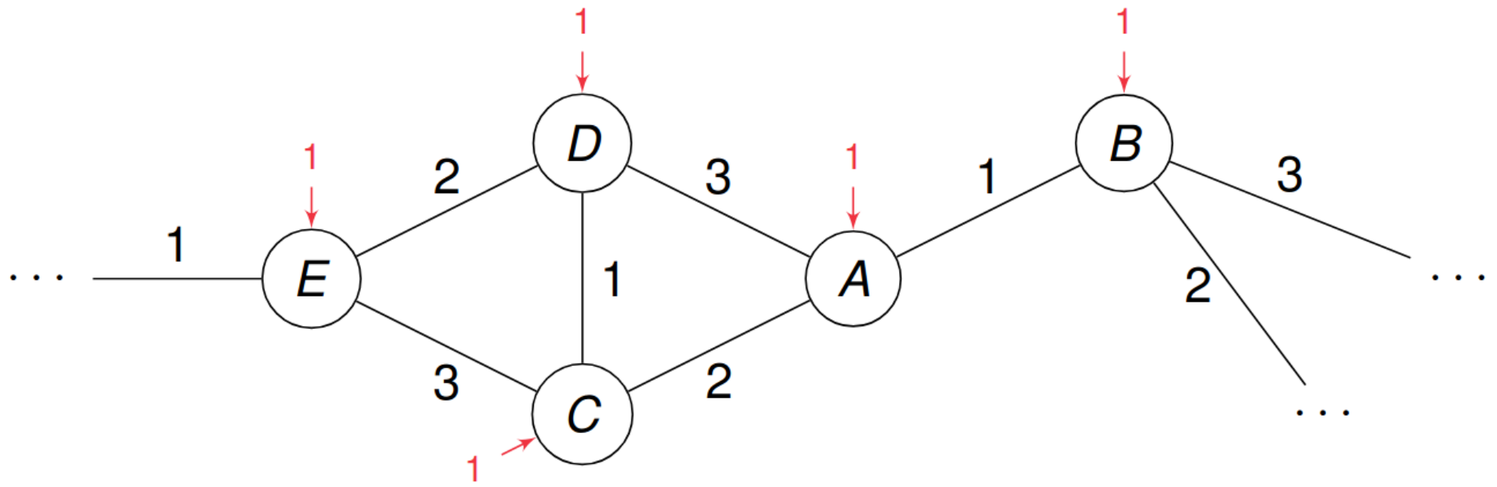
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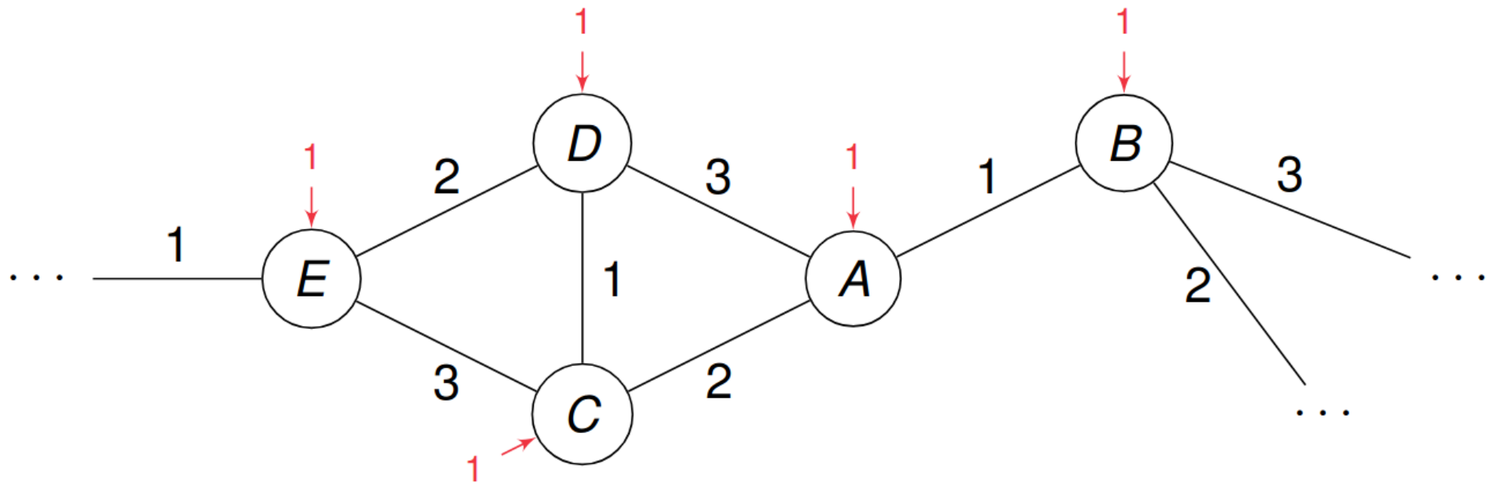


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- **Quantum upper bound?**

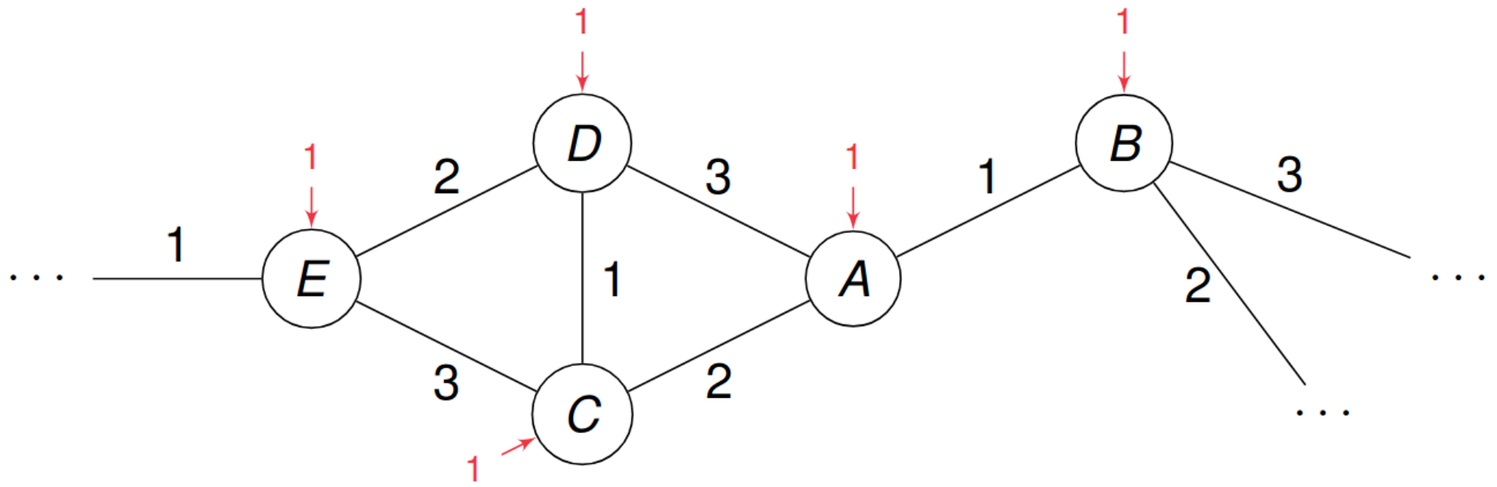
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- **Classical lower bound?**
  - $\Omega(\Delta)$ by round elimination [STOC '25b]
- **Quantum upper bound?**
  - winning prob. of single game is $\approx 83\%$
  - we need a better game (win 100%)

GHZ game

- **Greenberger-Horne-Zeilinger game**

Alice



Bob



Charlie



Inputs	$0,0,0$	$0,1,1$	$1,0,1$	$1,1,0$
Outputs	$0,0,0$	$1,0,0$	$1,0,0$	$1,0,0$
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Charlie



- Inputs always even number of 1

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GHZ game

- **Greenberger-Horne-Zeilinger game**

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Bob



Charlie



- Inputs always even number of 1
- **XOR** of **outputs** is 0 iff inputs are all 0, otherwise 1

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GHZ game

- **Greenberger-Horne-Zeilinger game**

Alice



Bob



Charlie



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THANKS! Questions?

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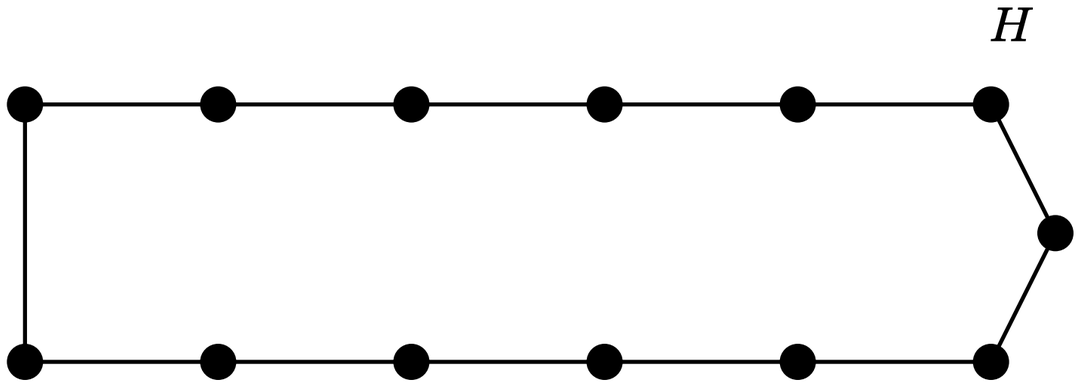
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Boosting failure probability: randomized-LOCAL

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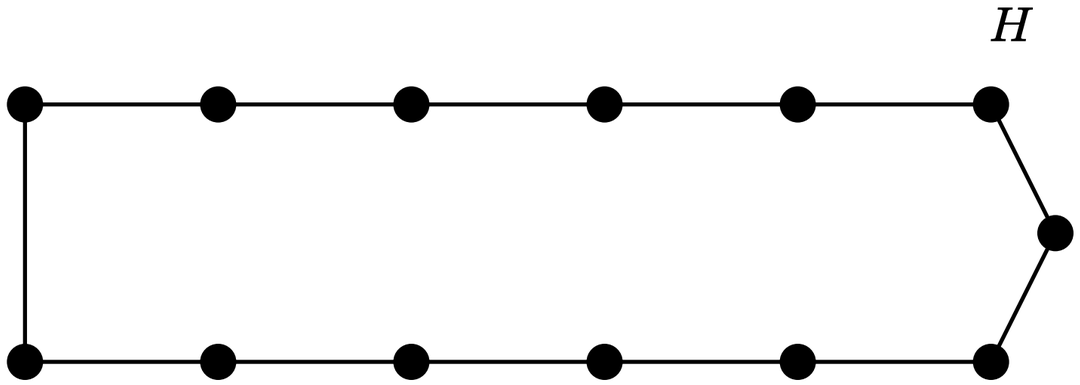
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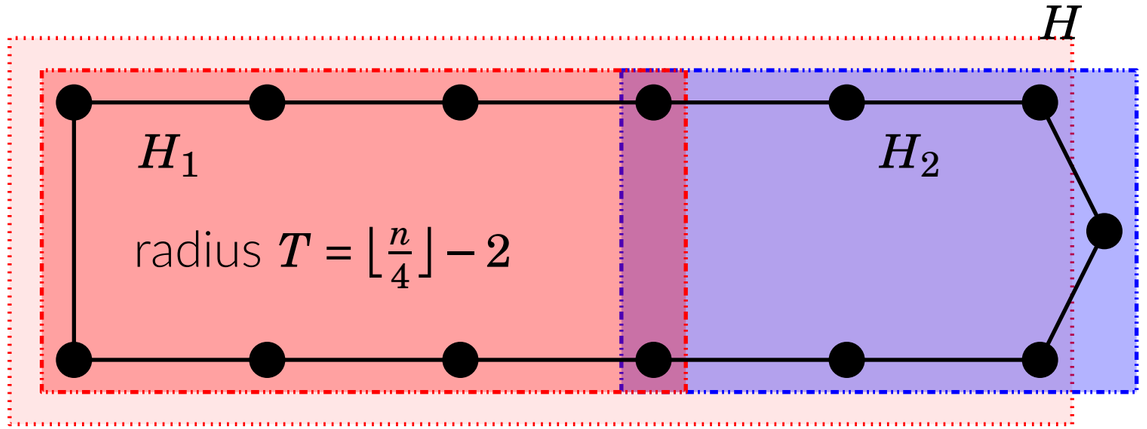
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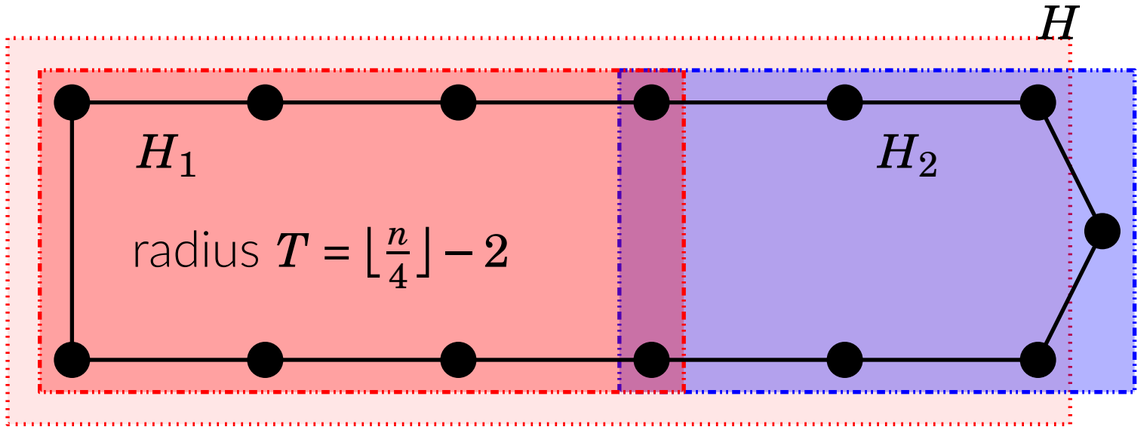
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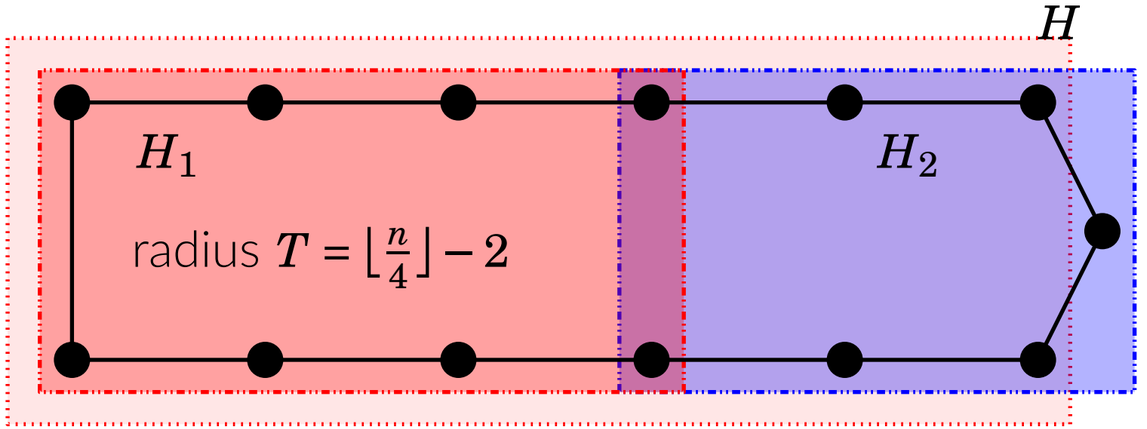
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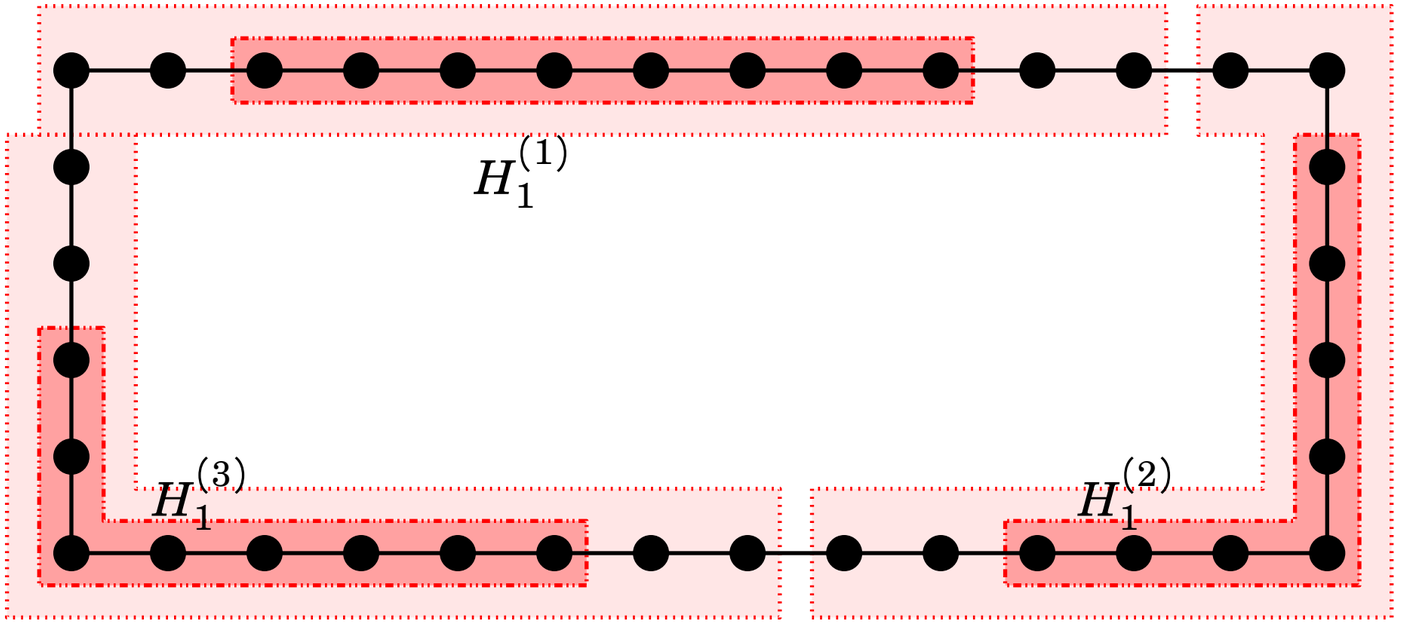
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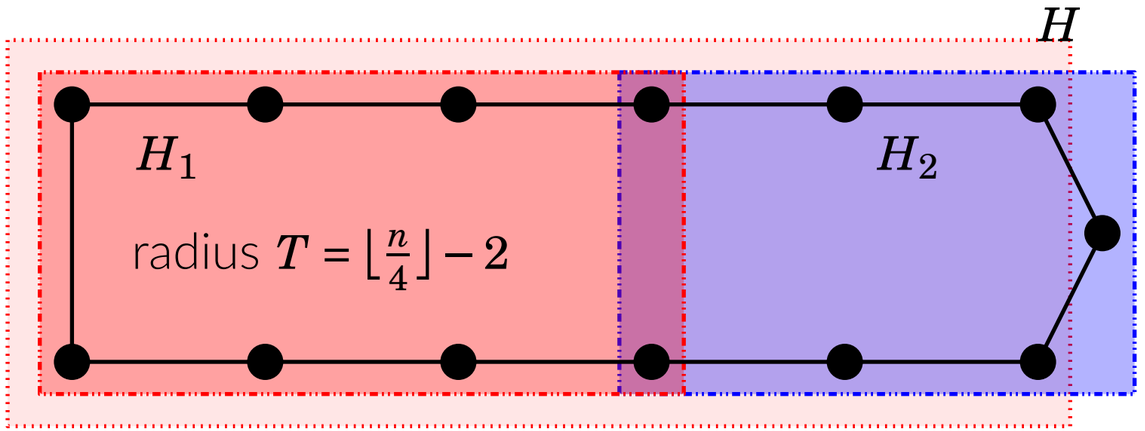
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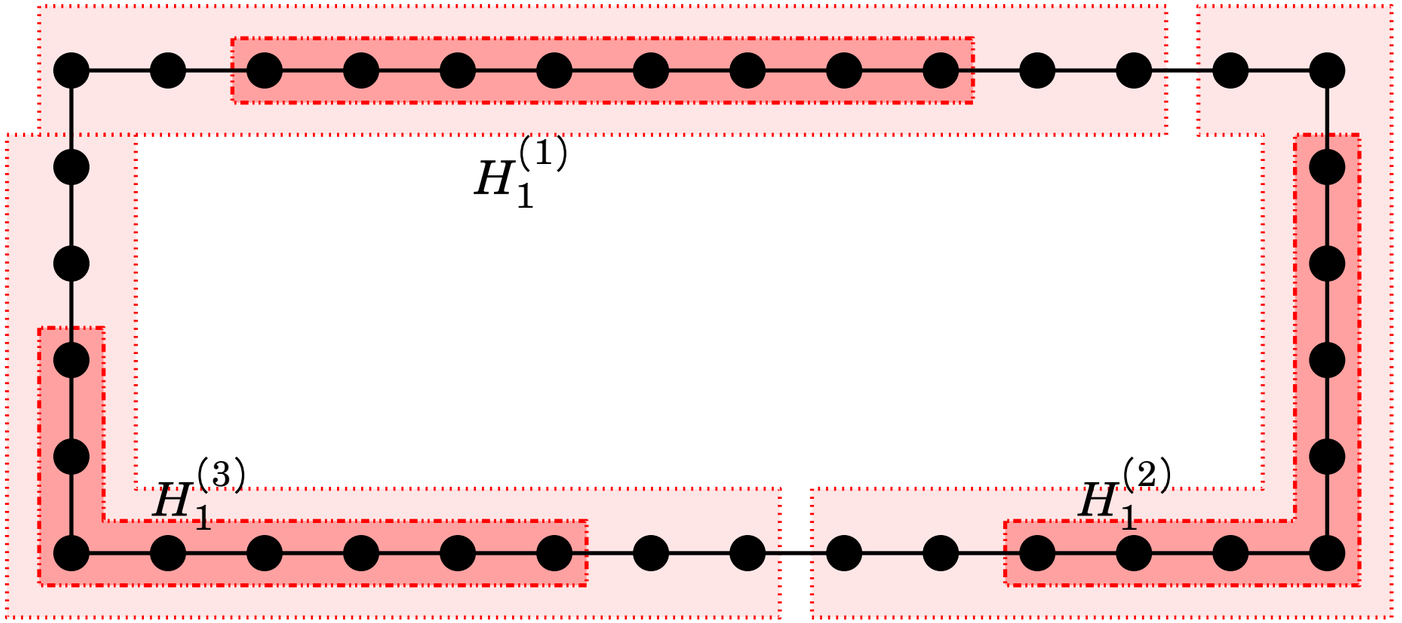
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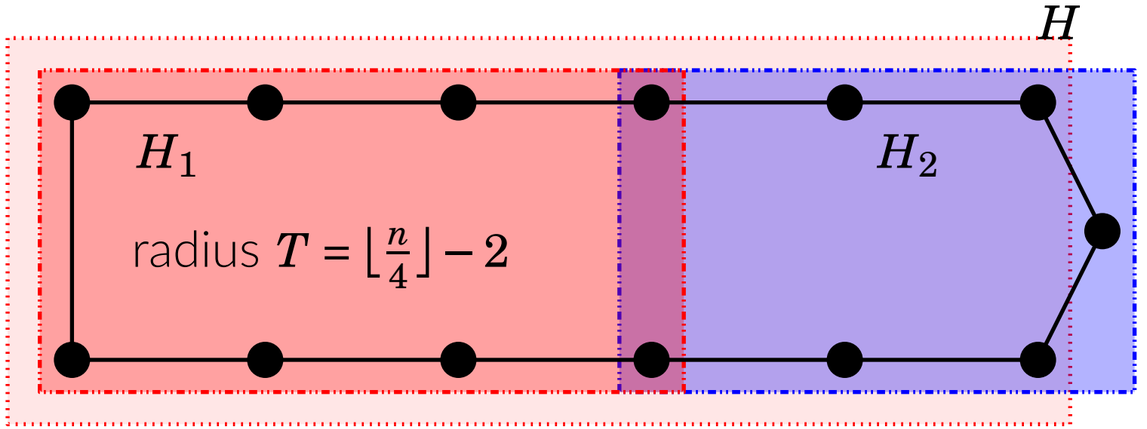
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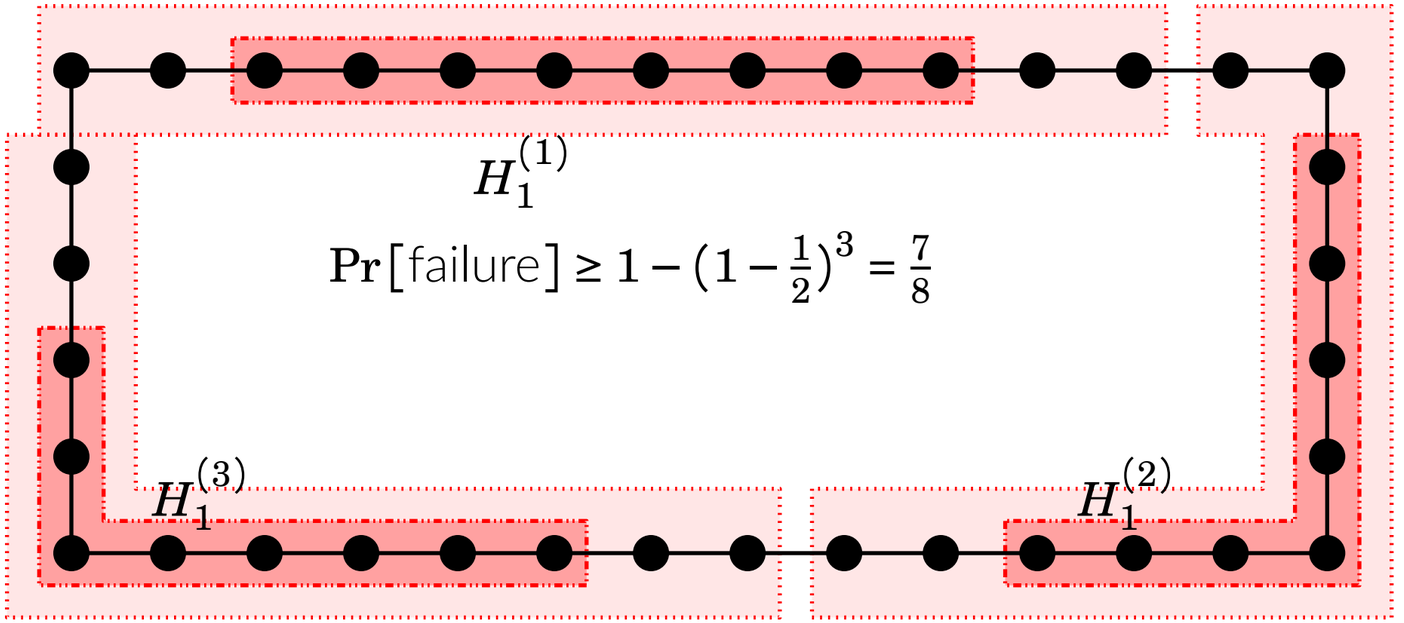
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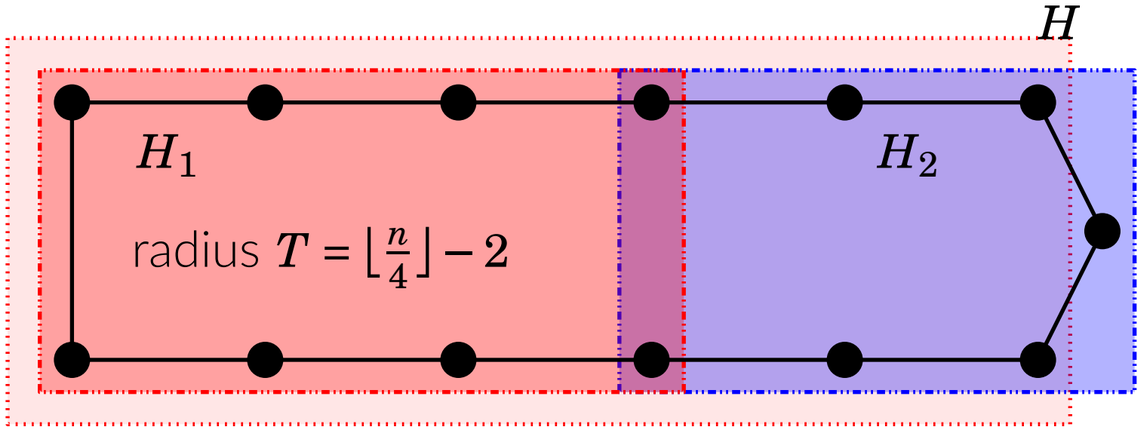
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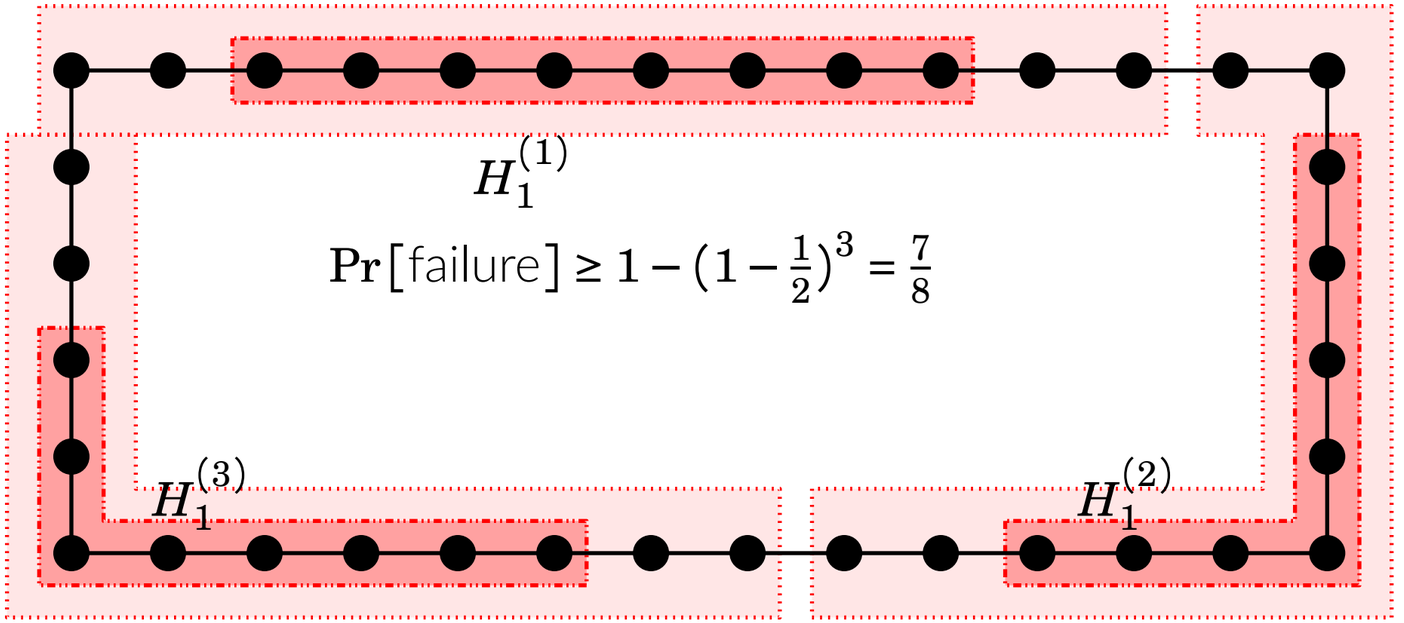
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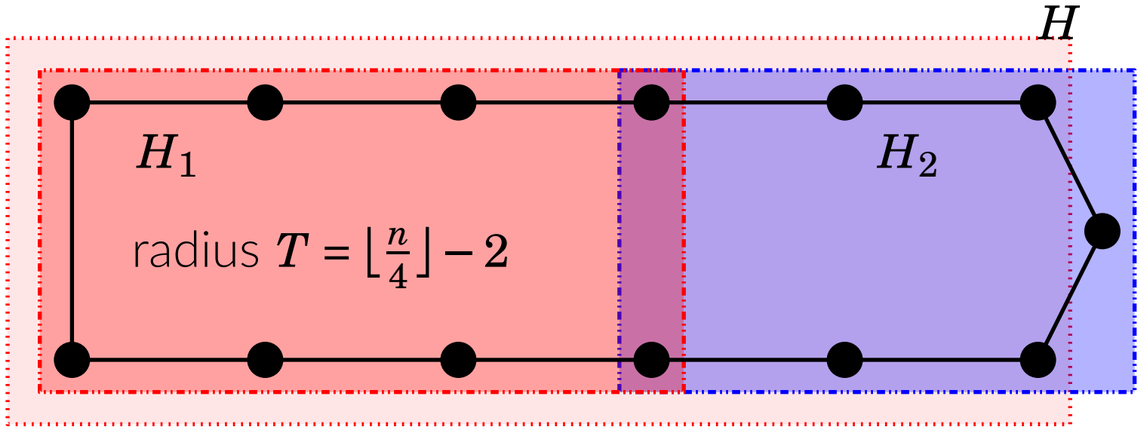
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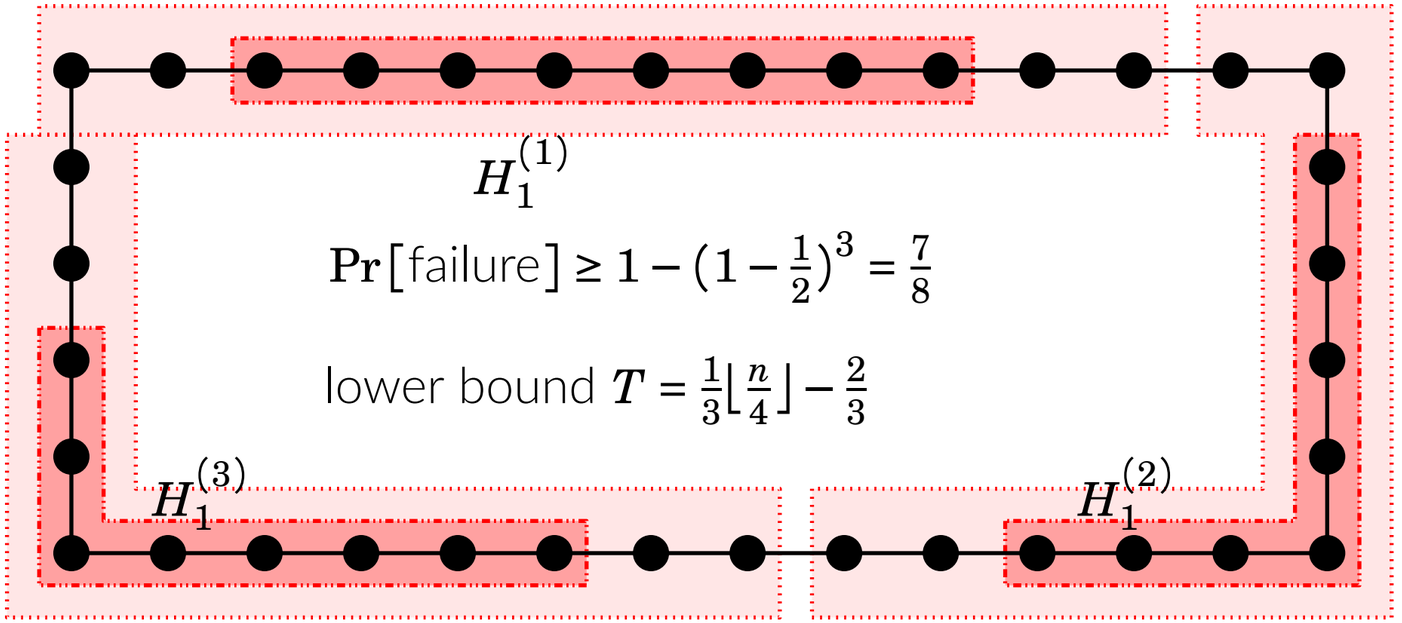
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The **non-signaling** model:

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  - outcome: function assigning to inputs (G, x) a distribution over labelings $\{(y_i, p_i)\}_{i \in I}$
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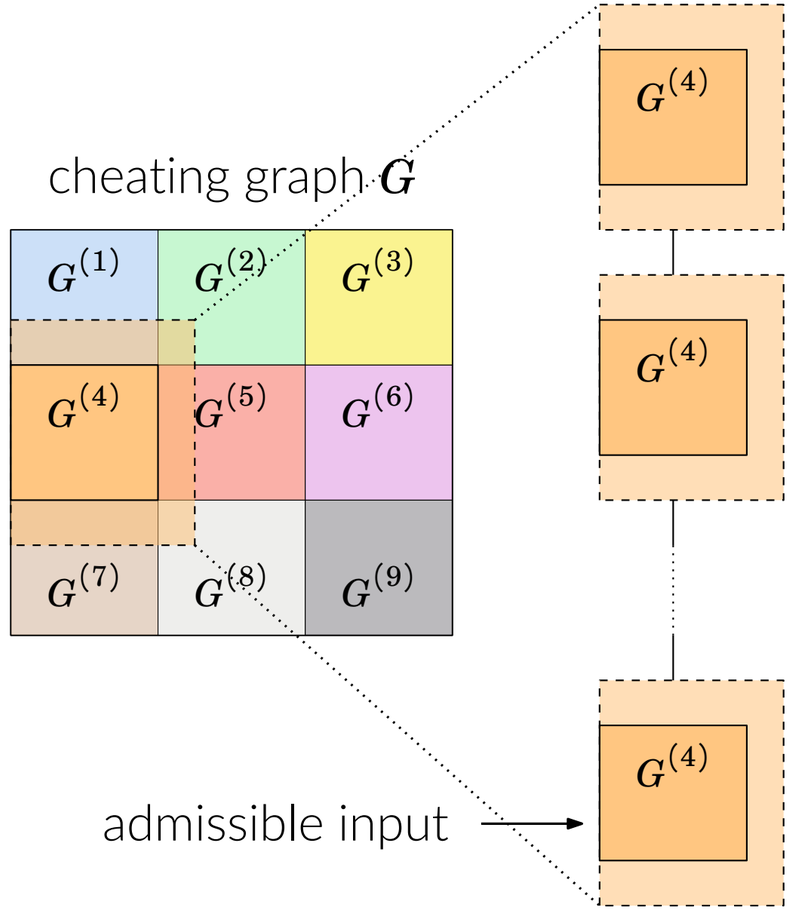
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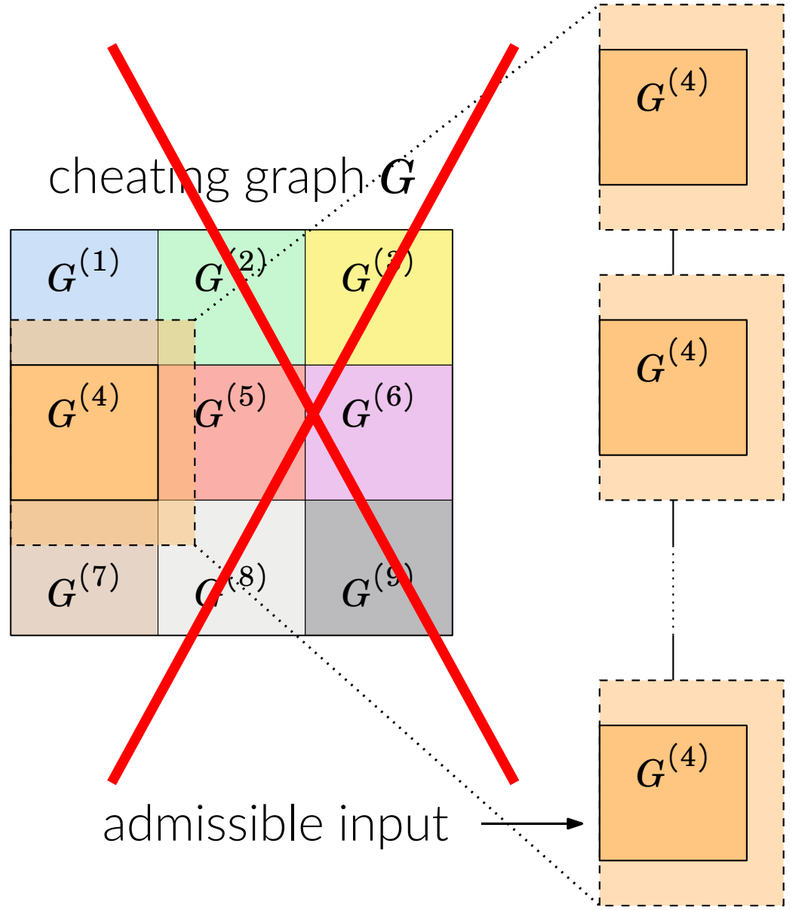
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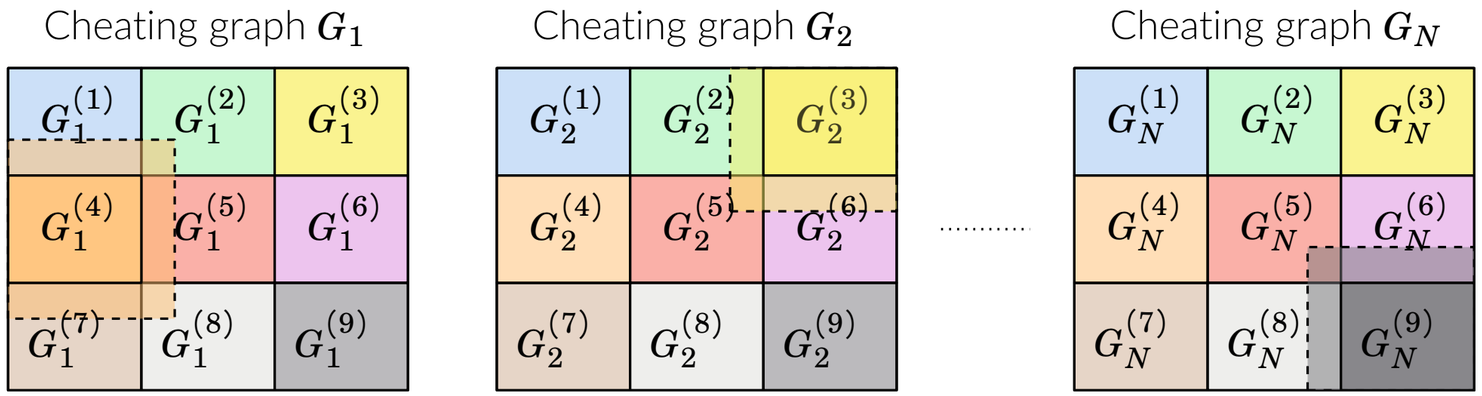
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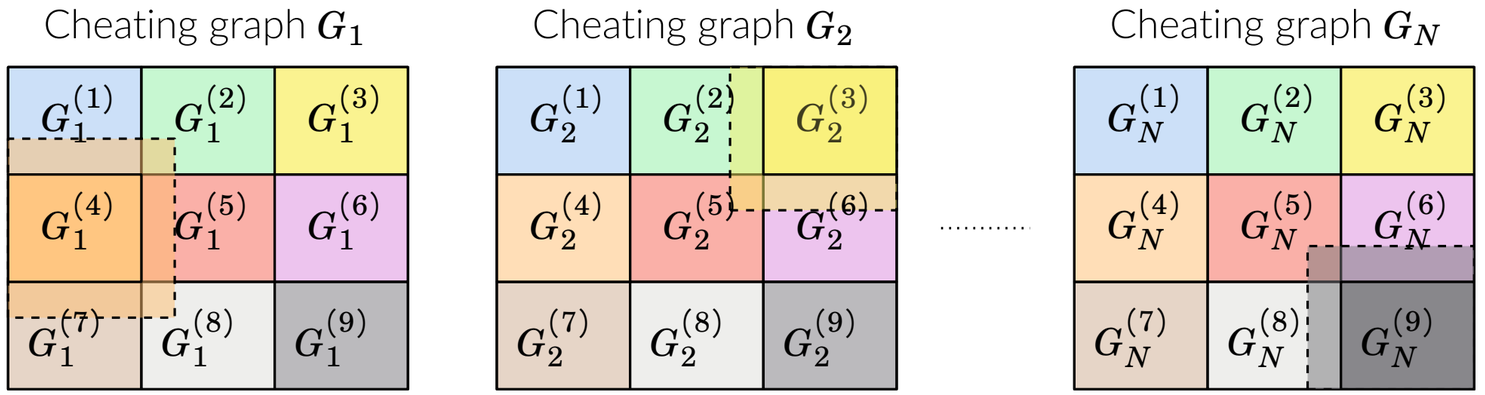


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